THE GEOLOGICAL SOCIETY OF AMERICA®

New geochronologic and paleomagnetic results from early Neoproterozoic mafic sills and late Mesoproterozoic to early Neoproterozoic successions in the eastern North China Craton, and implications for the reconstruction of Rodinia

Hanqing Zhao, Shihong Zhang†, Jikai Ding, Linxi Chang, Qiang Ren, Haiyan Li, Tianshui Yang, and Huaichun Wu
State Key Laboratory of Biogeology and Environmental Geology, China University of Geosciences, Beijing 100083, China

ABSTRACT

The interval from the late Mesoproterozoic to early Neoproterozoic is generally considered as a critical time for the amalgamation of Rodinia. The location of the North China Craton (NCC) in Rodinia remains contentious and demands greater paleomagnetic constraints. A combined geochronologic and paleomagnetic study was conducted on the late Mesoproterozoic to early Neoproterozoic rocks in the eastern NCC. Three sills were dated at ca. 945 Ma and one at ca. 920 Ma through use of the zircon U-Pb secondary ion mass spectroscopy method. Paleomagnetic investigation revealed no significant discrepancy between these sills. A positive baked-contact test, secular variation test and presence of reversals together support the primary origin interpretation for the remnant magnetization. A high-quality pole at (28.2 °S, 141.9 °E, A_{95} = 10.4°) was thus obtained by averaging our new results and a virtual geomagnetic pole previously reported for a ca. 920 Ma sill in the region. These Neoproterozoic sills intruded the successions that contain correlative strata that are named Nanfen, Xinxing, and Liulaobei formations in Liaoning, Jiangsu, and Anhui provinces, respectively. The Nanfen Formation and its equivalents are constrained between ca. 1120 and ca. 945 Ma by detrital zircons and the well-dated mafic sills. The paleomagnetic inclinations observed from the lower parts of the Nanfen, Xinxing, and Liulaobei formations are notably steep. The corresponding poles from these rock units are consistent and averaged at 38.6 °N, 136.7 °E (A_{95} = 3.2°). The quality of this pole is strengthened by a positive reversal test and its distinctiveness from the younger poles of the NCC. In the middle part of the Nanfen Formation, however, the paleomagnetic directions are characterized by moderate inclinations, being significantly different from those in the lower part of the Nanfen Formation and its equivalents. The calculated pole for the upper part of the Lower Member of the Nanfen Formation is at 8.0 °N, 128.5 °E (A_{95} = 7.9°). Another pole obtained from the Middle Member of the Nanfen Formation is at 11.2 °S, 127.7 °E (A_{95} = 8.5°). These two poles also differ from the younger poles of the NCC and likely represent the primary remanences. Our new results, together with the existing global paleomagnetic data and geological evidence, aided by the "right-way-up" connection model between Laurentia and Baltica in Rodinia, support a NCC–NW Laurentia connection between ca. 1120 and 890 Ma.

INTRODUCTION

The assembly of the Rodinia supercontinent during the late Mesoproterozoic to early Neoproterozoic has long been suggested, but its configuration remains debated (e.g., Hoffman, 1991; Pisarevsky et al., 2003; Li et al., 2008; Evans, 2009; Merdith et al., 2017). The paleogeographic location of the North China Craton (NCC) in Rodinia has been an issue of great contention. Numerous models have been proposed (Hoffman, 1991; Li et al., 1996, 2008; Zhang et al., 2000, 2006; Torsvik, 2003; Zhai et al., 2003; Peng et al., 2011a; Fu et al., 2015; Cederberg et al., 2016; Liu et al., 2017; Merdith et al., 2017; De Oliveira Chaves et al., 2019), but no consensus has been reached, at least partially due to the lack of high-quality paleomagnetic data from the late Mesoproterozoic to Neoproterozoic interval of the NCC (Fu et al., 2015). In this paper, we report new paleomagnetic and geochronologic results obtained from the ca. 945–920 Ma mafic sills and the late Mesoproterozoic to early Neoproterozoic strata exposed in the eastern NCC. These data provide new constraints for the position of NCC in Rodinia.

GEOLOGIC SETTING AND SAMPLING

The NCC had been considered as having a paucity of ca. 1.0 Ga strata or igneous rocks until recent time when the early Neoproterozoic mafic dikes and sills were identified from several locations in the craton (Liu et al., 2006; Gao et al., 2009; Peng et al., 2011a, 2011b; Wang et al., 2012; Fu et al., 2015; Zhang et al., 2016). In the eastern NCC, two group mafic sills (ca. 920 Ma and 890 Ma) were identified and classified based on their geochronologic, geochemical, and paleomagnetic characteristics (Wang et al., 2012; Fu et al., 2015; Zhang et al., 2016). They widely intruded into the Huaibei Group (Gp) in the northern Jiangsu Province (Fu et al., 2015) and the Xihe, Wuxingshan, and Jinxian groups in the eastern Liaoning Province (Zhang et al., 2016). U-Pb ages of the detrital zircons from the Huaibei and Xihe groups suggest that the strata are younger than ca. 1120 Ma (Yang et al., 2012). These strata are mainly composed of clastic and carbonate rocks. Baked contact margins between the Neoproterozoic sills and the carbonate rocks are often clearly indicated by thermal alterations that were called marblization in some of the literature (Liu et al., 2006; Wang et al., 2012; Zhang et al., 2016). The sills and the strata overall provide great promise for paleomagnetic research (Fu et al., 2015). We carried out the paleomagnetic sampling in four regions (Fig. 1). From north to south, they are the Benxi region in the eastern Liaoning Province (Figs. 1 and 2), Dalian region in the southern Liaoning Province (Figs. 1 and 3), Xuzhou region in the northern Jiangsu Province (Figs. 1, 4A, and 4B), and Huainan region in the northern Anhui Province (Figs. 1 and 4C).

†Corresponding author: shzhang@cugb.edu.cn.

GSA Bulletin; March/April 2020; v. 132; no. 3/4; p. 739–766; https://doi.org/10.1130/B35198.1; 16 figures; 3 tables; Data Repository item 2019298; published online 17 July 2019.

For permission to copy, contact editing@geosociety.org
© 2019 Geological Society of America

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

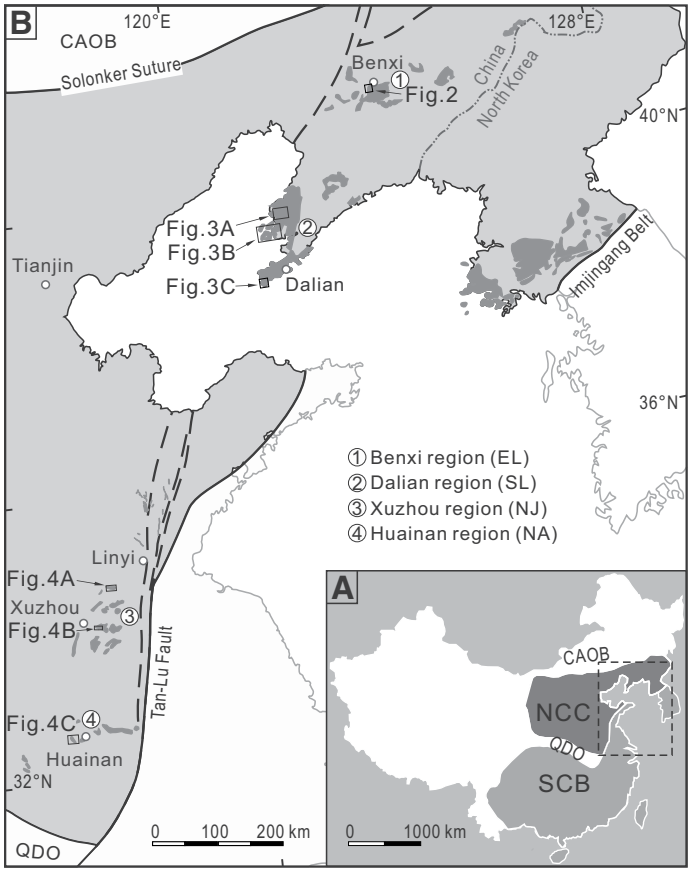


Figure 1. (A) Tectonic position of the North China Craton (NCC). (B) Distribution of the studied late Mesoproterozoic to Neoproterozoic successions in the eastern NCC. Modified from IGCAGS (2002); Chough (2013); Fu et al. (2015). CAOB—Central Asian Orogenic Belt; EL—eastern Liaoning Province; NA—northern Anhui Province; NJ—northern Jiangsu Province; QDO—Qinling-Dabie Orogen; SCB—South China Block; SL—southern Liaoning Province.

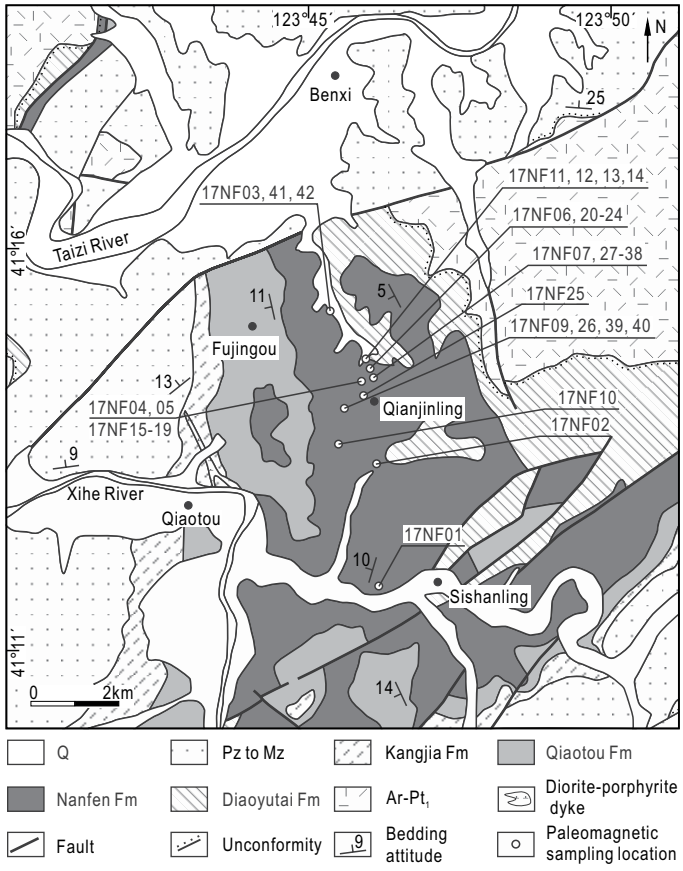


Figure 2. Simplified geological map for paleomagnetic sampling locations in the south of the Benxi region in the eastern Liaoning Province (modified from LBGMR, 1975). Ar—Archean; Fm—Formation; Mz—Mesozoic; Pt_1—Paleoproterozoic; Pz—Paleozoic; Q—Quaternary.

Benxi Region

The Benxi region is located in the eastern Liaoning Province in the northeastern segment of the NCC (Fig. 1B). Above the Archean Anshan Gp metamorphic basement, the Paleoproterozoic series, namely the Liaohe Gp, is composed of weakly metamorphosed rocks which are dominantly volcanic and Mg-rich detrital rocks. The late Mesoproterozoic to Neoproterozoic shallow marine siliciclastic and carbonate rocks are widely exposed in the Taizi River basin (Fig. 2) and are divided into the Xihe Gp and the Kangjia Formation (Fm) (Fig. 5A). The Xihe Gp is subdivided into three formations: in ascending order, the Diaoyutai Fm, Nanfen Fm, and Qiaotou Fm (LBGMR, 1989; Xing, 1989). The Nanfen Fm is conformably overlying the greyish-green quartz sandstone of the Diaoyutai Fm and is disconformably overlain by light-gray quartz sandstone of the Qiaotou Fm (Figs. 5A and 6A). The Nanfen Fm is variable in thickness (234–1084 m) and can be divided into three members (Fig. 6A). Our studied section is ~560 m thick, located in Qianjinling, ~10 km south of the downtown of Benxi city (Fig. 2). The Lower Member (~200 m) of the Nanfen Fm mainly consists of pale-blue to greenish-gray, thin- to medium-bedded micritic limestone and argillaceous limestone (Fig. 7A), with an ~10 m purplish-red thin-bedded limestone unit in the uppermost of this member (Fig. 7B). The Middle Member (~180 m) is composed of purplish-red thin-bedded calcareous mudstone intercalated with pale-blue thin-bedded limestone (Fig. 7C). The Upper Member (~180 m) is characterized by yellowish-green shale and sandy shale with thin-bedded argillaceous sandstone. The Upper Member of the Nanfen Fm yields abundant macroscopic carbonaceous compression fossils such as *Chuaria*, *Tawuia*, and *Shouhsienia* (Fig. 6A; LBGMR, 1989). In the region, the strata are generally flat-lying with westward dips up to 10°. Previous paleomagnetic study of the Nanfen Fm in the Benxi region was carried out for the Middle Member only (Lin, 1984). In this study, we collected 357 paleomagnetic samples from 40 sites in the Lower and Middle Members of the Nanfen Fm (Figs. 2 and 6A).

Dalian Region

The Dalian region is situated in the southern Liaoning Province, ~250–300 km SSW of the Benxi region (Fig. 1B). Late Mesoproterozoic to Neoproterozoic strata in this region are also a suite of siliciclastic and carbonate rocks (LBGMR, 1989; Xing, 1989; Hong et al., 1991). They are subdivided into 12 formations: in an ascending order,

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

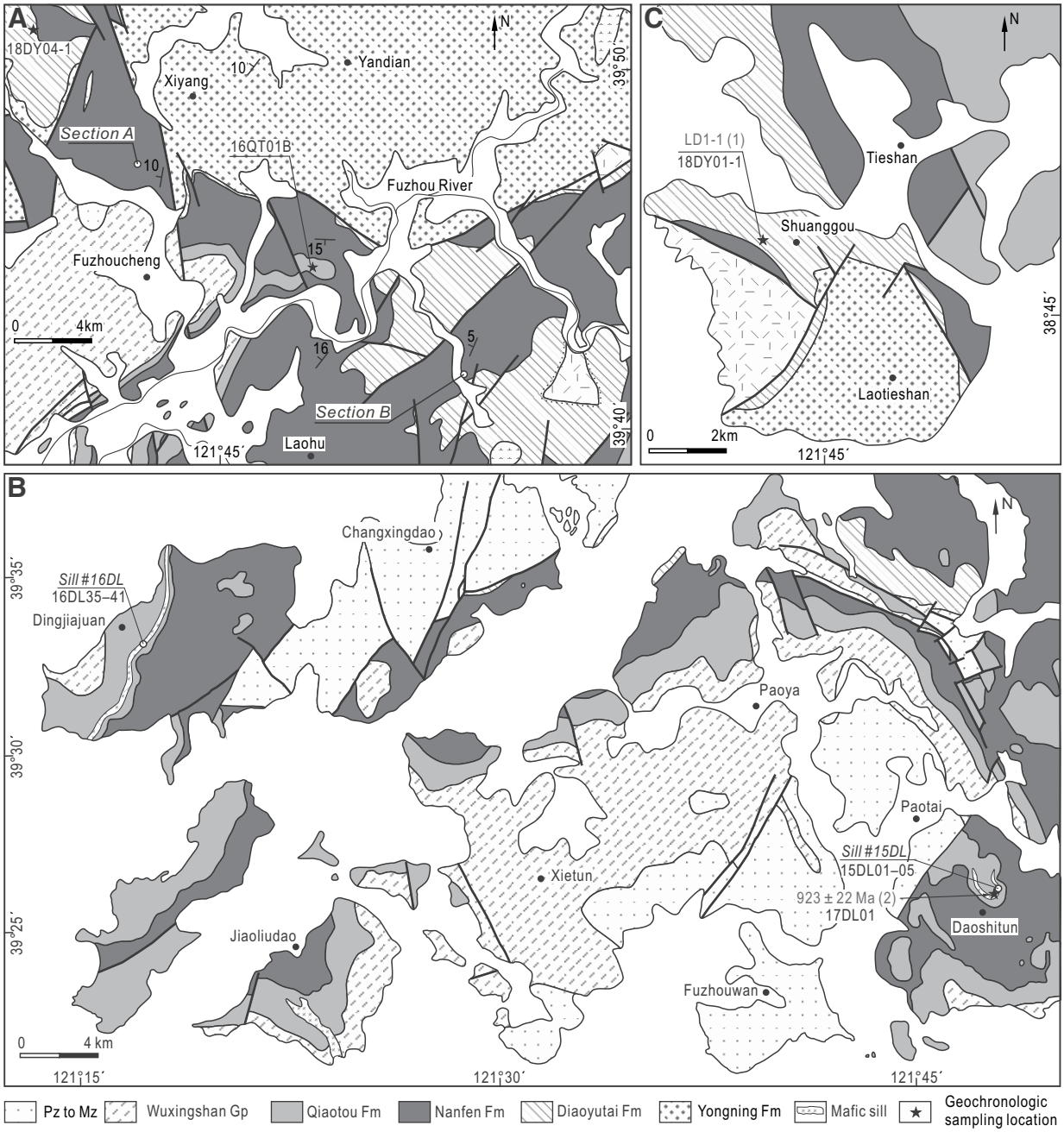


Figure 3. Simplified geological maps for paleomagnetic and geochronologic sampling locations: (A–B) Wafangdian area in the northern Dalian region, southern Liaoning Province (modified from LBGMR, 1972a); (C) Lushun area in the southern Dalian region, southern Liaoning Province (modified from LBGMR, 1972b). The locations of geochronology samples are shown with stars. References: (1) Yang et al. (2012) and (2) Zhang et al. (2016). Fm—Formation; Gp—Group; Mz—Mesozoic; Pz—Paleozoic. Other legends are as for Figure 2.

the Yongning, Diaoyutai, Nanfen, Qiaotou, Changlingzi, Nanguanling, Ganjingzi, Yingchengzi, Shisanlitai, Majiatun, Cuijiatun, and Xingmincun formations (Fig. 5B). The strata are disconformably overlain by Cambrian–Ordovician marine clastic and carbonate strata. The rock association of the Diaoyutai, Nanfen and Qiaotou formations in the region is generally similar to that in the Benxi region; meanwhile, the Changlingzi Fm is generally correlated to the Kangjia Fm in the Benxi region (LBGMR, 1989; Hong et al., 1991). There are abundant macroscopic carbonaceous compression fossils such as *Chuaria*, *Tawuia*, *Shouhsienia*, and *Proterocladus* sp. in the yellowish-green shales of the upper part of the Nanfen Fm (Fig. 6B; Duan, 1982; Lin and Xing, 1984; Hong et al., 1991; Liu et al., 2018; Xiao and Tang, 2018). Middle–Late Triassic deformation led to folding in the region (Hong et al., 1991; Yang et al., 2011). In the Wafangdian area, the northern part of the Dalian region, a total of 90 paleomagnetic samples from 11 sites were collected from the lower part of the Nanfen Fm near the Fuzhoucheng town (Figs. 3A and 6B).

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

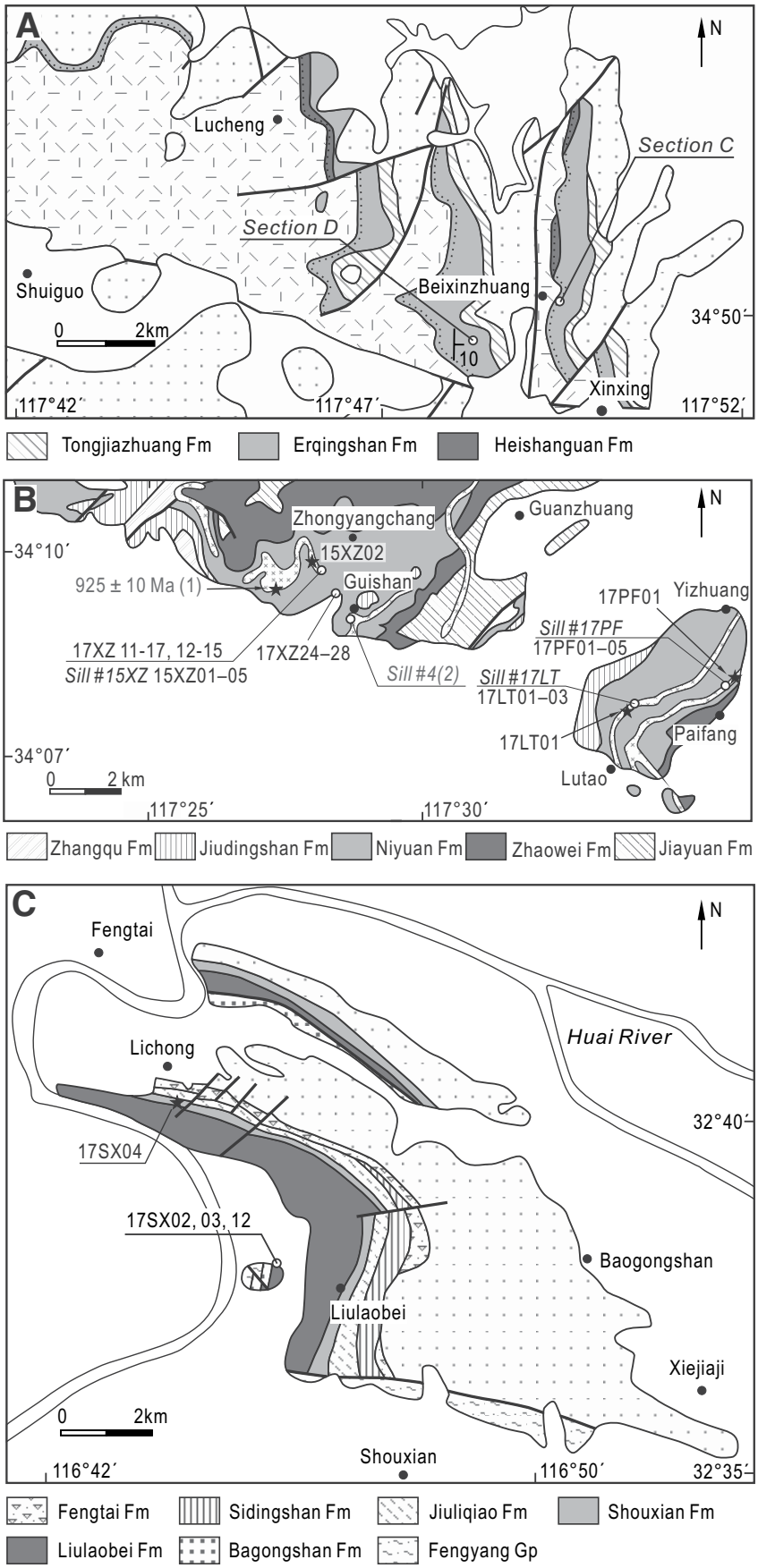


Figure 4. Simplified geological maps for paleomagnetic and geochronologic sampling locations in (A) the north of Xinxing town in the Xuzhou region (modified from SBGMR, 1961); (B) the east of Xuzhou region (modified from JBGMR, 1978, and Fu et al., 2015); (C) the north of Shouxian in the Huainan region, north Anhui Province (modified from ABGMR, 1979). As discussed in the text, the Lanling and Xinxing formations are collectively referred to as the Erqingshan Formation (Fm), and the Jushan Formation is correlated to the Tongjiazhuang Formation in Figure 4A. The locations of geochronology samples are shown with stars. (1) Isotopic ages from Liu et al. (2006) and recalculated by Fu et al. (2015). (2) Paleomagnetic analyses from Fu et al. (2015). Other legends are as for Figures 2 and 3. Gp—Group.

Among the samples, 65 samples from nine sites were from section A, and 25 samples from two sites were from section B (Fig. 3A). The strata in both sections are generally flat-lying, with northwestward dips up to 10° at some sampling sites (Figs. 3A and 7D; Table 1). Block samples for detrital zircon U–Pb geochronologic study were also collected from the Diaoyutai and Qiaotou formations in the Wafangdian area (locations of the samples 18DY04-1 and 16QT01B in Fig. 3A) and from the Diaoyutai Fm in the Lushun area, southern part of the Dalian region (location of the sample 18DY01-1 in Fig. 3C).

Xuzhou Region

The late Mesoproterozoic to Neoproterozoic strata, namely the Huaibei Gp, is a suite of siliciclastic and carbonate rocks exposed in the Xuzhou region, northern Jiangsu Province (Fig. 1B; Cao et al., 1985; Xing, 1989; Xiao et al., 2014; Fu et al., 2015). The Huaibei Gp is subdivided into 13 formations: from bottom to top, the Lanling, Xinxing, Jushan, Jiayuan, Zhaowei, Niyuan, Jiudingshan, Zhangqu, Weiji, Shijia, Wangshan, Jinshanzhai, and Gouhou formations (Fig. 5C). The Huaibei Gp is disconformably overlain by Cambrian to Ordovician sequences. Earlier paleomagnetic study of the Huaibei Gp mainly focused on the middle part of the successions (e.g., Fang et al., 1983; Fu et al., 2015). Since the exposures of the lowermost part of the Huaibei Gp—including Lanling and Xinxing formations in the north Jiangsu and Anhui provinces—are extremely limited, the standard sections of the Lanling and Xinxing formations that are studied are both located near the Xinxing town in the Linyi city, southwest Shandong Province (JBGMR, 1984; ABGMR,

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

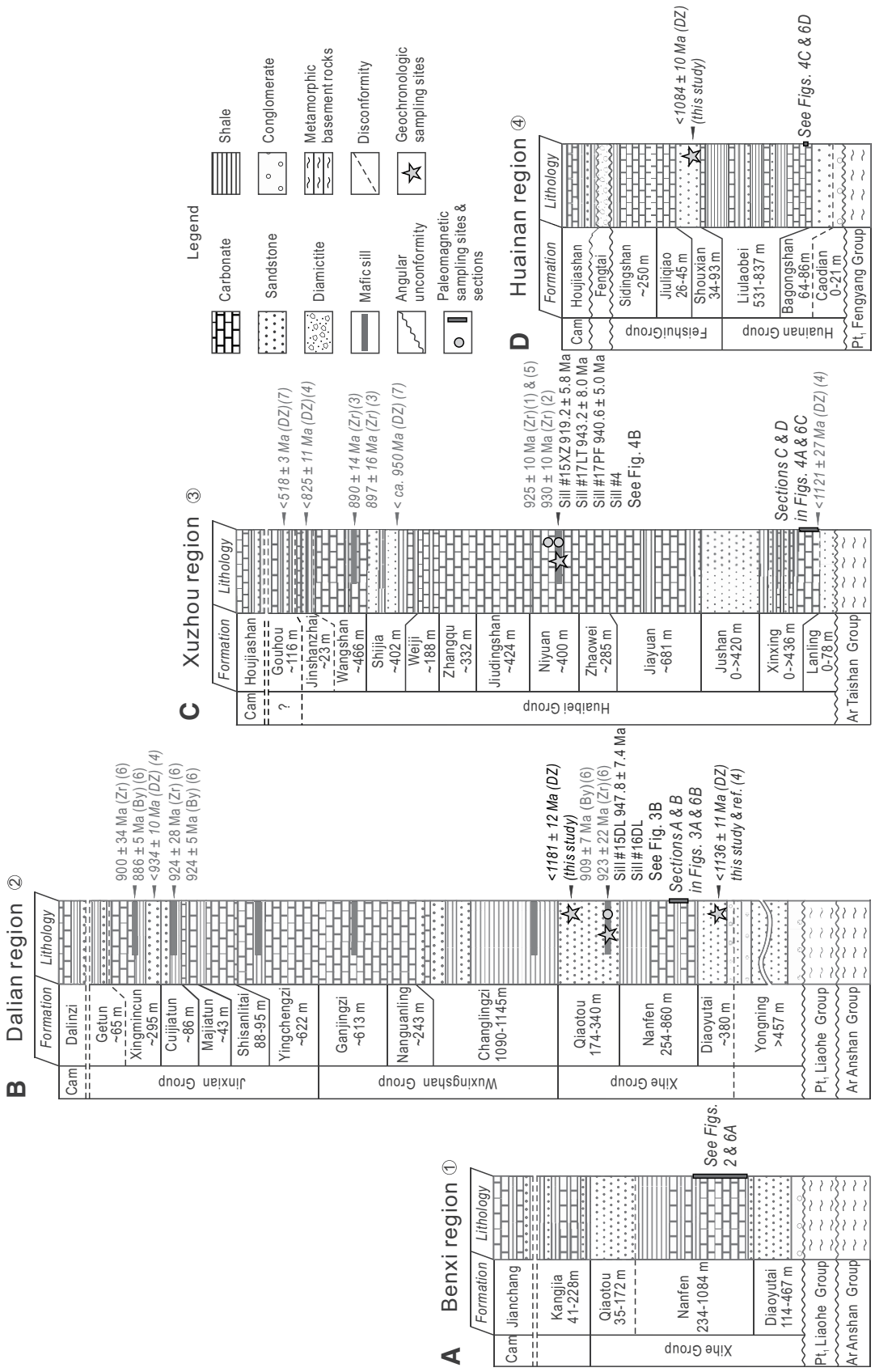


Figure 5. Simplified stratigraphic columns of the late Mesoproterozoic to Neoproterozoic successions in the four studied regions shown in Figure 1B. Modified from JBGMR (1984), Wang et al. (1984), LBGMR (1989), Hong et al. (1991), SBGMR (1991), Dong et al. (2008), Xiao et al. (2014), Fu et al. (2015), and Zhang et al. (2016). Columns are not to scale. Previous U-Pb geochronologic data are indicated in gray next to the stratigraphic columns. The paleomagnetic sampling information and the geochronologic results reported in this study are marked in black. Ar—Archean; By—baddeleyite; Cam—Cambrian; DZ—detrital zircon; Pt_1—Paleoproterozoic; Zr—magmatic zircon. References for radiogenic isotope ages: (1) Liu et al. (2006), (2) Gao et al. (2009), (3) Wang et al. (2012), (4) Yang et al. (2012), (5) Fu et al. (2015), (6) Zhang et al. (2016), and (7) He et al. (2017).

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

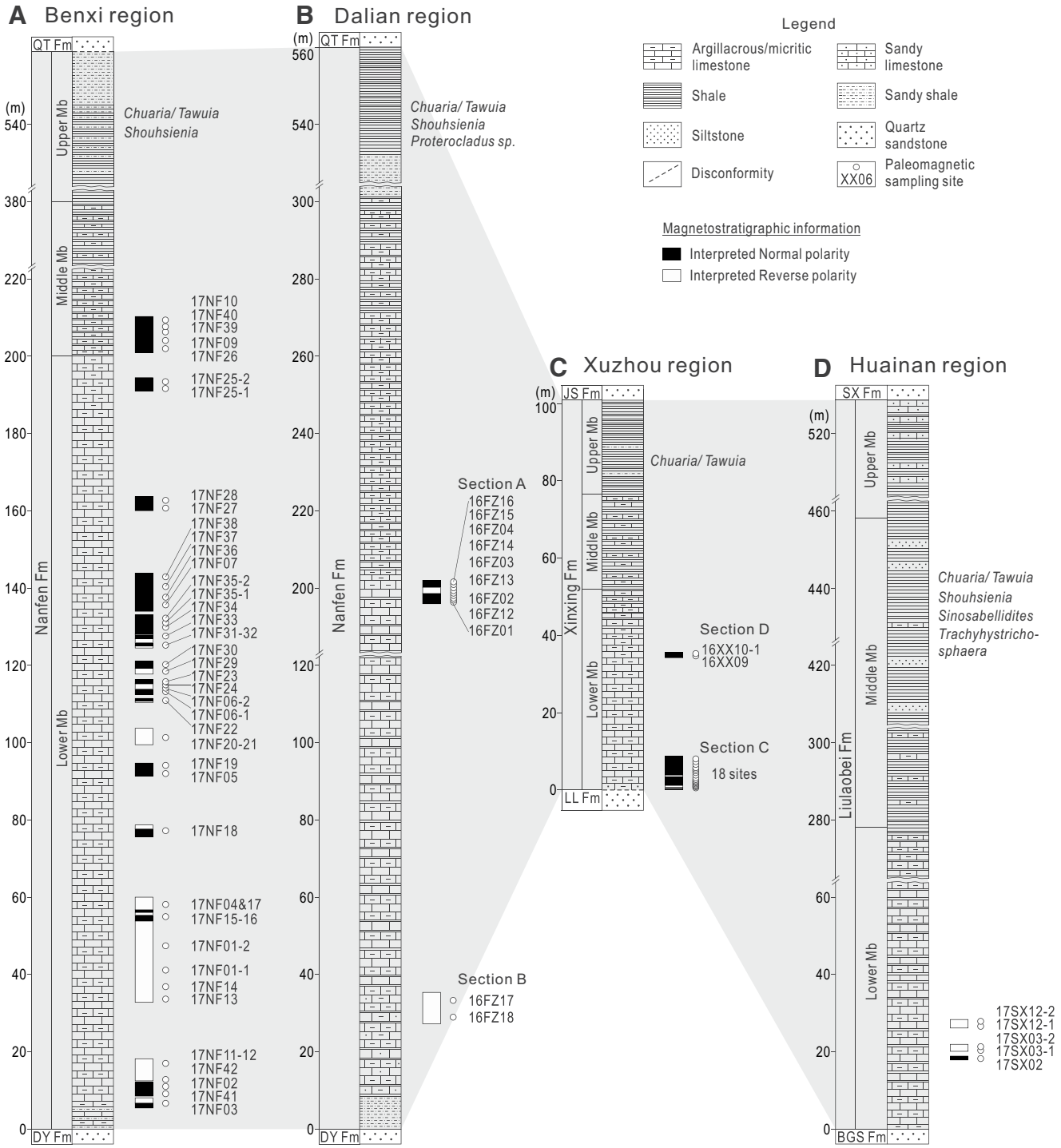


Figure 6. Lithologic strata of the studied sedimentary rock units for paleomagnetic sampling locations and magnetostratigraphic information. The suggested correlation for the Nanfen, Xinxing, and Liulaobei formations are highlighted with a gray polygon. Fossil occurrences are marked next to the stratigraphic columns (see text for references). Polarity-1/polarity-2 in this study is interpreted as normal polarity/reverse polarity. BGS—Bagongshan; DY—Diaoyutai; Fm—Formation; JS—Jushan; LL—Lanling; Mb—Member; QT—Qiaotou; SX—Shouxian.

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

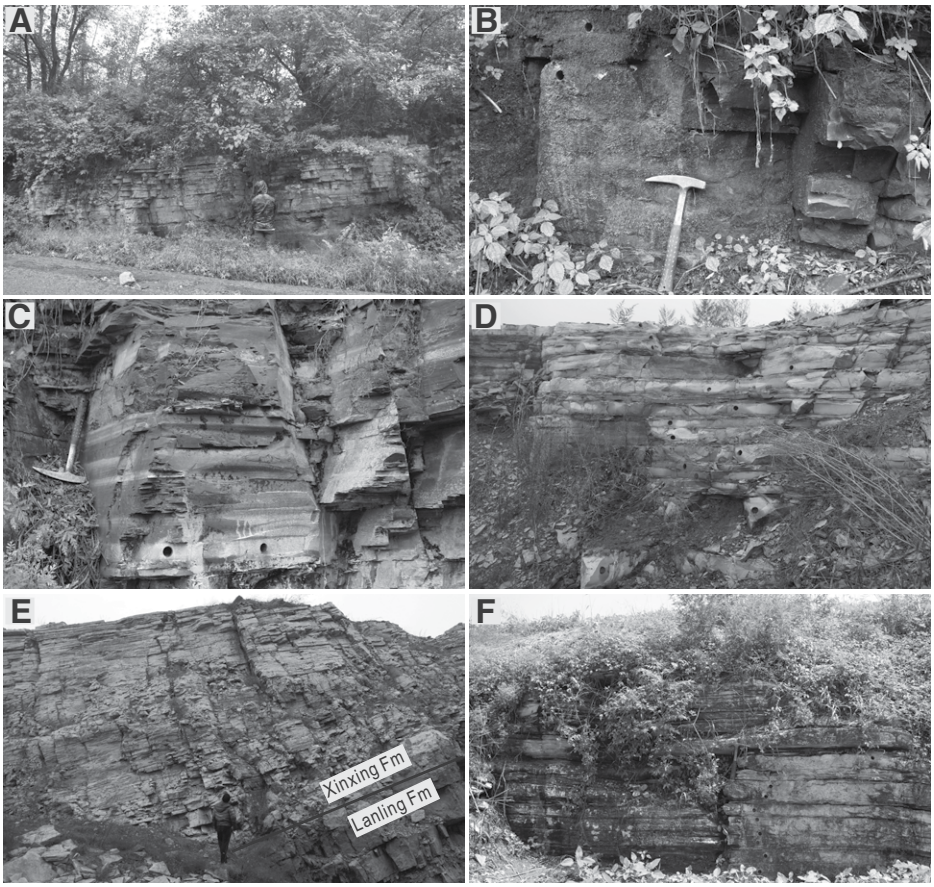


Figure 7. Representative photographs of sampling outcrops. (A) Pale blue to greenish-gray, thin- to medium-bedded limestone in the lower part of the Lower Member of the Nanfen Formation (Fm) in the Benxi region. (B) Purplish-red limestone in the upper part of the Lower Member of the Nanfen Fm in the Benxi region. (C) Purplish-red thin-bedded calcareous mudstone intercalated with pale blue thin-bedded limestone in the Middle Member of the Nanfen Fm in the Benxi region. (D) Pale blue limestone in the lower part of the Nanfen Fm in the section A in the Dalian region. (E) Quartz sandstone of the Lanling Fm and limestone from the lower part of the Xinxing Fm in the section C in the Xuzhou region. (F) Greenish-gray thin bedded micritic limestone in the lower part of the Liulaobei Fm in the Huainan region.

1987). The Lanling and Xinxing formations are collectively referred to as the Erqingshan Fm in Shandong Province; while the Jushan Fm is correlated to the Tongjiazhuang Fm (Fig. 4A), according to JBGMR (1984) and SBGMR (1991). The Xinxing Fm is conformably or disconformably overlying the gray-purple quartz sandstone of the Lanling Fm, and is conformably overlain by light-gray quartz sandstone of the Jushan Fm (Figs. 5C). The Xinxing Fm is variable in thickness (0 – >436 m) and can be divided into three members. The two studied sections (sections C and D) are located near the Beixinzhuang village (Fig. 4A). The Lower Member mainly consists of pale-blue to greenish-gray thin-bedded micritic limestone and argillaceous limestone, with gray-purple argillaceous limestone in the bottom of this member (Fig. 6C). The Middle Member is mainly composed of gray-purple to purplish-red sheet-like to thin-bedded micritic limestone and calcareous shale intercalated with pale-blue thin-bedded micritic limestone. The Upper Member is characterized by yellowish-green thin-bedded calcareous and sandy shale intercalated with thin-bedded siltstone (Fig. 6C). The Upper Member of Xinxing Fm yields abundant macroscopic carbonaceous compression fossils such as *Chuaria* and *Tawuia* (Fig. 6C; JBGMR, 1984; SBGMR, 1991). In sampling sites, the strata are moderately (10–20°) dipping to the east (Figs. 4A and 7E; Table 1). Preliminary paleomagnetic investigation for the Xinxing Fm was carried out by Fu (2016), but the sampling was limited to a single outcrop. In this study, 152 paleomagnetic samples from 18 sites were drilled from the lower part of the Xinxing Fm in the section C; while 17 samples from two sites were collected from the upper part of the Lower Member in the section D (Figs. 4A and 6C).

Huainan Region

The Huainan region is situated on the south bank of the Huai River drainage in the northern Anhui Province, southeastern part of the NCC (Fig. 1B). The late Mesoproterozoic to Neoproterozoic successions are sporadically exposed in the Huainan region, and are divided into the Huainan Gp, Feishui Gp, and Fengtai Fm (Figs. 4C and 5D). The Huainan Gp is subdivided into three formations (in ascending order, the Caodian, Bagongshan, and Liulaobei formations), while the Feishui Gp consists of the Shouxian, Jiuliuqiao, and Sidingshan formations. This succession unconformably overlies the metamorphic rocks of the Paleoproterozoic Fengyang Gp, and is unconformably overlain by the Fengtai diamictite or the Cambrian Houjiashan Fm (Wang et al., 1984; Sun et al., 1986; ABGMR, 1987; Dong et al., 2008). The Liulaobei Fm is conformably overlying the light-gray quartz sandstone of the Bagongshan Fm, and is conformably overlain by light-gray, thick-bedded, calcareous quartz sandstone of the Shouxian Fm (Figs. 5D and 6D). The Liulaobei Fm is variable in thickness (530–840 m) and can also be divided into three members. The Lower Member is composed of grayish-purple and greenish-gray, thin- to medium-bedded micritic limestone, argillaceous limestone, and dolomitic limestone intercalated with calcareous shale (Fig. 7F). The upper strata of this member are generally covered by Quaternary deposits. The Middle Member consists of gray and yellowish-green silty mudstone and shale with calcareous limestone and siltstone interbeds. The Upper Member is characterized by yellowish-green and gray shale with thin-bedded fine-grained calcareous siltstone and sandy limestone. The shales of the Liulaobei Fm contain well-preserved organic-walled microfossils and macroscopic carbonaceous compression fossils such as *Chuaria*, *Tawuia*, *Shouhsienia Sinosabellidites*, and *Trachyhystrichosphaera aimika* (Duan, 1982; Sun et al., 1986; Wang et al., 1984; Zang and Walter, 1992; Tang et al., 2013; Xiao et al., 2014; Tang et al., 2017). Previous paleomagnetic study of the Liulaobei Fm in the Huainan region mainly focused on the red calcareous shale of the Lower Member (Piper and Zhang, 1997), but their results have no verifiable details. We collected 47 greenish-gray limestone samples from five sites in the lower part of Liulaobei Fm near the Liulaobei village, ~5 km north of the Shouxian county (Figs. 4C and 6D). The sampled strata are moderately

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

TABLE 1. PALEOMAGNETIC RESULTS FROM THE NANFEN FORMATION IN THE BENXI AND DALIAN REGIONS, XINXING FORMATION IN THE XUZHOU REGION AND LIULAOBEI FORMATION IN THE HUAINAN REGION, EASTERN NORTH CHINA CRATON

Site	Location		n/n_0	Dip direction /Dip (°)	In situ		Tilt-corrected				Tilt-corrected		
	Lat (°N)	Long (°E)			*D* (°)	*I* (°)	*D* (°)	*I* (°)	*k*	α_{95} (°)	Plat (°N)	PLong (°E)	*dp*/*dm* A_{95} (°)
Benxi region													
Limestone in the lower part of the Lower Member of the Nanfen Fm (BNFa)													
17NF01-1	41.199	123.767	8/9	320/10	281.5	–67.7	259.3	–74.3	156.9	4.4	–40.0	342.7	7.2/8.0
17NF01-2	41.199	123.767	8/11	320/10	321.9	–70.1	336.0	–80.7	50.2	8.0	–24.4	311.8	14.8/15.4
17NF02	41.224	123.770	8/9	305/10	139.0	80.6	212.1	86.9	91.3	5.8	35.9	119.7	11.5/11.5
17NF03	41.258	123.762	5/9	250/8	55.5	79.2	22.6	86.4	59.7	10.0	47.8	127.9	19.8/19.9
17NF41	41.258	123.762	6/6	250/8	49.8	79.6	6.9	86.0	51.9	9.4	49.2	125.2	18.5/18.7
17NF42	41.258	123.762	10/13	250/8	241.8	–77.5	228.0	–85.3	86.4	5.2	–47.1	314.0	10.2/10.3
17NF11-12	41.246	123.773	7/11	250/7	246.9	–71.9	241.1	–78.4	58.9	7.9	–48.4	333.8	14.1/14.9
17NF13	41.245	123.772	7/8	268/6	251.5	–73.5	242.7	–79.2	100.7	6.0	–47.6	331.8	10.9/11.4
17NF14	41.245	123.772	8/10	268/6	245.1	–71.2	235.7	–76.6	116.4	5.2	–51.0	338.2	9.0/9.7
17NF15-16	41.241	123.771	6/15	271/4	96.4	80.4	100.1	84.4	34.8	11.5	38.4	137.8	22.4/22.7
17NF04&17	41.241	123.771	11/14	271/4	173.9	–79.9	153.6	–78.7	40.9	7.1	–59.6	284.8	12.7/13.4
17NF18	41.242	123.771	6/6	250/3	70.8	75.8	70.9	78.8	46.6	9.9	44.7	153.1	17.8/18.8
17NF05	41.241	123.771	7/8	211/7	91.6	81.6	142.2	82.2	124.0	5.4	28.6	134.4	10.2/10.5
17NF19	41.241	123.771	7/8	211/9	103.4	82.1	161.8	80.0	71.3	7.2	22.6	130.2	13.2/13.8
17NF20-21	41.242	123.772	10/18	260/7	186.9	–76.1	157.8	–76.5	50.1	6.9	–63.6	282.2	11.9/12.8
17NF22	41.242	123.773	7/7	255/6	83.0	79.5	93.2	85.4	37.8	9.9	40.1	135.7	19.4/19.6
17NF06-1	41.243	123.773	6/6	224/6	74.8	80.5	109.8	84.7	306.1	3.8	37.0	136.2	7.4/7.5
17NF06-2	41.243	123.773	5/5	224/6	280.9	–73.5	301.1	–75.9	234.8	5.0	–24.5	328.8	8.5/9.2
17NF24	41.243	123.773	6/6	224/6	91.3	80.6	130.5	83.1	76.0	7.7	31.7	135.9	14.8/15.1
17NF23	41.243	123.773	4/6	224/6	58.9	87.2	211.7	86.6	169.7	7.1	35.4	119.4	14.1/14.1
17NF29	41.243	123.774	10/11	250/4	299.9	–80.4	323.4	–82.3	51.2	6.8	–28.6	314.0	12.9/13.2
17NF30	41.243	123.774	9/9	250/4	82.7	79.7	90.5	83.6	206.6	3.6	39.9	140.4	6.9/7.1
17NF31-32	41.244	123.774	8/13	249/5	281.0	–80.2	306.4	–83.9	73.9	6.5	–33.5	315.4	12.6/12.8
17NF33	41.244	123.774	7/7	251/6	109.1	78.9	138.9	82.7	94.3	6.2	29.9	134.6	11.8/12.1
17NF34	41.244	123.774	8/10	251/6	115.8	78.5	145.7	81.6	31.2	10.1	27.2	134.1	19.0/19.6
17NF35-1	41.244	123.774	8/8	251/6	93.6	81.8	134.8	86.5	343.2	3.0	36.2	129.9	5.9/6.0
17NF35-2	41.244	123.774	9/9	251/6	119.1	85.7	203.5	85.6	38.2	8.4	33.1	119.6	16.5/16.7
17NF07	41.243	123.774	8/11	247/5	103.4	80.9	133.7	84.2	387.3	2.8	32.9	133.6	5.4/5.5
17NF36	41.243	123.774	6/8	247/5	68.4	75.7	69.1	80.7	108.4	6.5	45.2	148.2	12.1/12.5
17NF37	41.243	123.774	5/6	247/5	20.7	82.0	342.0	84.2	57.6	10.2	52.0	118.0	19.8/20.1
17NF38	41.243	123.774	5/7	247/5	65.2	80.7	63.1	85.7	240.3	4.9	44.6	134.5	9.6/9.7
17NF27	41.24	123.773	8/8	252/6	129.7	76.7	155.8	78.7	158.3	4.4	21.0	133.1	7.9/8.3
17NF28	41.24	123.773	7/8	252/6	140.0	76.2	165.3	77.3	88.9	6.4	17.6	130.1	11.2/12.0
Mean of 33 sites*					86.7	80.3	102.5	85.6	112.5	2.4	38.8	134.6	A_{95} = 4.6
Purplish red limestone in the upper part of the Lower Member of the Nanfen Fm (BNFb)													
17NF25-1	41.239	123.772	8/8	240/7	140.6	71.8	162.0	71.7	111.0	5.3	8.9	133.7	8.2/9.3
17NF25-2	41.239	123.772	8/8	240/7	160.1	73.7	181.0	71.2	42.4	8.6	7.0	123.2	13.1/15.0
Mean of 16 samples					149.8	73.0	171.5	71.7	60.0	4.8	8.0	128.5	A_{95} = 7.9
Purplish red thin-bedded calcareous mudstone intercalated with pale blue limestone in the Middle Member of the Nanfen Fm (BNFc1)													
17NF26	41.238	123.771	7/7	237/6	152.0	66.9	165.4	65.7	158.6	4.8	0.1	133.5	6.4/7.8
17NF09	41.238	123.764	9/10	282/13	148.9	55.9	167.5	63.1	189.2	3.8	–3.4	132.6	4.7/6.0
17NF39	41.236	123.764	8/10	290/5	141.0	56.3	145.4	60.5	52.2	7.7	–1.6	149.0	8.9/11.7
17NF40	41.236	123.764	9/9	290/5	146.8	50.8	150.9	54.7	90.4	5.4	–9.0	147.5	5.4/7.6
17NF10	41.233	123.764	5/5	280/8	164.6	55.2	176.0	57.9	46.2	11.4	–10.1	126.9	12.3/16.8
Mean of 5 sites					150.5	57.3	160.7	60.9	112.8	7.2	–4.9	137.9	A_{95} = 10.2
Dalian region													
Limestone in the lower part of the Nanfen Fm (WNFa)													
Section A													
16FZ01	39.785	121.706	10/10	332/2	118.0	81.6	108.7	83.2	129.5	4.3	34.4	137.2	8.3/8.4
16FZ12	39.785	121.706	8/9	332/2	110.6	77.9	103.6	79.3	169.2	4.3	32.3	145.7	7.8/8.2
16FZ02	39.785	121.706	8/8	332/2	103.1	78.4	94.9	79.6	100.6	5.5	35.3	146.6	10.0/10.5
16FZ13	39.785	121.706	7/7	332/2	89.3	70.4	84.1	71.2	60.4	7.8	35.0	164.8	11.9/13.6
16FZ03	39.785	121.706	8/9	332/2	273.0	–75.6	265.8	–76.6	25.3	11.2	–37.0	334.2	19.3/20.8
16FZ14	39.785	121.706	7/8	332/2	265.6	–84.3	245.0	–84.8	117.5	5.6	–43.4	314.6	10.9/11.1
16FZ04	39.785	121.706	4/7	332/2	79.9	83.4	62.2	83.8	78.6	10.4	44.5	137.0	20.1/20.4
16FZ15	39.785	121.706	7/9	332/2	143.1	56.2	142.6	58.2	65.5	7.5	–4.2	150.0	8.2/11.1
16FZ16	39.785	121.706	4/6	332/2	103.1	81.3	91.7	82.4	135.9	7.9	37.8	140.7	15.0/15.4
Section B													
16FZ17	39.695	121.905	8/9	306/8	240.6	–69.7	218.6	–72.7	130.0	4.9	–59.3	342.2	7.7/8.7
16FZ18	39.695	121.905	6/8	306/8	281.7	–78.4	251.8	–85.5	78.1	7.6	–41.9	313.4	14.9/15.1
Mean of 11 sites†					102.7	77.5	93.7	79.8	59.7	6.0	36.7	145.8	A_{95} = 9.9
Xuzhou region													
Limestone in the lower part of the Xinxing Fm (XXa)													
Section C													
XX01	34.832	117.830	8/8	107/17	274.4	85.7	122.2	79.9	481.3	2.5	23.0	135.8	4.6/4.8
XX02	34.832	117.830	9/10	107/17	316.4	84.1	99.4	81.1	325.4	2.9	30.3	137.8	5.4/5.6
XX03	34.832	117.830	10/10	107/17	341.5	80.6	66.1	79.9	154.9	3.9	40.5	141.6	7.1/7.5
XX04-1	34.832	117.830	6/6	107/17	304.7	69.4	326.0	80.2	87.6	7.2	49.6	101.5	13.2/13.8
XX05-1	34.832	117.830	7/7	107/17	350.4	85.5	84.8	79.3	161.6	4.8	34.1	143.0	8.7/9.1
XX05-2	34.832	117.830	9/9	107/17	323.3	80.2	51.7	82.9	93.8	5.3	42.6	132.8	10.1/10.4
XX06	34.832	117.830	11/11	107/17	266.1	85.2	119.9	82.3	67.9	5.6	26.4	132.5	10.6/10.9

(*Continued*)

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

TABLE 1. PALEOMAGNETIC RESULTS FROM THE NANFEN FORMATION IN THE BENXI AND DALIAN REGIONS, XINXING FORMATION IN THE XUZHOU REGION AND LIULAOBEI FORMATION IN THE HUAINAN REGION, EASTERN NORTH CHINA CRATON (*Continued*)

Site	Location		n/n_0	Dip direction /Dip	In situ		Tilt-corrected				Tilt-corrected		
	Lat (°N)	Long (°E)		(°)	*D* (°)	*I* (°)	*D* (°)	*I* (°)	*k*	α_{95} (°)	Plat (°N)	PLong (°E)	*dp*/*dm* A_{95} (°)
Xuzhou region													
Limestone in the lower part of the Xinxing Fm (XXa)													
Section C													
XX14-1	34.832	117.830	6/7	107/17	145.0	–72.0	184.9	–78.8	276.8	4.0	–56.3	301.1	7.2/7.6
XX14-2	34.832	117.830	6/7	107/17	135.0	–75.3	188.3	–83.1	586.5	2.8	–48.3	300.8	5.4/5.5
XX14-3	34.832	117.830	6/6	107/17	155.7	–72.8	198.9	–77.2	145.8	5.6	–57.3	312.2	9.8/10.5
XX15-1	34.832	117.830	9/9	107/17	149.1	–81.7	243.0	–82.0	162.3	4.1	–40.6	316.3	7.7/8.0
XX15-2	34.832	117.830	10/10	107/17	135.6	–84.1	264.5	–82.6	72.4	5.7	–34.9	315.6	10.9/11.1
XX16	34.832	117.830	10/11	107/17	214.3	–82.7	260.4	–74.2	83.9	5.3	–34.4	333.9	8.7/9.6
XX17-1	34.832	117.830	7/7	107/17	20.0	–84.9	310.6	–77.2	142.7	5.1	–17.4	317.0	8.9/9.5
XX17-2	34.832	117.830	7/8	107/17	88.9	–84.6	300.7	–82.9	160.7	4.8	–26.9	311.3	9.2/9.4
XX18	34.832	117.830	7/8	107/17	65.2	–87.0	298.6	–80.0	110.2	5.8	–24.1	316.5	10.6/11.1
XX19	34.832	117.830	8/9	107/17	314.0	78.8	38.9	84.6	154.8	4.5	42.8	127.0	8.8/8.9
XX20	34.832	117.830	8/9	107/17	309.7	75.4	2.0	84.2	106.6	5.4	46.3	118.4	10.5/10.6
Section D													
16XX09	34.824	117.811	11/12	92/9	134.9	83.4	110.1	75.5	65.5	5.7	22.2	145.6	9.6/10.5
16XX10-1	34.824	117.811	5/5	92/9	150.7	86.4	107.9	78.7	391.1	3.9	25.9	140.9	7.0/7.4
Mean of 20 sites§					316.2	83.2	77.7	83.2	113.9	3.1	36.6	134.1	A_{95} = 5.9
Huainan region													
Limestone in the lower part of the Liulaobei Fm (LLBa1)													
17SX02	32.629	116.758	10/14	49/12	336.7	89.0	44.5	77.7	27.8	9.3	47.3	141.1	16.4/17.5
17SX03-1	32.629	116.758	7/8	49/12	273.9	–86.1	239.7	–75.0	65.5	7.5	–42.5	330.4	12.5/13.7
17SX03-2	32.629	116.758	9/9	49/12	154.0	–83.0	202.6	–74.6	48.2	7.5	–57.9	317.2	12.4/13.6
17SX12-1	32.629	116.757	6/7	49/12	162.2	–84.7	209.7	–75.1	318.2	3.8	–55.1	320.7	6.3/6.9
17SX12-2	32.629	116.757	8/9	49/12	11.8	–83.8	257.4	–82.0	27.6	10.7	–34.7	315.5	20.2/20.8
Mean of 5 sites#					337.5	88.7	43.4	77.5	212.8	5.3	47.6	141.1	A_{95} = 9.7
NFL (mean pole by averaging the 69 site VGPs from the lower parts of the Nanfen, Xinxing and Liulaobei formations)**											38.6	136.7	A_{95} = 3.2

Note: n/n_0—number of samples used to calculate mean/total samples demagnetized; *D*—declination; *I*—inclination; *k*—precision parameter of Fisher (1953); α_{95}—radius of 95% confidence cone of the mean direction; Plat, Plong—latitude, longitude of the virtual geomagnetic pole (VGP); *dm, dp*—semi-minor and semi-major axes of 95% polar error ellipse. A_{95}—radius of 95% confidence cone of the paleomagnetic pole.

*All the 33 sites jointly pass a reversal test (McFadden and McElhinny, 1990), class B, with a critical angle 5.1°, and the angular difference is 3.8°.

†All the 11 sites jointly pass a reversal test (McFadden and McElhinny, 1990), class C, with a critical angle 12.0°, and the angular difference is 7.1°.

§All the 20 sites jointly pass a reversal test (McFadden and McElhinny, 1990), class B, with a critical angle 6.7°, and the angular difference is 1.6°.

#All the 5 sites jointly pass a reversal test (McFadden and McElhinny, 1990), class C, with a critical angle 16.4°, and the angular difference is 0.4°.

**The two polarity groups are consistent with an antipodal common mean at the 95% confidence level and pass the bootstrap reversal test (Tauxe, 2010).

(10–20°) dipping to northeast. One block sample for detrital zircon U-Pb geochronologic study was collected from the Shouxian Fm near the Lichong town (sample 17SX04; Fig. 4C).

Early Neoproterozoic Mafic Sills in the Dalian and Xuzhou Regions

In the previous paleomagnetic investigation of the early Neoproterozoic mafic sills in the Xuzhou region, Fu et al. (2015) reported a high-quality paleomagnetic pole obtained from the ca. 890 Ma sills. However, the paleomagnetic data for the ca. 920 Ma sills remain limited. Only a virtual geomagnetic pole (VGP) from sill #4 (Fig. 4B) was reported. We conducted a combined geochronologic and paleomagnetic study for the older group of sills in the Dalian and Xuzhou regions. Samples were collected from five diabase sills: two intruding the Qiaotou Fm in the Dalian region (sills #15DL and #16DL; Fig. 3B) and three intruding the Niyuan Fm in the Xuzhou region (sills #15XZ, #17LT and #17PF; Fig. 4B). The sills sometimes display spheroidal weathering structure on the surface and columnar joints at the edges. Their thicknesses vary from dozens to hundreds meters and the lengths range from hundreds of meters to over ten kilometers. The mafic sills display typical diabasic texture with a mineral assemblage of clinopyroxene and plagioclase (Fig. DR1[1]). In the Xuzhou region, the carbonate host rocks of the Niyuan Fm near the sills have been partially altered to marble or skarn, and the alteration diminishes as the distance increases. This provides the possibility to conduct a baked-contact test.

GEOCHRONOLOGY

Sample Preparation and Imaging

Four fresh diabase block samples from the sill #15XZ, sill #17LT, and sill #17PF in the Xuzhou region (Fig. 4B) and the sill #15DL in the Dalian region (Fig. 3B) were collected to separate zircon grains for secondary ion mass spectroscopy (SIMS) U-Pb geochronologic analyses. Four sandstone block samples (18DY01-1, 18DY04-1, 16QT01B, and 17SX04) were used to collect the detrital zircon grains for laser ablation–inductively coupled plasma–mass spectrometry (LA-ICP-MS) U-Pb analyses. With conventional density and magnetic separation techniques, the zircon grains were separated from the block samples at the mineral separation laboratory of the Bureau of Geology and Mineral Resources, Langfang, Hebei Province. The zircon grains were mounted with the standard grains in transparent epoxy and the mounts were polished to expose the interior of the crystals. The zircons were documented with transmitted and reflected light micrographs and cathodoluminescence (CL) images to reveal their internal structures and find suitable positions for in situ measurements.

LA-ICP-MS Detrital Zircon U-Pb Analytical Methods and Results

The detrital zircon U-Pb analyses were performed using a New Wave Up 193 nm laser

[1]GSA Data Repository item 2019298, Figures DR1–DR5 and Tables DR1–DR3, is available at http://www.geosociety.org/datarepository/2019 or by request to editing@geosociety.org.

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

ablation system (New Wave Research Inc., Fremont, California, USA) and LA-ICP-MS (PlasmaQuant®; Analytik Jena AG, Jena, Germany) at Beijing Createch Testing Technology Co., Ltd. The analyses were carried out with a beam diameter of ~25 μm, a repetition rate of 10 Hz and an energy density of 4 J/cm^2. The analytical procedures followed the method described by Hou et al. (2009). The ICPMSDataCal 8.4 program was used for fractionation correction and calculation of the dating results (Liu et al., 2010). Zircon standard 91500 (Wiedenbeck et al., 1995) was used as an external standard for U-Pb dating. Representation of statistical distribution for visualization of data was made with probability density plots (PDPs; Ludwig, 2003) and kernel density estimates (KDEs; Vermeesch, 2012). Weighted mean age calculation was performed using Isoplot program (Ludwig, 2003), after the exclusion of analyses with discordance >5%. The $^{206}Pb/^{238}U$ ages are preferred for zircons with ages younger than 1500 Ma, and $^{207}Pb/^{206}Pb$ ages are preferred for grains older than 1500 Ma, following recommendations by Spencer et al. (2016). The uncertainties on individual analyses in the data table and on concordia plots are presented at 1σ, while the uncertainties in weight mean ages are given at the 95% confidence level. The LA-ICP-MS detrital zircon U-Pb data are summarized in Table DR1 (see footnote 1).

Diaoyutai Formation in the Dalian Region (Samples 18DY01-1 and 18DY04-1)

A gray quartz sandstone sample (18DY01-1) was collected from the Shuanggou section in the Lushun area (Fig. 3C). More than 2000 detrital zircon grains were separated from this sample, of which 142 grains were used to conduct U-Pb geochronologic analyses. The sizes of the zircons are generally 100–120 μm (Fig. DR3A; see footnote 1). Subrounded and rounded shapes of the zircon grains imply a long transport distance from their source. The CL images reveal oscillatory growth zoning (Fig. DR3A) for most grains, indicating their magmatic origin (Corfu et al., 2003). 128 grains with discordance ≤5% yield five age populations (Table DR1): 1757 ± 14 Ma (n = 22), 1615 ± 10 Ma (n = 21), 1430 ± 22 Ma (n = 21), 1205 ± 10 Ma (n = 9), and 1134 ± 12 Ma (n = 14) (Fig. DR4A; see footnote 1).

The zircon grains separated from gray glauconite-bearing quartz sandstone of the Diaoyutai Fm in the Wafangdian area (sample 18DY04-1; Fig. 3A) range from 100 to 150 μm in size. Their CL images display oscillatory growth zoning (Fig. DR3B), together with the Th/U ratios (0.2–2.5), indicating their magmatic origin (Corfu et al., 2003; Hoskin and Schaltegger, 2003). The zircon grains are well rounded, which suggests they were transported a long distance and/or experienced a high degree of abrasion. Ninety-four analyses out of 101 grains were generally concordant (Fig. DR4B) and only seven analyses that were >5% discordant were excluded from further analyses (Table DR1). The 94 analyses yield five age populations that are generally similar to the sample 18DY01-1: 1797 ± 18 Ma (n = 22), 1617 ± 16 Ma (n = 16), 1473 ± 27 Ma (n = 6), 1197 ± 8 Ma (n = 8), and 1156 ± 8 Ma (n = 10) (Fig. DR4B).

The new results of the detrital zircon analyses are consistent with the data previously published by Yang et al. (2012; sample LD1-1 in Fig. 3C). We thus put the data together and estimated the age of the youngest zircon grains. The youngest peak with a weighted mean age of 1136 ± 11 Ma (N = 33, 2σ, mean square of weighted deviates [MSWD] = 4.3) is considered as the maximum depositional age of the Diaoyutai Fm, and of course also being a constraint for the overlying Nanfen Fm (Fig. 5B).

Qiaotou Formation in the Dalian Region (Sample 16QT01B)

A quartz sandstone sample (16QT01B) was collected from the Qiaotou Fm near the town of Fuzhoucheng (Fig. 3A). Zircon grains from this sample are generally 120–200 μm in size and most of them are subrounded to rounded with oscillatory zoning (Fig. DR3C). The Th/U ratios range from 0.2 to 2.4, which is also a characteristic feature of magmatic zircons (Hoskin and Schaltegger, 2003). Of the 120 grains analyzed, 115 ones provide ages with discordance ≤5% (Table DR1). The probability density plot and kernel density estimate of this sample show similar pattern with two predominant age peaks of 1609 ± 11 Ma (n = 48, 42%) and 1181 ± 12 Ma (n = 36, 31%) (Fig. DR4C).

Shouxian Formation in the Huainan Region (Sample 17SX04)

Most zircon grains separated from the quartz sandstone sample (17SX04) from the Shouxian Fm are 70–150 μm in size. The zircons display oscillatory growth zoning under CL (Fig. DR3D). This characteristic and the Th/U ratios (0.3–2.0) indicate their magmatic origin (Hoskin and Schaltegger, 2003). The zircon grains are well rounded, which is an indication of long-distance transport and/or a high degree of abrasion. One hundred out of 110 grains analyzed give ages with discordance ≤5% (Table DR1). Those analyses yield four age populations: 1592 ± 18 Ma (n = 14), 1427 ± 19 Ma (n = 23), 1194 ± 20 Ma (n = 14), and 1084 ± 10 Ma (n = 24) (Fig. DR4D).

SIMS Zircon U-Pb Dating of the Mafic Sills

Zircon U-Pb analyses for the mafic sills were conducted on a Cameca IMS-1280HR large-radius ion microprobe at the Institute of Geology and Geophysics, Chinese Academy of Sciences (IGGCAS) in Beijing following the method of Li et al. (2009). The ellipsoidal spot is ~20 μm × 30 μm in size. Measured Pb/U ratios were calibrated relative to the zircon standard Plešovice (Sláma et al., 2008). U and Th concentrations were calibrated against zircon standard 91500 (Wiedenbeck et al., 1995). For Pb isotope analyses, each measurement lasted 12 min, with 70 cycles. Non-radiogenic Pb was subtracted from the measured Pb isotopic composition using the measured ^{204}Pb and the present-day average terrestrial Pb isotopic composition in the model of Stacey and Kramers (1975). The uncertainties of the ages for individual analyses are cited at the 1σ level, and the weighted mean ages are quoted at the 95% confidence level. Data reduction was carried out using the Isoplot program (Ludwig, 2003). The SIMS zircon U-Pb data are presented in Table DR2 (see footnote 1).

Zircons from four diabase samples are 80–120 μm in length and have aspect ratios of ~2:1–3:1. The CL imaging reveals that all the zircons analyzed are quite homogeneous and exhibit weak oscillatory zoning (Fig. 8). High Th/U ratios (1.00–5.00) also indicate a magmatic origin of the diabase zircons (Table DR2; Heaman et al., 1990).

Sample 15XZ02, collected from the sill #15XZ in the Xuzhou region (Fig. 4B), is coarse-grained. Thirty-three spots on 33 zircon grains from this sample were analyzed, and the results are plotted in Figure 8A. U and Th concentrations of the zircons vary from 354 to 778 ppm and 893 to 3398 ppm, respectively. Apart from four analyses (plots 6, 7, 17, and 20) with high common Pb, low total Pb ($^{206}Pb/^{204}Pb < 1000$) and one analysis (plot 29) not concordant due to minor amounts of Pb loss, the remaining 28 analyses are concordant and define a concordant age of 921.3 ± 5.1 Ma (2σ, MSWD = 2.4), which is identical to the $^{207}Pb/^{206}Pb$ weighted mean age of 919.2 ± 5.8 Ma (2σ, MSWD = 1.18).

Samples 17LT01 and 17PF01 were collected from sill #17LT and sill #17PF, respectively, in the Xuzhou region (Fig. 4B). For zircons from these two samples, analyses with U concentration exceeding 1200 ppm and those with high proportions of common Pb were excluded from

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

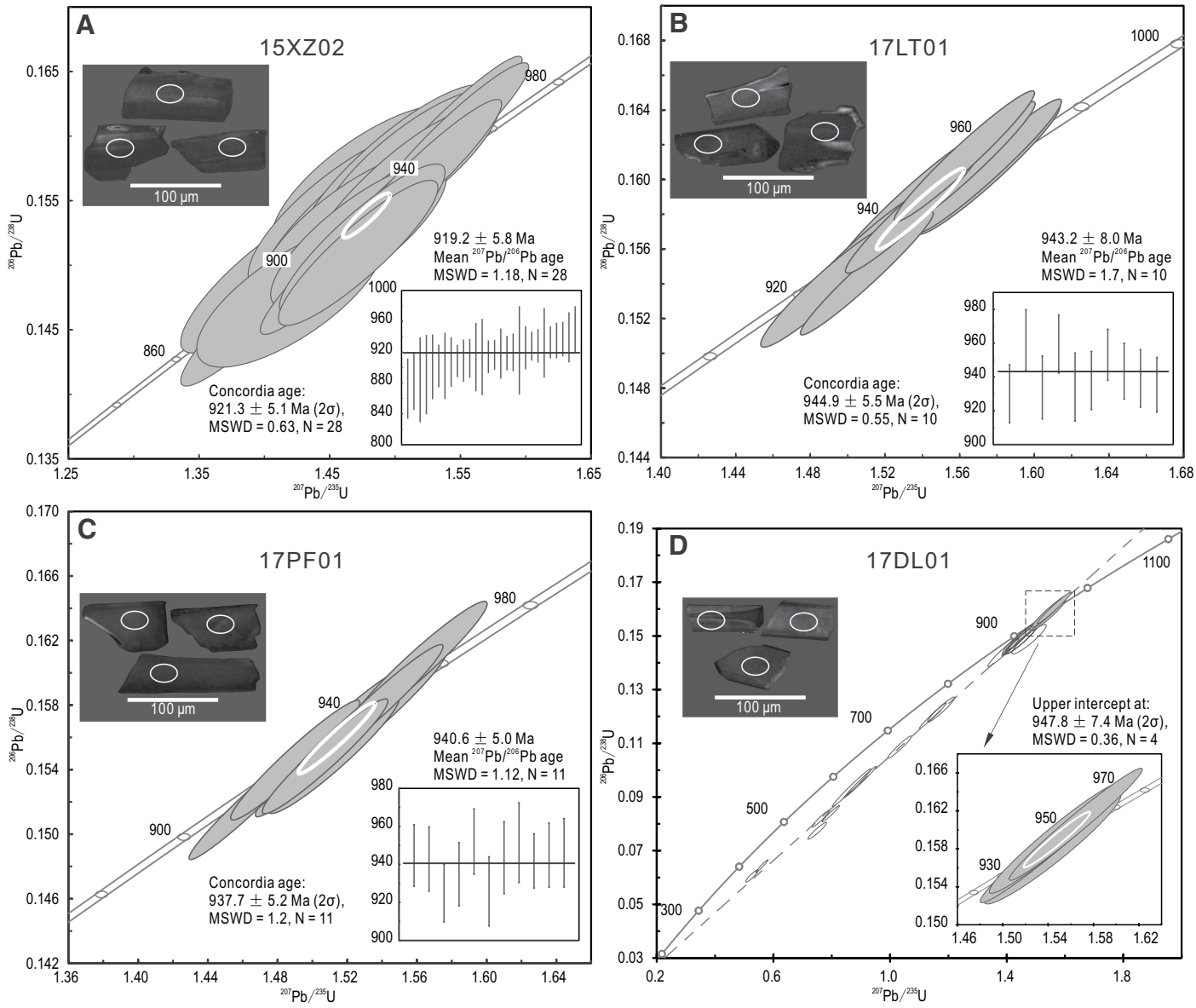


Figure 8. Selected cathodoluminescence images of zircons, U-Pb concordia plots and $^{207}Pb/^{206}Pb$ age plots: (A) sample 15XZ02 (sill #15XZ); (B) sample 17LT01 (sill #17LT); (C) sample 17PF01 (sill #17PF); (D) sample 17DL01 (sill #15DL). Data-point error crosses are 2σ. The errors on averages of multiple analyses are given at the 95% confidence level. MSWD—mean square of weighted deviates. The white circles on zircon grains show the location of secondary ion mass spectroscopy (SIMS) U-Pb analyses (spots size of 20 × 30 μm).

age calculations. For the sample 17LT01, apart from two analyses, 10 analyses are concordant and define a concordant age of 944.9 ± 5.5 Ma (2σ, MSWD = 0.55), and the $^{207}Pb/^{206}Pb$ weighted mean age is 943.2 ± 8.0 Ma (2σ, MSWD = 1.7) (Fig. 8B). As for the sample 17PF01, 11 analyses define a concordant age of 937.2 ± 5.2 Ma (2σ, MSWD = 1.2). The $^{207}Pb/^{206}Pb$ weighted mean age of 17PF01 is 940.6 ± 5.0 Ma (2σ, MSWD = 1.12) (Fig. 8C).

Sample 17DL01 is a coarse-grained diabase sample collected from the sill #15DL in the Dalian region. Twenty analyses performed on 20 zircon grains from the sample yield U contents of 523–2109 ppm, Th contents of 606–3869 ppm, and Th/U ratios from 1.16 to 1.99. As illustrated in Figure 8D, the majority of analyses are discordant due to Pb loss. The calibration of Pb/U fractionation is biased, due to an efficiency difference between Pb and U ionization that existed during SIMS measurements on high-U zircons (e.g., White and Ireland, 2012; Gao et al., 2014). The upper intercept age is 947.8 ± 7.4 Ma (2σ, MSWD = 0.36), which indicates that the sill #15DL was most likely emplaced at the time of ca. 945 Ma.

Before this study, several zircon and baddeleyite U-Pb ages in a range of ca. 930 Ma to ca. 920 Ma of the sills in the Xuzhou and Dalian regions were published (Liu et al., 2006; Gao et al., 2009; Zhang et al., 2016). We provide four new SIMS zircon U-Pb ages for the sills that we drilled for paleomagnetic research (Table 2). Although it could not be ignored the possibility of outlier dates due to insuperable radiogenic lead loss, all the above U-Pb dating results demonstrate that the emplacement ages of the mafic sills are extending from ca. 945 Ma to ca. 920 Ma.

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

TABLE 2. PALEOMAGNETIC RESULTS OF THE CA. 945–920 MA SILLS IN THE XUZHOU AND DALIAN REGIONS, EASTERN NORTH CHINA CRATON

Sill/Site	Age (Ma)	Location		n/n_0	Dip direction/ Dip (°)	In situ		Tilt-corrected				Tilt-corrected		
		Lat (°N)	Long (°E)			*D* (°)	*I* (°)	*D* (°)	*I* (°)	*k*	α_{95} (°)	Plat (°N)	Plong (°E)	*dp*/*dm* A_{95} (°)
Xuzhou region														
Sill #15XZ	919.2 ± 5.8*; 925 ± 10†	34.164	117.463	26/35	134/12	152.0	61.9	147.2	50.3	103.3	2.8	–17.8	146.7	2.5/3.8
Sill #17LT	943.2 ± 8.0*	34.122	117.568	23/30	338/38	325.9	–0.6	322.6	–37.6	50.0	4.3	24.3	336.0	3.0/5.1
Sill #17PF	940.6 ± 5.0*	34.137	117.587	31/54	332/24	332.2	–7.6	332.2	–31.6	62.6	3.3	32.3	329.4	2.1/3.7
Sill #4§	-	34.148	117.480	18/19	291/24	152.1	22.8	162.7	39.7	108.0	3.3	–30.9	136.2	2.4/4.0
Dalian region														
Sill #15DL	947.8 ± 7.4*, 923 ± 22#	39.438	121.806	32/42	266/17	350.1	–34.2	1.8	–34.2	61.4	3.3	31.8	299.8	2.2/3.8
Sill #16DL	-	39.556	121.273	23/49	262/35	323.6	–21.5	341.5	–33.6	34.4	5.2	29.5	321.5	3.4/5.9
Mean of 6 sills**	**ca. 945–920**	-	-	-	-	-	-	-	-	-	-	**–28.2**	**141.9**	**A_{95} = 10.4**
Baked-contact test: dolostone of the Niyuan Fm to the sill #15XZ														
17XZ11-17 (baked dolostone)		34.166	117.463	8/10	135/14	152.3	58.9	147.6	45.3	38.5	9.0	–21.7	148.4	7.2/11.4
17XZ12-15 (baked dolostone)		34.166	117.463	7/10	135/14	139.8	54.2	138.6	40.3	20.3	13.7	–20.6	158.0	10.0/16.5
17XZ24-25 (unbaked dolostone)		34.159	117.472	7/13	183/0	183.1	–30.2	183.1	–30.4	24.6	12.4	–72.0	107.8	7.7/13.8
17XZ26 (unbaked dolostone)		34.159	117.472	7/8	154/2	187.5	–32.0	188.2	–33.6	20.4	13.7	–72.6	90.5	8.9/15.6
17XZ27 (unbaked dolostone)		34.159	117.472	6/9	315/3	183.8	–41.4	181.9	–39.4	143.9	5.6	–78.1	109.0	4.0/6.7
17XZ28 (unbaked dolostone)		34.159	117.472	5/8	140/2	180.0	–35.9	181.0	–37.5	60.0	10.0	–76.8	113.4	6.9/11.8
Mean of the unbaked dolostone (4 sites)		-	-	-	-	183.6	–34.9	183.6	–35.3	284.4	5.5	–75.0	104.3	3.7/6.4

Note: n/n_0—number of samples used to calculate mean/total samples demagnetized; *D*—declination; *I*—inclination; *k*—precision parameter of Fisher (1953); α_{95}—radius of 95% confidence cone of the mean direction; Plat, Plong—latitude, longitude of the virtual geomagnetic pole; *dm, dp*—semi-minor and semi-major axes of 95% polar error ellipse. A_{95}—radius of 95% confidence cone of the paleomagnetic pole.

*Age from this study.

†Age from Liu et al. (2006), recalculated by Fu et al. (2015).

§After Fu et al. (2015).

#Age from Zhang et al. (2016). The upper intercept age is 953 ± 27/–25 Ma.

**The two polarity groups of the 6 sills are consistent with an antipodal common mean at the 95% confidence level, but the classification level of the reversal test is indeterminate (McFadden and McElhinny, 1990; critical angle γ_c = 25.0° and observed angular difference γ_o = 5.7°).

PALEOMAGNETISM

Paleomagnetism Laboratory Conditions and Procedures

Standard paleomagnetic core samples were collected from the outcrops using a portable gasoline-powered drill. The orientations were measured by using magnetic and solar compasses. We collected a total of 931 samples from the Nanfen Fm, Xinxing Fm, Liulaobei Fm, five mafic sills, and the Niyuan Fm. The oriented core samples were cut into 1–2 2.2-cm-long cylindrical specimens using a nonmagnetic dual-blade water-cooled rock saw. In general, one specimen from each sample was selected for the paleomagnetic experiments. Most of the specimens underwent stepwise thermal demagnetization (TD) in an ASC-TD 48 furnace with an internal residual field less than 10 nT. Demagnetization temperature intervals ranged generally from a maximum 80 °C for lower temperatures to minimum 3 °C for higher temperatures up to 585 °C or 685 °C. Some specimens from the sill #16DL were subjected to stepwise alternating field demagnetization (AFD) up to 120 mT using a D-2000 alternating field demagnetizer. The measurements of 35 specimens from the sill #15XZ were conducted on the JR-6A spinner magnetometer. While the remanent magnetizations of all other specimens were measured using a 2G-755-4K cryogenic magnetometer. The ASC TD-48 furnace, the D-2000 alternating field demagnetizer and two magnetometers were installed within a magnetically shielded room with a residual field less than 200 nT. Isothermal remanent magnetization (IRM), back-field demagnetization of saturation IRM (SIRM), and thermal demagnetization of the three-axis IRMs (Lowrie, 1990) were measured for representative specimens. Fields of 2.4 T, 0.4 T, and 0.12 T were applied in turn along the Z, Y, and X axes of the specimens using an ASC IM-10-30 impulse magnetizer, respectively. Afterward, stepwise thermal demagnetization up to 690 °C was conducted. The IRM was measured using an AGICO JR-6A spinner magnetometer. Temperature-dependent low field susceptibility was measured continually from room temperature up to 700 °C and back to room temperature in an argon atmosphere using an AGICO KLY-4S Kappabridge with CS-3 high-temperature device. An AGICO MFK1-FA Kappabridge was used to measure the anisotropy of magnetic susceptibility (AMS) in order to analyze the magnetic fabric of the rocks. All the above magnetic measurements were conducted in the Paleomagnetism and Environmental Magnetism Laboratory (PMEML) of the China University of Geosciences, Beijing. Magnetic hysteresis parameters (i.e., saturation magnetization, Ms; saturation remanence, Mrs; coercive force, Hc; and coercivity of remanence, Hcr) were measured for representative samples using a MicroMag 3900 Vibrating Sample Magnetometer with maximum magnetic field of 1.0 T in the Paleomagnetic and Geochronology Laboratory of the IGGCAS. Magnetic components of all the specimens were identified by principal component analysis (Kirschvink, 1980), and interval-mean directions were calculated using Fisher statistics (Fisher, 1953). Paleomagnetic data were analyzed using Enkin (1990) and Cogné (2003) computer program packages. Paleogeographic reconstruction was carried out using GPlates software (Boyden et al., 2011).

Rock Magnetic Results

The AMS of the specimens from the Nanfen Fm in both the Benxi and Dalian regions in the stratigraphic coordinates shows that the most minimum susceptibility axes (K_3) are approximately perpendicular to the bedding plane, while the intermediate axes (K_2) and maximum (K_1) lie in the bedding plane (Figs. DR2A and DR2B; see footnote 1). Combined with the low lineation and foliation values, the AMS data imply that these rocks still preserve the primary sedimentary magnetic fabrics (Tarling and Hrouda, 1993). The AMS of the specimens from

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

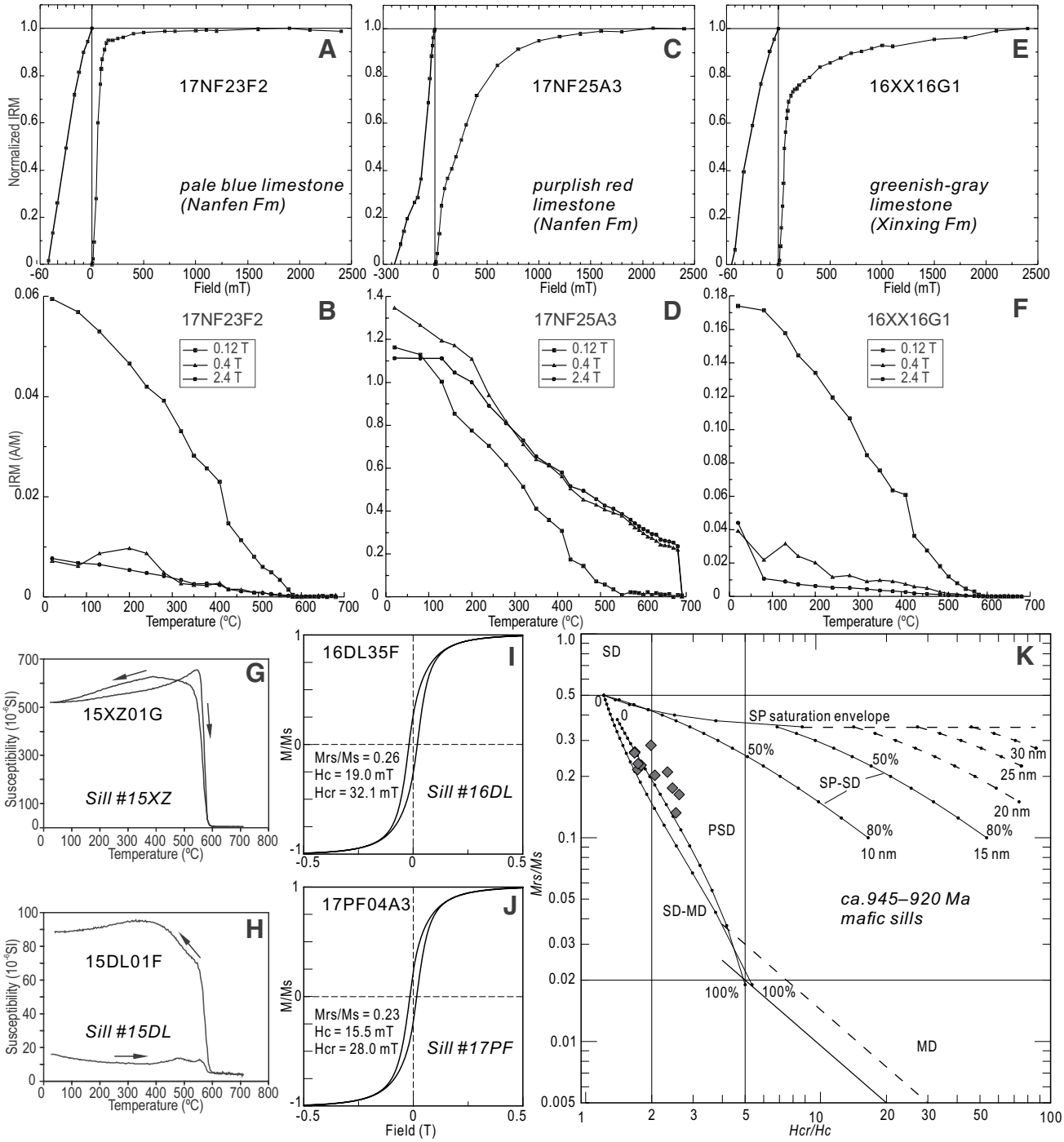


Figure 9. Rock magnetic results of representative specimens. (A, C, and E) Isothermal remanent magnetization (IRM) and back-field demagnetization of saturation IRM (SIRM) acquisition curves of representative specimens from the Nanfen, Xinxing, and Liulaobei formations. (B, D, and F) Thermal demagnetization of the three-axis IRM for representative specimens showing unblocking temperatures around 580 °C or 680 °C. (G–H) Thermomagnetic susceptibility curves of representative specimens of the ca. 945–920 Ma mafic sills. (I–J) Magnetic hysteresis loops. (K) Day plot (Dunlop, 2002) of representative specimens of the ca. 945–920 Ma sills. Ms—saturation magnetization; Mrs—saturation remanence; Hcr—coercivity of remanence; Hc—coercivity; SD—single domain; PSD—pseudo-single domain; MD—multidomain; SP—superparamagnetic.

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

the Xinxing and Liulaobei formations, however, shows that the directions of the K_1, K_2, and K_3 axes are irregularly scattered (Figs. DR2C and DR2D).

The IRM acquisition curve of the pale-blue limestone specimen (17NF23F2) from the Nanfen Fm shows a rapid increase below 120 mT and saturation is fully reached at ~400 mT. The progressive back-field demagnetization curve of SIRM illustrates that the coercivity of remanence is ~50 mT (Fig. 9A). These results reveal that the predominant magnetic carriers are low-coercivity magnetic minerals. Stepwise TD of three orthogonal IRM components (Lowrie, 1990) shows that each component—hard (2.4 T), medium (0.4 T) and soft (0.12 T)—has a similar unblocking temperature around 580 °C (Fig. 9B), demonstrating that the major magnetic carrier of the pale-blue limestone is magnetite.

The IRM acquisition curve of the purplish-red limestone specimen (17NF25A3) from the Nanfen Fm shows a rapid increase below 90 mT and acquires 40% of the SIRM. The increase of IRM intensity becomes slower above 90 mT, and the saturation is reached at ~2.1 T. These phenomena reveal the presence of both low- and high-coercivity magnetic minerals. The progressive back-field demagnetization of SIRM also indicates the existence of two magnetic carriers, and the coercivity of remanence is ~240 mT (Fig. 9C). Stepwise TD of three orthogonal IRM components (Lowrie, 1990) shows that the soft component (0.12 T) was unblocked around 580 °C, while both medium component (0.4 T) and hard component (2.4 T) were unblocked at ~680 °C (Fig. 9D). These results indicate that both hematite and magnetite are abundant in the purplish-red limestone of the Nanfen Fm.

The IRM acquisition curve of the greenish-gray limestone specimen (16XX16G1) from the Xinxing Fm shows a rapid increase below 100 mT and acquires 70% of the SIRM, which suggests that low-coercivity magnetic minerals are dominant. The increase of IRM intensity becomes slower above 100 mT, and the saturation is not fully reached at 2.4 T, revealing the presence of high-coercivity magnetic carriers (Fig. 9E). The progressive back-field demagnetization curve of SIRM indicates that the coercivity of remanence is less than 60 mT, indicating that the presence of low-coercivity minerals is highlighted. Stepwise TD of three orthogonal IRM components (Lowrie, 1990) shows that both soft component (0.12 T) and medium component (0.4 T) were unblocked around 580 °C, while the intensity of the hard component (2.4 T) dropped by 80% at ~100 °C (Fig. 9F). These results demonstrate that both magnetite and goethite are present, but magnetite is the predominant magnetic carrier in this rock unit.

Thermomagnetic curves obtained from the mafic sills exhibit an abrupt decrease in susceptibility at ~580 °C, indicating that magnetite is the major magnetic carrier (Dunlop and Özdemir, 1997; Figs. 9G–9H). The magnetic susceptibility values for the cooling curves of some specimens are higher than the values for the heating curves at temperatures below ~580 °C, a feature indicative of changes in strong magnetic minerals during heating (Fig. 9H). Since TDs were performed within a no-field furnace, the newly transformed magnetite has no effect on the characteristic remanent magnetization (ChRM) directions. All hysteresis loops from the sills are narrow-waisted with low coercivity (Figs. 9I–9J). The typical grain size of magnetic phases in the sills falls in pseudo-single domain (PSD) field on the Day diagram (Dunlop, 2002), and some distribute along the theoretical single-domain (SD)–multidomain (MD) magnetite mixing curves (Fig. 9K). These characteristics indicate that PSD magnetite is the main magnetic mineral of the sills.

Paleomagnetic Results

The Nanfen Formation in the Benxi Region

In the Benxi region, of the 357 specimens (also representing 357 samples) that underwent stepwise TD, 294 specimens revealed stable paleomagnetic signals (Table 1), whereas the remaining specimens were not accepted because they became unstable or broke up during heating. The natural remanent magnetization (NRM) intensities of the pale-blue to greenish-gray limestone specimens from the Lower Member of the Nanfen Fm ranged from 0.02 to 50 mA/m, mostly from 0.1 to 5.0 mA/m. The NRM intensities of the remaining purple-hued limestone specimens from the middle part of the Nanfen Fm ranged from 1.2 to 50 mA/m. Representative Zijderveld plots (after Zijderveld, 1967) are shown in Figures 10A–10E. A low temperature component (LTC) was identified for most specimens between room temperature and 300 °C. The mean direction of the LTCs in situ is close to that of the recent geomagnetic field in the region (D = 0°, I = 60.3°, calculated based on the geocentric axial dipole [GAD] model; Fig. DR5A [see footnote 1]), suggesting that the LTC is a viscous remanent magnetization (VRM) of the recent geomagnetic field. The dispersed low-stability moderate temperature component (MTC) was also cleaned generally between 350 °C and 450 °C. After removal of the LTC and MTC, stable high temperature components (HTCs) decayed toward the origin near the unblocking temperatures of ~540–575 °C or ~665–680 °C. Furthermore, the HTC directions are defined as the characteristic remnant magnetization (ChRM) directions. As for the pale-blue to greenish-gray limestone specimens from the Lower Member of the Nanfen Fm, the ChRM with notably steep inclinations was generally isolated between ~450 °C and 540–575 °C (Figs. 10A–10C). The ChRM is dual-polarity (Fig. 6A), with magnetization directed eastward and down being termed as polarity-1 and westward and up as polarity-2. The site-mean direction for the 33 sites is D = 86.7°, I = 80.3°, k = 107.7, α_{95} = 2.4° in geographic coordinates and D = 102.5°, I = 85.6°, k = 112.5, α_{95} = 2.4° in stratigraphic coordinates (Table 1; Fig. 11A). The antipodal directions of the ChRM passed a positive reversal test (class B; McFadden and McElhinny, 1990) at the 95% probability level: γ_o (observed angular difference) = 3.8° < γ_c (critical angle) = 5.1°. The mean pole 38.8 °N/134.6 °E (A_{95} = 4.6°) for the lower part of the Lower Member of the Nanfen Fm in the Benxi region were obtained by averaging the VGPs calculated for each site (pole "BNFa"; Table 1).

The ChRM of the purplish-red limestone specimens, from the uppermost part (~10 m in thickness) of the Lower Member of the Nanfen Fm, was defined in temperatures up to 665–680 °C (Fig. 10D). This demagnetization behavior together with the rock magnetic test, indicate that both magnetite and hematite carried the remanent magnetizations. The ChRM directs southeast with positive inclinations (Fig. 10D). The mean direction from 16 specimens that represented 16 samples is D = 149.8°, I = 73.0°, k = 60.0, α_{95} = 4.8° in geographic coordinates and D = 171.5°, I = 71.7°, k = 60.0, α_{95} = 4.8° in stratigraphic coordinates (Table 1; Fig. 11B). The mean pole at 8.0 °N, 128.5 °E, A_{95} = 7.9° (pole "BNFb"; Table 1) was obtained by averaging the VGPs calculated for each sample.

As for the purplish-red thin-bedded calcareous mudstone intercalated with pale blue limestone specimens from the Middle Member of the Nanfen Fm, the ChRM was generally determined between 550 °C and 680 °C, directing southeast with moderate positive inclinations (Fig. 10E). The site-mean direction of the five sites is D = 150.5°, I = 57.3°, k = 111.6, α_{95} = 7.3° in geographic coordinates and D = 160.7°, I = 60.9°, k = 112.8, α_{95} = 7.2° in stratigraphic coordinates (Table 1; Fig. 11C). The mean pole at 4.9 °S, 137.9 °E (A_{95} = 10.2°) was determined by averaging all site VGPs (pole "BNFc1"; Table 1). Notably, it is close to the position of the published paleo-pole (16.5 °S, 121.1 °E, *dp*/*dm*

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

= 8.8°/13.0°) also derived from the Middle Member of the Nanfen Fm in the region (Lin, 1984). We thus further average all site VGPs (N = 13) to get a mean pole for this rock unit, which falls at: 11.2 °S, 127.7 °E (A_{95} = 8.5°, Table 3; pole "BNFc" in Fig. 12).

Paleomagnetism of the Nanfen Formation in the Dalian Region

In the Dalian region, of the 90 specimens (also representing 90 samples) from the lower part of the Nanfen Fm that underwent stepwise TD, 77 revealed stable paleomagnetic signals (Table 1). The NRM intensities of the specimens ranged from 0.1 to 2.0 mA/m. Typical demagnetization behavior of specimens is shown in Figures 10F–10G. A well-defined LTC was yielded from most specimens between room temperature and 300 °C. The LTCs in situ cluster around the recent geomagnetic field direction in the region (D = 0°, I = 59.0°; calculated based on the GAD model; Fig. DR5B), revealing that the LTC is a VRM of the recent geomagnetic field. The ChRM directions toward the origin were generally isolated between ~450 °C and 520–575 °C (some up to 680 °C). The ChRM is considerably steep and directs east down and west up (Figs. 10F–10G), possessing dual polarity (Fig. 6B). The site-mean direction for the 11 sites from the two sections is D = 102.7°, I = 77.5°, k = 63.1, α_{95} = 5.8° in geographic coordinates and D = 93.7°, I = 79.8°, k = 59.7, α_{95} = 6.0° in stratigraphic coordinates (Table 1; Fig. 11D). The antipodal directions of the ChRM also passed a positive reversal test at the 95% probability level: γ_o = 7.1° < γ_c = 12.0° (class C; McFadden and McElhinny, 1990). We hence obtained the mean pole at 36.7 °N/145.8 °E (A_{95} = 9.9°) by averaging the site VGPs (pole "WNFa"; Table 1).

Paleomagnetism of the Xinxing Formation in the Xuzhou Region

Of the 169 specimens (also representing 169 samples) from the limestone of the lower part of Xinxing Fm that underwent stepwise TD, 160 specimens revealed stable paleomagnetic signals (Table 1). The NRM intensities of the specimens ranged from 1.0 to 10 mA/m. Typical demagnetization behavior of specimens is shown in Figure 10H. A well-defined LTC was yielded from most specimens between room temperature and 300 °C. The LTCs in situ distribute around the recent geomagnetic field direction in the region (D = 0°, I = 54.3°; calculated based on the GAD model; Fig. DR5C), suggesting that the LTC is a VRM of the recent geomagnetic field. The ChRM directions were generally isolated between ~400 °C and 525–575 °C (some up to ~680 °C). Similar to that of the Nanfen Fm, the ChRM of Xinxing Fm has notably steep inclinations, with dual polarity (Fig. 11E), but the magnetostratigraphy is predominated by the steep down direction (Fig. 6C). The site-mean direction of the 20 sites from two sections is D = 316.2°, I = 83.2°, k = 102.1, α_{95} = 3.2° in geographic coordinates and D = 77.7°, I = 83.2°, k = 113.9, α_{95} = 3.1° in stratigraphic coordinates (Table 1; Fig. 11E). The antipodal directions of the ChRM passed a positive reversal test at the 95% probability level: γ_o = 1.6° < γ_c = 6.7° (class B; McFadden and McElhinny, 1990). We obtained the mean pole at 36.6 °N/134.1 °E (A_{95} = 5.9°) by averaging the site VGPs (pole "XXa"; Table 1). It is worth mentioning that the sampling sites in this study are stratigraphically lower than those sites of Xinxing Fm in Fu (2016); we did not include the previous data for an average.

Paleomagnetism of the Liulaobei Formation in the Huainan Region

In the Huainan region, 47 specimens from the limestone of the lower part of the Liulaobei Fm were subjected to stepwise TD, in which 40 specimens revealed stable magnetic signals (Table 1). The NRM intensities of the specimens ranged from 1.0 to 10 mA/m. Typical demagnetization behavior is shown in Figure 10I. Although the intensity of some specimens reduced to less than 30% below 80 °C, a well-defined LTC could be yielded from all specimens between ~80 °C and 350 °C. The mean direction of the LTCs in situ is close to that of the recent geomagnetic field in the region (D = 0°, I = 52.0°; calculated based on the GAD model; Fig. DR5D). The ChRM directions were isolated between ~450 °C and 550–575 °C (Fig. 10I; Table DR3 [see footnote 1]). This indicates that the magnetite is the dominant magnetic carrier. The ChRM directions are notably steep (Fig. 11F) with dual polarity (Fig. 6D). The site-mean direction of the five sites is D = 337.5°, I = 88.7°, k = 212.8, α_{95} = 5.3° in geographic coordinates and D = 43.4°, I = 77.5° °, k = 212.8, α_{95} = 5.3° in stratigraphic coordinates (Table 1; Fig. 11F). The antipodal directions of the ChRM passed a positive reversal test at the 95% probability level: γ_o = 0.4° < γ_c = 16.4° (class C; McFadden and McElhinny, 1990). The mean pole at 47.6 °N/141.1 °E (A_{95} = 9.7°) was determined by averaging site VGPs (pole "LLBa1"; Table 1). Our resulting pole obviously overlaps with the published pole (42.6 °N, 149.6 °E, dp/dm = 11.9°/13.0°; pole "LLBa2" in Table 3) acquired from this formation (Piper and Zhang, 1997). We did not combine them because there are no statistic details in the published results.

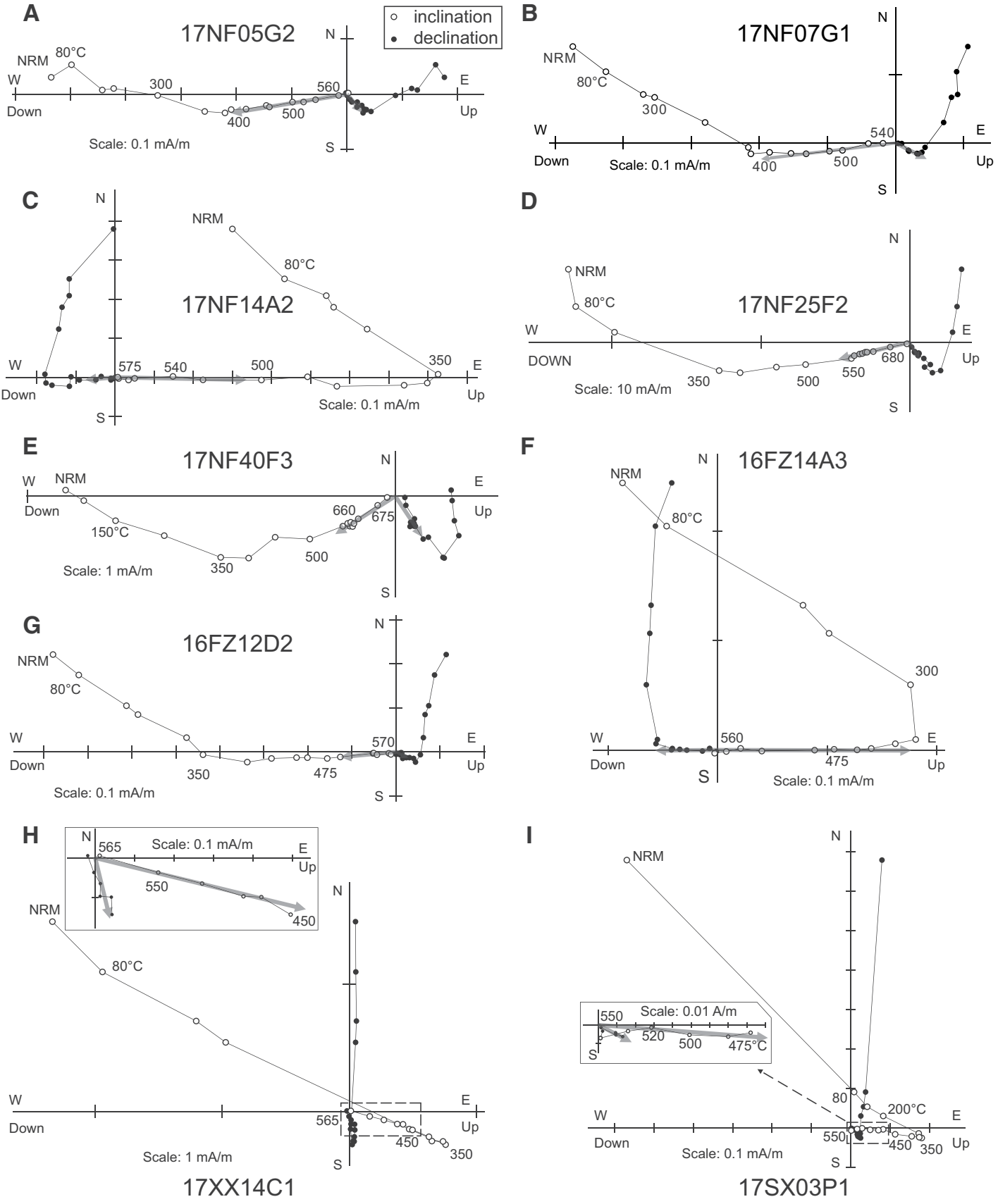
Age Constraints of the Nanfen, Xinxing, and Liulaobei Formations for the Paleomagnetic Study

The stratigraphic sequences and lithologic associations of the Xihe Gp in the Benxi and Dalian regions, the lowermost part of the Huaibei Gp (including the Lanling, Xinxing and Jushan formations) in the Xuzhou region, as well as the Huainan Gp to the Shouxian Fm in the Huainan region are strikingly similar (Figs. 5 and 6). These similarities were originally used as evidence for regional correlation (JBGMR, 1984; ABGMR, 1987; LBGMR, 1989; Xing, 1989; SBGMR, 1991; Dong et al., 2008). Previous studies also suggest that the Nanfen, Xinxing, and Liulaobei formations are correlative rock units based on the similar fossil assemblages observed in the upper part of each unit, such as the presence of *Chuaria-Tawuia* (Duan, 1982; Wang et al., 1984; Hong et al., 1991; Liu et al., 2018; Figs. 6A–6D). These characteristic fossil assemblages are more likely pre-Cryogenian Neoproterozoic or late Mesoproterozoic in age (Tang et al., 2013, Xiao et al., 2014, and references therein). Although there are no volcanic ashes available in the studied regions that allow for precise zircon U-Pb dating, recent zircon U-Pb analyses from the mafic sills intruding the strata and the detrital zircon U-Pb geochronologic data from the quartz sandstones do provide age constraints for the studied formations (Fig. 5; Liu et al., 2006; Gao et al., 2009; Yang et al., 2012; Zhang et al., 2016; this study). We consider that the lower part (from the Xinxing Fm to Zhaowei Fm) of the Huaibei Gp should be older than the ca. 945 Ma sills that intruded the Niyuan Fm (Fig. 5C) and younger than 1121 ± 27 Ma (Fig. 5C; the weighted mean age of the youngest detrital zircon population of the sandstone from the Xinxing Fm; n = 9; Yang et al., 2012). Similarly, the Nanfen Fm should be older than the ca. 945 Ma mafic sills intruding the overlying Qiaotou Fm (Fig. 5B) and younger than 1136 ± 11 Ma (Fig. 5B; the weighted mean age of the youngest detrital zircon population from the underlying Diaoyutai Fm). In summary, the Nanfen, Xinxing, and Liulaobei formations are correlative rock units with ages younger than ca. 1120 Ma and older than ca. 945 Ma.

Magnetostratigraphic Pattern of the Reversals

Because the sampling outcrops are not continuous enough, the magnetostratigraphic implications of the reversals revealed from the strata are still limited at this stage. From the most continuous sampling section in the Benxi region, we can confirm two major polarity intervals (Fig. 6A). The lower one is in the lower part of the Lower Member of the

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user



Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

Figure 10. Typical demagnetization trajectories of representative specimens from the Nanfen, Xinxing and Liulaobei formations. (A–C) Pale blue to greenish-gray limestone in the lower part of the low member of the Nanfen Formation (Fm) in the Benxi region; (D) purplish-red limestone in the upper part of the Lower Member of the Nanfen Fm in the Benxi region; (E) purplish-red thin-bedded calcareous mudstone intercalated with pale blue thin-bedded limestone in the Middle Member of the Nanfen Fm in the Benxi region; (F–G) limestone in the lower part of the Nanfen Fm in the Dalian region; (H) limestone in the lower part of the Xinxing Fm in the Xuzhou region; (I) limestone in the lower part of the Liulaobei Fm in the Huainan region. The gray arrow indicates the least-squares fit direction of the characteristic remanent magnetization (ChRM). The directions of remanence are all in geographic coordinates. NRM—natural remanent magnetization.

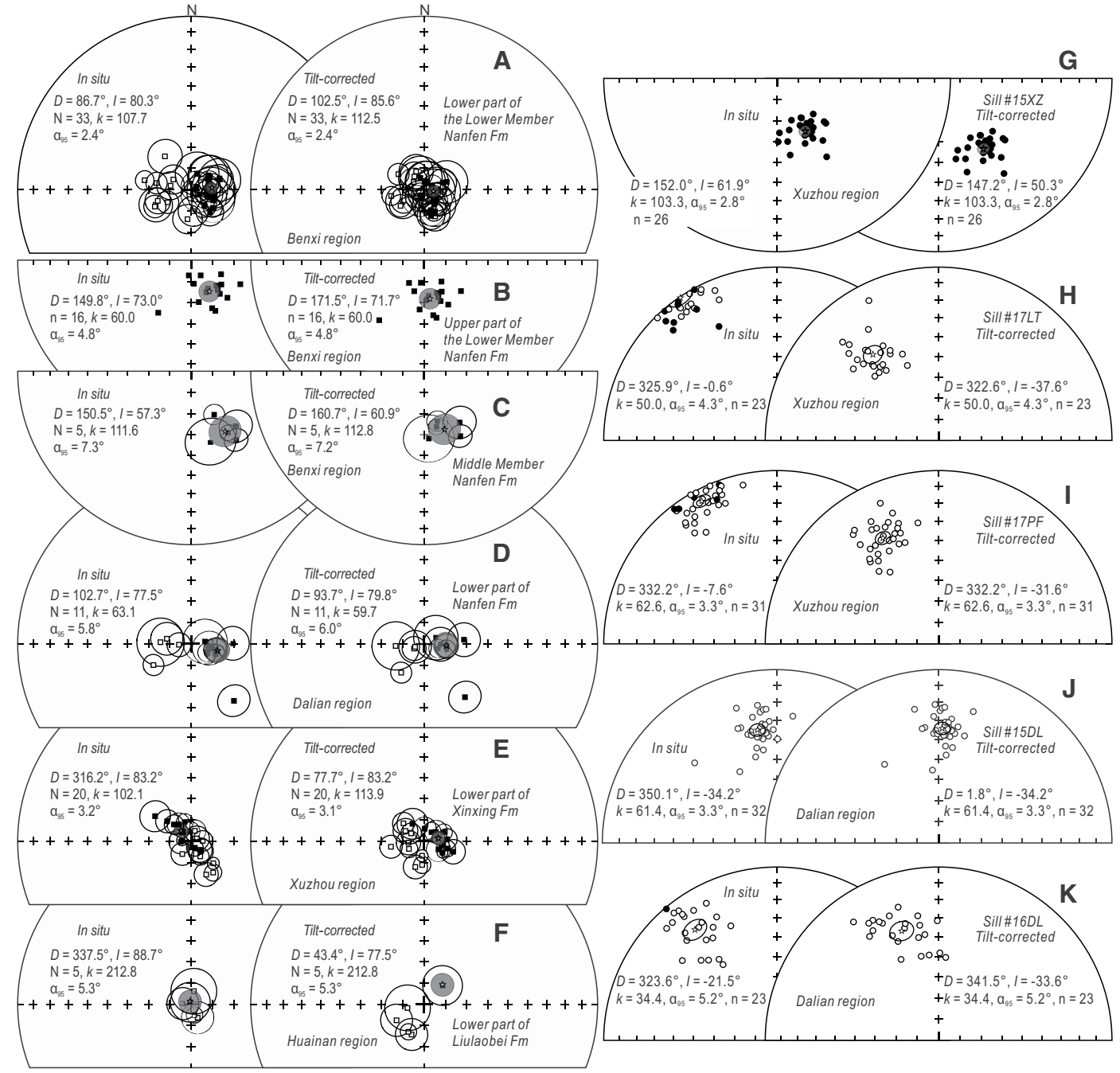


Figure 11. Equal-area stereographic projections of the characteristic remanent magnetizations (ChRMs) isolated from the Nanfen Formation (Fm), Xinxing Fm, Liulaobei Fm, and the ca. 945–920 Ma mafic sills for in situ and tilt-corrected coordinates. Figures 11B and 11G–K show sample-mean directions. All others show site-mean directions. The star with 95% confidence circle indicates Fisher statistic direction. Closed and open symbols show the downward and upward paleomagnetic inclinations. *D*—declination; *I*—inclination; α_{95}—radius of 95% confidence cone of the mean direction; N—site number; n—specimen number; *k*—precise parameter.

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

TABLE 3. SELECTED CA. 1120–850 MA PALEOMAGNETIC POLES FROM NORTH CHINA CRATON, BALTICA, LAURENTIA AND SIBERIA

Pole	Rock-unit	Age (Ma)	Plat (°N)	Plong (°E)	A_{95} (°)	1234567 (Q)	References
North China Craton							
NC890	ca. 890 Ma sills in the Huaibei region	890 ± 14, Uz	52.6	330	5.3	1111111 (7)	Fu et al. (2015)
NC930	ca. 945–920 Ma sills	ca. 945–920, Uz	–28.2	141.9	10.4	1111110 (6)	Fu et al. (2015), this study
WS	Wangshan Fm	950–890, DZ, S	26.1	320.3	5.2	0111100 (4)	Fu et al. (2015), He et al. (2017)
LHB	Middle part of Huaibei Group (mean pole)	<XXa, S	–50.6	107	3.8	0110111 (5)	Fu et al. (2015)
BNFc	Middle Member of Nanfen Fm, Benxi	<BNFb, >945, S	–11.2	127.7	8.5	0110101 (4)	Lin (1984), this study
BNFb	Upper part of the Lower Member of Nanfen Fm, Benxi	<BNFa, >945, S	8	128.5	7.9	0010101 (3)	This study
BNFa	Lower part of Nanfen Fm, Benxi	=WNFa, S	38.8	134.6	4.6	0110111 (5)	This study
WNFa	Lower part of Nanfen Fm, Dalian	1136–945, DZ, Uz	36.7	145.8	9.9	0110111 (5)	Yang et al. (2012), this study
XXa	Lower part of Xinxing Fm, Xuzhou	1121–945, DZ, Uz, S	36.6	134.1	5.9	0110111 (5)	Yang et al. (2012), this study
LLBa2	Lower part of Liulaobei Fm (old), Huainan	=XXa, S	42.6	149.6	12.4	0010101 (3)	Piper and Zhang (1997)
LLBa1	Lower part of Liulaobei Fm (new), Huainan	=XXa, S	47.6	141.1	9.7	0110111 (5)	This study
NFL	Lower part of the Nanfen Fm and its equivalents*	1120–945, DZ, Uz, S	38.6	136.7	3.2	0110111 (5)	Yang et al. (2012), this study
Baltica							
B1	Hunnedalen dykes	848 ± 27, Ar, 855 ± 59, SN	–41	222	10.5	1110101 (5)	Walderhaug et al. (1999)
B2	Rogaland Ingneous Complex	869 ± 14, Ar	–46	238	18.1	1010101 (4)	Walderhaug et al. (2007)
B3	Egersund-Onga anorthosite	ca. 900, CA	–42.1	200.4	9	0110100 (3)	Brown and McEnroe (2004)
B4	Bjerkreim-Sokndal intrusion	ca. 916, CA	–35.9	217.9	6	1110101 (5)	Brown and McEnroe (2015)
B5	Mean 951–935 Ma Baltica pole	951–935, Ub, Ar	–2.6	239.6	5.8	1111111 (7)	Gong et al. (2018)
B6	971 Ma Blekinge-Dalarna dolerite dykes VGP	971 ± 7, Ub	–27	230.4	14.9	1010111 (5)	Gong et al. (2018)
B7	Laanila Ristijärvi	1042 ± 50, SN	–2.1	212.2	16.5	0011101 (4)	Mertanen et al. (1996)
Laurentia							
L1	Adirondack microcline gneiss	ca. 960, CA	–18.4	151.1	10.5	1110010 (4)	Brown and McEnroe (2012)
L2	Adirondack metamorphic anorthosites and other rocks	ca. 970, CA	–25.1	149	11.6	1110010 (4)	Brown and McEnroe (2012)
L3	Adirondack fayalite granite	ca. 990, CA	–28.4	132.7	6.9	1110010 (4)	Brown and McEnroe (2012)
L4	Haliburton instrusion A	1015 ± 15, Ar	–32.6	141.9	6.3	1110000 (3)	Warnock et al. (2000)
L5	Chequamegon Fm	ca. 1035?	–12.3	177.7	4.6	0110100 (3)	McCabe and Van der Voo (1983), Evans et al. (2016a)
L6	Jacobsville Fm	ca. 1040?	–10	184	4.2	0110110 (4)	Roy and Robertson (1978), Evans et al. (2016a)
L7	Freda Fm	ca. 1070?	2.2	179	4.2	0110100 (3)	Henry et al. (1977), Swanson-Hysell et al. (2019)
L8	Nonesuch Shale	ca. 1080, RO	7.3	174.7	3	0110100 (3)	Henry et al. (1977), Cumming et al. (2013), Symons et al. (2013), Fairchild et al. (2017)
L9	Michipicoten Island Fm	1084.4–1083.5, Uz	17	174.7	4.4	1110101 (5)	Palmer and Davis (1987), Fairchild et al. (2017)
L10	Lake Shore Traps	1091–1084, Uz	23.1	186.4	4	1111101 (6)	Kulakov et al. (2013), Fairchild et al. (2017), Swanson-Hysell et al. (2019)
L11	Cardenas basalts and intrusions	1090.6±4.5, Ar	32	185	8	1110101 (5)	Weil et al. (2003)
L12	Portage Lake Volcanics	1093.4–1091.6, Uz	26.7	177.9	4.7	1111101 (6)	Hnat et al. (2006), Davis and Paces(1990), Swanson-Hysell et al. (2019)
L13	Chengwatana Volcanics	1094.6±2.1, Uz	30.9	186.1	8.2	1110111 (6)	Kean et al. (1997), Zartman et al. (1997)
L14	North Shore Volcanics (upper SW sequence)	1096.8–1093.9, Uz	36.9	179.3	2.1	1110111 (6)	Tauxe and Kodama (2009), Swanson-Hysell et al. (2019)
L15	North Shore Volcanics (upper NE sequence)	1098–1092, Uz	31.1	181.7	4.2	1110111 (6)	Tauxe and Kodama (2009), Davis and Green, (1997), Fairchild et al. (2017)
L16	Mamainse upper normal	1100–1090, Uz	31.2	183.2	2.5	1111111 (7)	Swanson-Hysell et al. (2014a)
L17	Osler reversed (upper)	1105.5–1104.8, Uz	42.3	203.4	3.7	1111111 (7)	Swanson-Hysell et al. (2014b, 2019)
L18	Osler reversed (lower)	1110–1105.15, Uz	40.9	218.6	4.8	1110111 (6)	Swanson-Hysell et al. (2014b, 2019)
L19	Logan sills	1108 ± 1, Uz	49	220	4	1111111 (7)	Halls and Pesonen (1982), Davis and Green (1997)
L20	Mamainse lower normal and upper reversed	1100.4 ± 0.3, Uz	36.1	189.7	4.9	1111111 (7)	Swanson-Hysell et al. (2014a)
L21	Mamainse lower reversed 2	1110.4–1109, Uz	37.5	205.2	4.5	1111111 (7)	Swanson-Hysell et al. (2014a)
L22	Mamainse lower reversed 1	1112–1106, Uz	49.5	227	5.3	1111111 (7)	Swanson-Hysell et al. (2014a)
Siberia							
SMl	Malgina Fm†	1120–1005, Uz, DZ	15	70	3	0111111 (6)	Gallet et al. (2000), Khudoley et al. (2015)
SL1-2	Lower and middle Linok Fm	=SMl?, S	14.3	80.8	4	0111111 (6)	Pavlov et al. (2019)
SL3	Upper Linok Fm	<SL1-2	12.5	62.6	6	0111101 (5)	Pavlov et al. (2019)
SKart	Kartochka Fm	ca. 1050?	19	36	12	0111101 (5)	Gallet et al. (2012), Evans et al. (2016a)
SMilk	Milkon subformation†	ca. 1050?	–6	39	3	0111101 (5)	Pavlov and Gallet (2010), Evans et al. (2016a)
SKd	Kandyk Fm†	ca. 990, Ub	–9	19	4	1111100 (5)	Rainbird et al. (1998), Pavlov et al. (2002)

Notes: Dating method of age: Ar—^{40}Ar-^{39}Ar; CA—Cooling ages; Fm—Formation; Uz—U-Pb zircon; Ub—U-Pb Beddeleyite; Up—U-Pb perovskite; DZ—detrital zircon U-Pb age constrained; RO—Re-Os; SN—Sm-Nd; S—stratigraphic correlation; Plat, Plong—latitude, longitude of the virtual geomagnetic pole (VGP). The seven quality criteria and "Q" factor are described by Van der Voo (1990).

*Mean pole by averaging the site VGPs from the lower parts of the Nanfen, Xinxing and Liulaobei formations obtained in this study.

†Siberia poles from the Aldan block are rotated to Anabar according to Evans (2009).

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

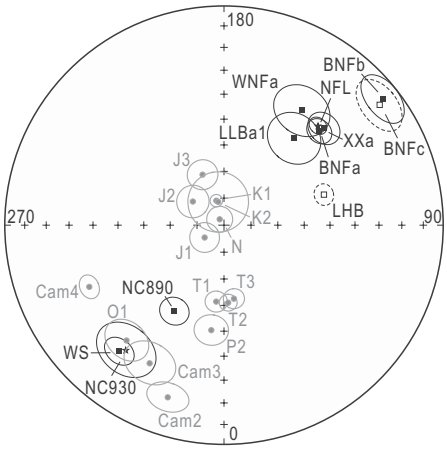


Figure 12. Equal-area projection of the North China Craton's (NCC's) ca. 1120–890 Ma paleomagnetic poles with their confidence circles (A_{95}) (black) and the younger poles of the NCC with their A_{95} (gray). Solid/open symbol indicates the pole in Northern/Southern Hemisphere. Precambrian pole abbreviations are identified in Table 3. Cam2, Cam3, Cam4, O1, P2, T1, T2, T3, J1, J2 and N data are after Huang et al. (2008). J3 data are from Ren et al. (2016). K1 data are from Ren et al. (2018). K2 data are from Zhao et al. (1996). Geologic time abbreviations: Cam2/Cam3/Cam4—Series 2/3/4 of Cambrian; O1—Early Ordovician; P2—Late Permian; T1—Early Triassic; T2—Middle Triassic; T3—Late Triassic; J1—Early Jurassic; J2—Middle Jurassic; J3—Late Jurassic; K1—Early Cretaceous; K2—Late Cretaceous; N—Cenozoic. The stratigraphic correlation of the Cambrian is after Zhu et al. (2019).

Nanfen Fm, being characterized by steep-up inclination. The upper one is in the uppermost of the Lower Member of the Nanfen Fm and Middle Member of the Nanfen Fm, being characterized by steep-down inclination. Multiple reversals were also observed between the two major polarity intervals (Fig. 6A). The Nanfen Fm in the Dalian region shows similar polarity pattern (Fig. 6B), but the Nanfen Fm in this region has not been sub-classified yet. Based on the lithological correlation and the ChRM direction similarities, we correlated "Section A" in Figure 6B to the lower of the upper polarity interval and correlated "Section B" to the lower polarity interval. All the strata of the Xinxing Fm sampled in the Xuzhou region seemingly record only the upper polarity interval (Fig. 6C), while the strata of Liulaobei Fm we sampled record only the lower polarity interval (Fig. 6D). As we will discuss in the last part of the paper, an apparent polar wander path (APWP) comparison between the NCC and Laurentia may permit correlating the upper polarity interval identified in this study to the Portage Lake normal-polarity zone (after ca. 1099 Ma; Swanson-Hysell et al., 2019).

Quality of the Paleomagnetic Results from the Sedimentary Rocks

The stratigraphic sequence and ChRM similarities may suggest that the strata that preserve the notably steep remanence in the study region are coeval. We calculated a mean pole for the lower part of the Nanfen Fm and its equivalents by averaging the 69-site VGPs from the lower parts of the Nanfen, Xinxing, and Liulaobei formations, which falls at: 38.6 °N, 136.7 °E (A_{95} = 3.2°, N = 69, pole "NFL" in Table 1). The two polarity groups are consistent with an antipodal common mean at the 95% confidence level and pass the bootstrap reversal test (Tauxe, 2010).

Several lines of evidence support the primary origin interpretation for the ChRMs isolated from the Nanfen, Xinxing, and Liulaobei formations in the four studied regions: (1) The AMS measurements indicate that the specimens from the Nanfen Fm still preserve primary sedimentary fabrics (Figs. DR2A–B). (2) Three distinguishable paleo-poles (poles "BNFa," "BNFb," and "BNFc") were isolated from different parts of the Nanfen Fm, which could rule out any strong regional remagnetization. (3) The presence of reversals. (4) All the paleo-poles acquired from the four studied formations could distinct obviously from other obtained younger paleo-poles of the NCC (see Fig. 12). (5) The poles "BNFa," "WNFa," "XXa," and "LLBa1" are consistent across the four distant correlative stratigraphic sections (Table 1; see Fig. 12), which could rule out the possibility of local tectonic rotation.

Paleomagnetism of the Sills in the Xuzhou Region

In the Zhongyangchang section, 35 oriented core samples were collected from the sill #15XZ (Fig. 4B). A total of 35 specimens, each representing one sample, underwent stepwise TD. The NRM intensities of the specimens are 0.9–1.6 A/m. The LTC remanence was identified and removed in temperatures below 350 °C in most cases (Fig. 13A). The LTC directions in situ cluster near that of the recent geomagnetic field in this region (Fig. DR5E) and are hence interpreted as a VRM of the recent geomagnetic field. The ChRM directions were usually identified in the fine-grained specimens, but could be occasionally identified in the coarse-grained ones. The ChRM was generally isolated between ~540 and 570 °C (Fig. 13A), suggesting that the main magnetic carrier is magnetite. The sample-mean (n = 26) direction is D = 152.0°, I = 61.9°, k = 103.3, α_{95} = 2.8° in geographic coordinates and D = 147.2°, I = 50.3°, k = 103.3, α_{95} = 2.8° in stratigraphic coordinates (Fig. 11G), corresponding to a VGP located at 17.8 °S, 146.7 °E (dp/dm = 2.5°/3.8°) in stratigraphic coordinates (Table 2).

In the Lutao section, 30 core samples were collected from the sill #17LT (Fig. 4B). A total of 30 specimens were subjected to stepwise TD. The NRM intensities of the specimens are in the range of 1.0–10 mA/m. Two remanence components were isolated from most cases. An LTC could be isolated between room temperature and 300 °C (Fig. 13B). The mean direction of the LTCs in situ is close to that of the recent geomagnetic field (D = 0°, I = 53.6°; calculated based on the GAD model; Fig. DR5F), suggesting an origin of viscous magnetization acquired in recent times. The ChRM was defined from 23 specimens, generally between ~500 and 570 °C. The sample-mean direction of the sill #17LT is D = 325.9°, I = –0.6°, k = 50.0, α_{95} = 4.3° in geographic coordinates and D = 322.6°, I = –37.6°, k = 50.0, α_{95} = 4.3° in stratigraphic coordinates (Fig. 11H; Table 2), corresponding to a VGP located at 24.3 °N, 336.0 °E (dp/dm = 3.0°/5.1°) in stratigraphic coordinates.

In the Paifang section, 54 core samples were collected from the sill #17PF (Fig. 4B). A total of 54 specimens underwent stepwise TD. The NRM intensities of the specimens are in the range of 0.1–1.5 A/m. An LTC could be isolated between room temperature and 300 °C, and the in situ mean direction is D = 346.7°, I = 41.5°, α_{95} = 8.5° (Fig. DR5G). This indicates that the LTC is a VRM of the recent geomagnetic field. The ChRM was generally well identified from 31 specimens with unblocking temperature up to ~575 °C (Fig. 13C), suggesting that the ChRM is carried by magnetite. The sample-mean direction of the sill #17PF is D = 332.2°, I = –7.6°, k = 62.6, α_{95} = 3.3° in geographic coordinates and D = 332.2°, I = –31.6°, k = 62.6, α_{95} = 3.3° in stratigraphic coordinates (Fig. 11I; Table 2), corresponding to a VGP located at 32.3 °N, 329.4 °E (dp/dm = 2.1°/3.7°) in stratigraphic coordinates.

Baked-Contact Test between the Sill #15XZ and Host Niyuan Fm

Sites 17XZ12-15 and 17XZ11-17 are situated ~0.5–2.2 m above the sill #15XZ (Figs. 4B and 14D). These whitish carbonate rocks were baked by the sill and presented partial alteration to marble. The NRM intensities of the specimens are 0.1–2.0 mA/m. An LTC was identified in temperatures below 300 °C in most cases. The LTC remanence directions in situ are close to that of the recent geomagnetic field (Fig. DR5H),

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

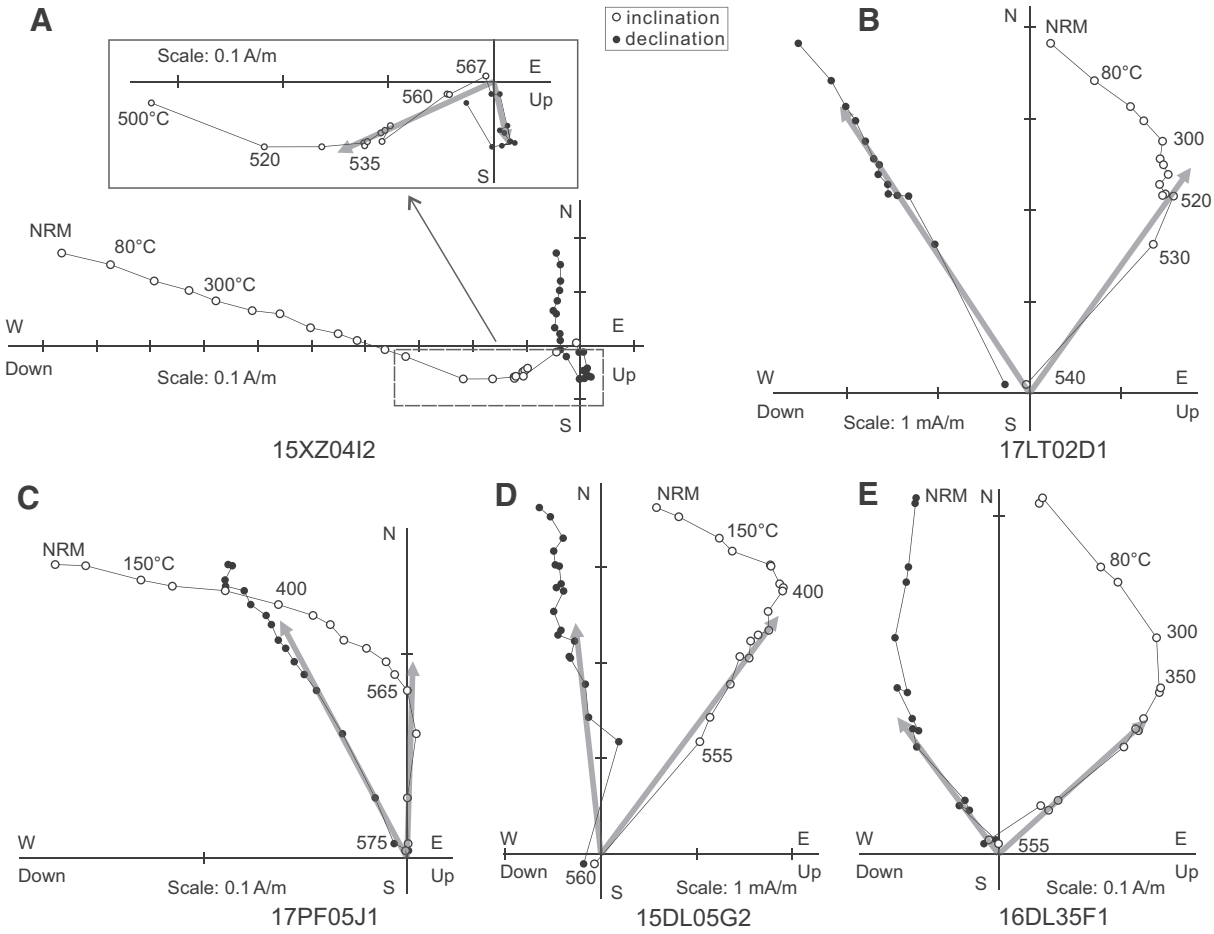


Figure 13. Typical demagnetization trajectories of representative specimens from the ca. 945–920 Ma sills. (A) Sill #15XZ, (B) sill #17LT and (C) sill #17PF in the Xuzhou region. (D) Sill #15DL and (E) sill #16DL in the Dalian region. The directions of remanence are all in geographic coordinates. Symbols are as for Figure 10. NRM—natural remanent magnetization.

suggesting an origin of viscous magnetization acquired in recent times. A stable HTC could generally be defined in temperatures between ~270 °C and ~350 °C (Fig. 14B), indicating that the HTC is likely carried by pyrrhotite. The site-mean directions are overlap with the ChRM direction of the sill #15XZ at the 95% confidence level (Fig. 14C; Table 2).

Five sites (17XZ24-28) were collected from unbaked dolostone of the Niyuan Fm for a contact test (Fig. 4B). These sites are far from (~200 m above) the sill #15XZ (Fig. 14D). The NRM intensities of the specimens are 0.2–1.2 mA/m. Two components of the remanent magnetizations were identified. An LTC was isolated in temperatures below 300 °C (Fig. 14A). The LTC remanence directions in situ are close to that of the recent geomagnetic field (Fig. DR5I), which suggests that the magnetization is a VRM of the recent geomagnetic field. The HTC was generally defined in temperatures between ~400 °C and 520–560 °C (Fig. 14A), indicating that the HTC is carried by magnetite. The site-mean direction of the Niyuan Fm is: D = 183.6°, I = –35.3°, k = 284.4, α_{95} = 5.5° in stratigraphic coordinates (Fig. 14C; Table 2). This south upward direction is significantly different from the ChRM directions of the sill #15XZ and the baked dolostone of the Niyuan Fm. Therefore, it is a positive baked-contact test (Everitt and Clegg, 1962) which indicates that the sill #15XZ carries a primary thermal remanence from the time of initial cooling.

Paleomagnetism of the Sills in the Dalian Region

In the Daoshitun section, 42 core samples were collected from the sill #15DL (Fig. 3B). A total of 42 specimens underwent TD. The NRM intensities of those paleomagnetic specimens are in the range of 3.7–5.8 mA/m. The LTC remanence was identified and removed in temperatures below 350 °C (Fig. 13D) in most cases. The LTC directions in situ distribute around the recent geomagnetic field direction in the region (Fig. DR5J), representing a VRM of the recent geomagnetic field. Apart from some weathered specimens, the ChRM was well defined from 32 specimens, generally between ~460 °C and ~570 °C (Fig. 13D), suggesting that the ChRM is carried by magnetite. The sample-mean direction of the sill #15DL is D = 350.1°, I = –34.2°, k = 61.4, α_{95} = 3.3° in geographic coordinates and D = 1.8°, I = –34.2°, k = 61.4, α_{95} = 3.3° in stratigraphic coordinates (Fig. 11J; Table 2), corresponding to a VGP located at: 31.8 °N, 299.8 °E (dp/dm = 2.2°/3.8°) in stratigraphic coordinates.

In the Dingjiajuan section on the Changxingdao island, 49 core samples were collected from the sill #16DL (Fig. 3B). Forty-four specimens underwent stepwise TD, and five were subjected to stepwise AFD. The NRM intensities of the specimens are in the range of 1.00–400 mA/m. The dispersed LTC or low coercivity component could be identified from most samples before ~350 °C or 15 mT (Fig. DR5K). The ChRMs were isolated between ~425 °C and 520–565 °C or between ~30 mT and 120 mT. Typical demagnetiza-

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

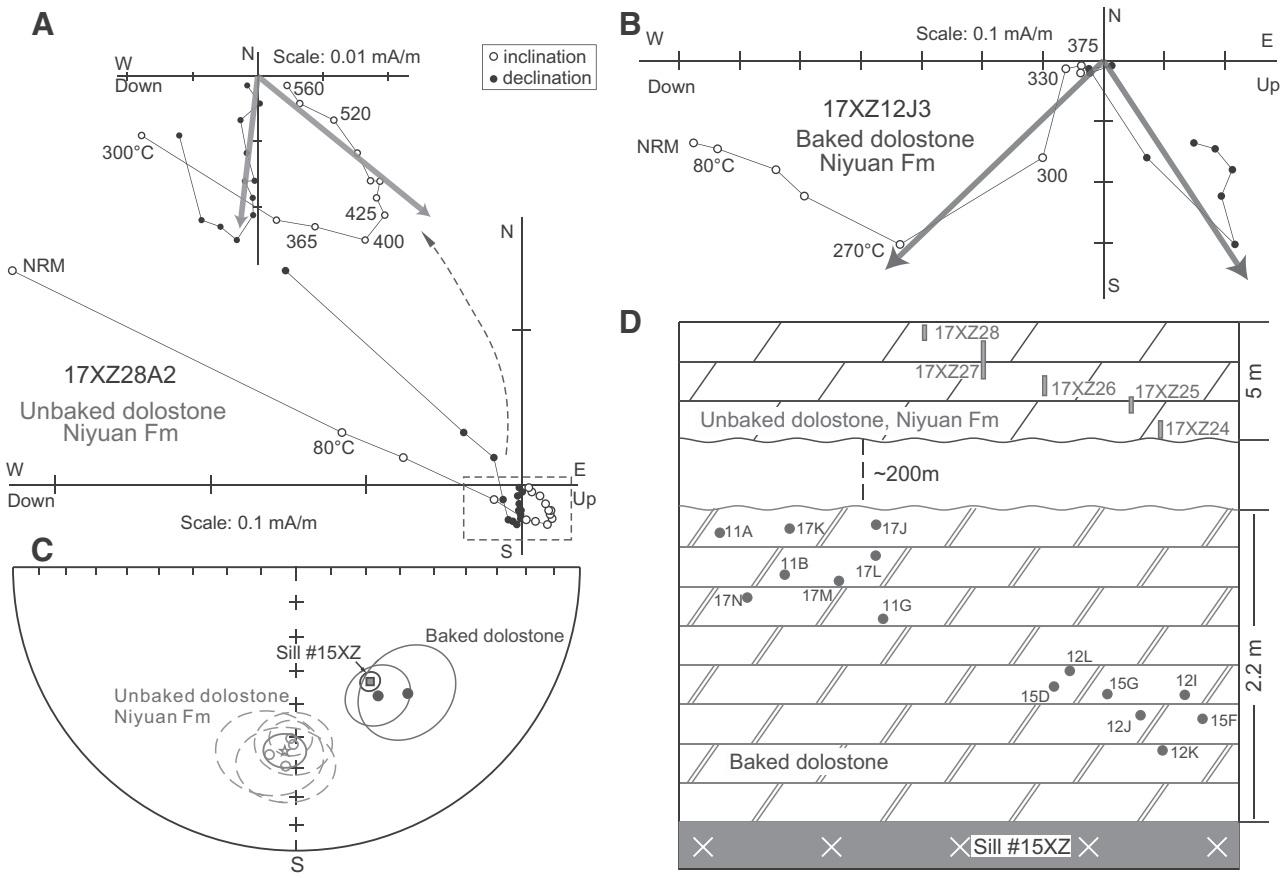


Figure 14. Baked-contact test between the host rocks of the Niyuan Formation (Fm) and the sill #15XZ. (A–B) Typical demagnetization trajectories of the representative specimens from the dolostone of the Niyuan Fm. (C) Equal area projections (after tilt correction). The high temperature component (HTC) directions of two sites (17XZ12-15 and 17XZ11-17) of the baked dolostone (dark-gray circles) from the Niyuan Fm are close to the mean direction of the sill #15XZ (dark-gray squares); while they are obviously different from the mean direction of the Niyuan Fm in this region and the mean direction of the unbaked sites 17XZ24–17XZ28 (in light gray), suggesting that the HTC of the baked dolostone was a thermal overprint acquired during the intrusion of the sill. Closed (open) symbols represent downward/upward magnetizations. The open star with 95% confidence circle represents mean direction of sites 17XZ24–17XZ28. (D) The distributions of the sill #15XZ, the baked dolostone specimens (from sites 17XZ12-15 and 17XZ11-17) and the five sites (17XZ24–17XZ28) of unbaked dolostone. Dark-gray dots represent paleomagnetic cores of the baked specimens; light-gray elongate rectangles represent paleomagnetic sites of the unbaked dolostone. Cross patterns for diabase, double slashes for baked dolostone and single slashes for unbaked dolostone.

tion behavior is shown in Figure 13E. The sample-mean direction of the sill #16DL is D = 323.6°, I = –21.5°, k = 34.4, α_{95} = 5.2° in geographic coordinates and D = 341.5°, I = –33.6°, k = 34.4, α_{95} = 5.2° in stratigraphic coordinates (Fig. 11K; Table 2), corresponding to a VGP located at: 29.5 °N, 321.5 °E (dp/dm = 3.4°/5.9°) in stratigraphic coordinates.

Secular Variation and Mean Pole of the Mafic Sills

The ChRMs of sills #15XZ, #17LT, and 17PF in the Xuzhou region are of dual polarity (Figs. 11G–11I), and the ChRMs identified from sill #15XZ yield a positive baked-contact test. Those new paleomagnetic results are consistent with the reported results from the Guishan mafic sill (sill #4 in Table 2) and significantly different from the ca. 890 Ma results reported by Fu et al. (2015) (see Fig. 12). No systematic differences have been observed between the VGPs acquired from those two sills in the Dalian region and the VGPs in the Xuzhou sills (Table 2). The similarity of the VGPs from the two sampled regions implies that the possibility of local tectonic rotation could be ruled out. The paleomagnetic directions from six ca. 945–920 Ma sills manifest a dual-polarity magnetization. The mean of two polarity-1 sills yields a longitude of 141.7 °E and a latitude of 24.4 °S, and four polarity-2 sills yields a mean longitude of 322.1 °E and a latitude of 30.1 °N. The two polarity groups are consistent with an antipodal common mean at the 95% confidence level, but the classification level of the reversal test is indeterminate (McFadden and McElhinny, 1990; critical angle γ_c = 25.0° and observed angular difference γ_o = 5.7°; Table 2).

Paleomagnetic records from one sill alone provide a spot reading of geomagnetic field behavior. The accuracy of the paleomagnetic results rests on the ability of the sills to time-average secular variation sufficiently (Tauxe, 1993). Therefore, the method of Biggin et al. (2008) was employed to evaluate the secular variation of the sills and the confidence interval (1σ) was calculated using the n-1 jackknife method (Efron, 1982). The VGP scatter S_B of the six sills is 12.2°, with 95% confidence interval (8.4°, 16.0°), which overlaps with the interval (9.5°, 12.7°) for paleolatitude 23.5°, according to the model for 0.5–1.5 Ga

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

(Veikkolainen and Pesonen, 2014). Coupled with the positive reversal test, we interpret the newly obtained ca. 945–920 Ma pole as a time-averaged paleomagnetic pole. We calculated a mean pole from the six sills (including the one from Fu et al., 2015) for the NCC (28.2 °S, 141.9 °E, A_{95} = 10.4°). Actually, this new ca. 945–920 Ma pole (pole "NC930") is roughly similar to the pole from the carbonate rocks of the Wangshan Fm (pole "WS"; Table 3), which passed a fold test and an inverse contact test (Fu at al., 2015). We consider that the Wangshan Fm is older than ca. 890 Ma (zircon U-Pb age of the mafic sills intruding the strata; Fig. 5C; Wang et al., 2012) and should be younger than ca. 950 Ma, which is the weight mean age of the youngest detrital zircon population from the underlying Shijia Fm (Fig. 5C; He et al. 2017). Although both poles "NC930" and "WS" are close to the Cambrian Series 3 (Cam_3) and Early Ordovician (O_1) poles of the NCC (Fig. 12), the positive baked-contact test between the sill #15XZ and the Niyuan Fm, the presence of dual polarity, and the consistency of the results in the two sampled regions strongly suggest that the paleomagnetic data are acceptable. Considering that there was no Cambrian–Ordovician tectonic event in the NCC that might have caused a remagnetization, we agree with Fu et al. (2015) and suggest that the ChRMs of both the Wangshan Fm and ca. 945–920 Ma sills are the primary paleomagnetic records of the NCC at ca. 945–920 Ma. The resemblance of the pole "NC930" to the younger Cam_3 and O_1 poles of the NCC could also be the self-intersection of the APWP (e.g., Pivarunas et al., 2018). Consequently, we consider the pole "NC930" as a key pole (Q = 6; Van der Voo, 1990) of the NCC at ca. 945–920 Ma.

DISCUSSION

Paleolatitudinal Variations of the NCC between circa 1120 and circa 890 Ma

In this study, we have achieved a series of poles from the NCC (summarized in Table 3; Fig. 12), including several poles from the Nanfen Fm and its equivalents, and the pole from the ca. 945–920 Ma mafic sills. The new pole "NFL" (Table 3), which ranges from ca. 1120 to 945 Ma in age, indicates a polar region position for the NCC. Two other poles from the middle part of the Nanfen Fm (poles "BNFb" and "BNFc" in Table 3), however, indicate a moderate paleolatitudinal region for the NCC. Moreover, the poles from the younger rock units (poles "LHB," "NC930," and "NC890" in Table 3) indicate that the NCC wandered in the low latitudinal regions in most time of the early Neoproterozoic. The paleomagnetic data obtained thus far may strongly suggest a large-scale equator-ward drift of the NCC occurred in the transition between Mesoproterozoic and Neoproterozoic.

The NCC's Paleogeographic Position in Supercontinent Rodinia

In a recent study, Fu et al. (2015) proposed a kinematic relationship between the NCC and Baltica in the early Neoproterozoic time based on an APWP comparison. Together with the broadly accepted "right-way-up" model for the connection of Laurentia and Baltica in Rodinia (Cawood and Pisarevsky, 2006; Li et al., 2008; Evans, 2009), the NCC was placed along the northwest side of Laurentia (present coordinates) in the Rodinia reconstruction. Our new geochronologic and paleomagnetic results have strengthened the quality of the pole obtained from the ca. 945–920 Ma mafic sills in the eastern NCC and still support the NCC–NW Laurentia connection proposed by Fu et al. (2015) (Fig. 15B). Furthermore, our new finding may extend the connection to ca. 1100 Ma. The pole from the lower part of the Nanfen Fm and its equivalents (pole "NFL" in Table 3) could match well with the ca. 1110–1100 Ma poles from the Midcontinent Rift volcanics in Laurentia (poles L17–L22 in Table 3; Fig. 15A), and two younger poles "BNFb" and "BNFc" (Table 3) from the middle part of the Nanfen Fm are consistent with the ca. 1095 Ma and ca. 1085 Ma poles of Laurentia (Table 3; Fig. 15A). The similarities between the APWPs also permit the correlation of the single polarity zone in the middle part of the Nanfen Fm with the Portage Lake normal-polarity zone (after ca. 1099 Ma; Swanson-Hysell et al., 2019).

The NCC–NW Laurentia connection model makes some testable predictions. Both our and previous studies have identified lots of ca. 1.2–1.0 Ga detrital zircons from the Diaoyutai, Qiaotou, Xinxing, and Shouxian formations in the eastern NCC (Figs. 1B and 5; Fig. DR4; Luo et al., 2006; Li et al., 2007; Hu et al., 2012; Yang et al., 2012), and the Hujiertu Fm in the northern margin of the NCC (Liu et al., 2017). However, magmatic rocks of this age range are rarely reported in the NCC, except the ca. 1195 Ma hornblende-bearing granites in a limited area of the northern Nangrim block at the northeastern part of the NCC (Zhao et al., 2006). Some researchers thus suggested that those Grenville-age detrital zircons should not be sourced from the NCC but from other cratons once juxtaposed to the NCC (e.g., Hu et al., 2012; Liu et al., 2017). Those detrital zircons isolated from the Diaoyutai, Qiaotou, Xinxing, and Shouxian formations in the eastern margin of the NCC also display age peaks at 1.68–1.58 Ga, 1.50–1.34 Ga, and 1.20–1.00 Ga (Figs. 16D–16G), which are consistent with the age peaks of the late Mesoproterozoic to early Neoproterozoic strata in the Mackenzie and Amundsen basins, northwest Canada (Figs. 16A and 16B; Turner et al., 2016; Rainbird et al., 2017). It is interpreted that the Grenville-age detrital zircons in NW Canada were possibly transported from the Grenville Province through a big river system across the Laurentian Craton (Rainbird et al., 2012, 2017). Whether the detrital zircons of similar ages in the NCC Precambrian strata also come from the Grenville Province or whether a Grenville-age magmatic belt once existed close to the NW Laurentia and NCC needs more investigations.

Notably, existing paleomagnetic data support a mid-Proterozoic location of Siberia near the northern margin of Laurentia (Gallet et al., 2000; Pavlov et al. 2002; Didenko et al., 2009; Wingate et al., 2009). A controversial issue is whether the two cratons were separated by several thousand kilometers (Pisarevsky and Natapov, 2003; Pisarevsky et al., 2008; Gladkochub et al., 2013, 2019; Merdith et al., 2017) or tightly juxtaposed (Pavlov et al., 2002; Metelkin et al., 2007; Evans and Mitchell, 2011; Ernst et al., 2016; Evans et al., 2016a). If we consider adopting the "tight connection" model of Laurentia and Siberia (Ernst et al., 2016; Evans et al., 2016a; Fig. 15), which is also supported by other geological observations, such as U-Pb-Hf detrital zircon data (Rainbird et al., 1998; Khudoley et al., 2001, 2015; Priyatkina et al., 2017; Fig. 16C) and correlation of igneous suites (Frost et al., 1998; Ernst et al., 2016; Donskaya et al., 2018), the paleomagnetic pole "NFL" of the NCC (Table 3) can fall close to the poles of the middle Riphean Malgina Fm and lower-middle part of Linok Fm in Siberia (Gallet et al., 2000; Pavlov et al., 2019; Table 3; Fig. 15C). This model allows for the northwest margin of the NCC to be facing the southwest side of Siberia (Fig. 15), which is similar to what was suggested by Li et al. (1996) based mainly on tectonostratigraphic correlations. In light of geochronologic data, some studies suggested the presence of Grenville-age (ca. 1.2–0.9 Ga) granitoid and metamorphic rocks in the Yenisey Ridge, southwestern Siberia (Fig. 15E; Likhanov et al., 2014, 2015; Nozhkin et al., 2015), but no Grenville-age orogenic events on the northern margin of the NCC have been reported (e.g., Lu et al., 2008; Zhang et al., 2008). It is easier to understand if we consider accepting a "loose connection"

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

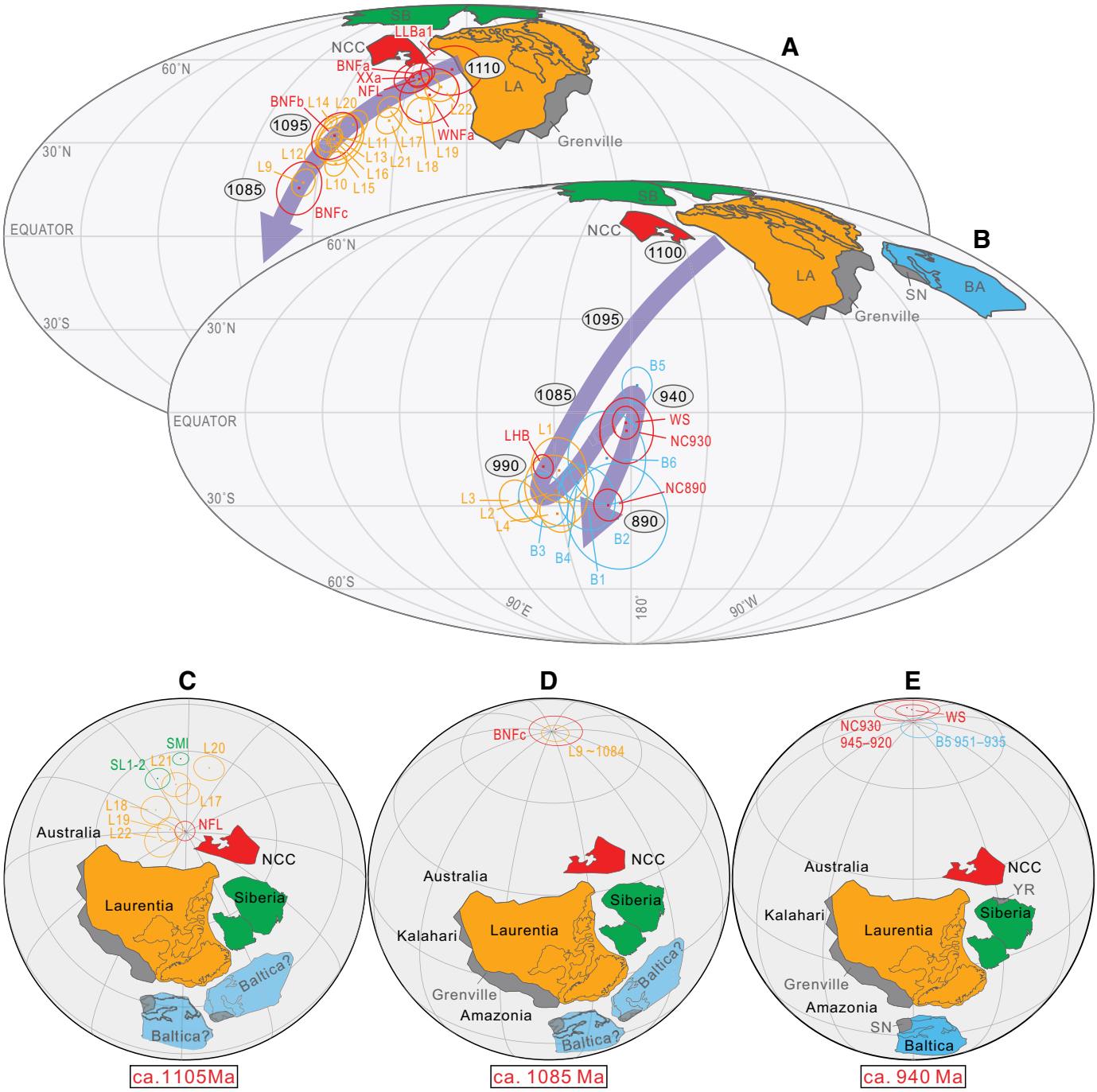


Figure 15. (A–B) Possible connection between the North China Craton (NCC) and Laurentia in Rodinia based on the paleomagnetic data available (in present North American coordinates). Arrows indicate the younging directions of the apparent polar wander path. The NCC and its poles are rotated to Laurentia using Euler rotation (50.6°, 19.2°, 56.2°). Baltica and its poles are rotated to Laurentia according to the "right-way-up" model of Laurentia–Baltica connection (Cawood and Pisarevsky, 2006) after ca. 990 Ma using Euler rotation (75.8°, –95.8°, –59.2°) after Li et al. (2008). Aldan is rotated to Anabar using Euler rotation (60°, 115°, 25°) after Evans (2009). Anabar is rotated to Laurentia using Euler rotation (77°, 98°, 137°) after Evans et al. (2016a). (C–E) Paleogeographic reconstructions of the NCC, Laurentia, Baltica and Siberia in Rodinia at ca. 1105, ca. 1085, and ca. 940 Ma, respectively. (C) Circa 1105 Ma (based on the poles "NFL" and "L19" in Table 3). (D) Circa 1085 Ma (based on the poles "L9" and "BNFc" in Table 3). In Figures 15C–D, Baltica are shown in both its pre–1265 Ma position using Euler rotation (45.0°, 7.5°, 44.9°) after Cawood et al. (2010) and post–990 Ma position, due to the opening time of the Asgard sea is not well paleomagnetically constrained (Cawood and Pisarevsky, 2017). (E) Circa 940 Ma (based on the poles "NC930," "WS," and "B5" in Table 3). Absolute longitudes for the entire assemblage are arbitrary. Cratons and their poles are matched in color. For pole symbols and more information see the text and Table 3. BA—Baltica; SB—Siberia; SN—Sveconorwegian Belt; LA—Laurentia; YR—Yenisey Ridge.

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

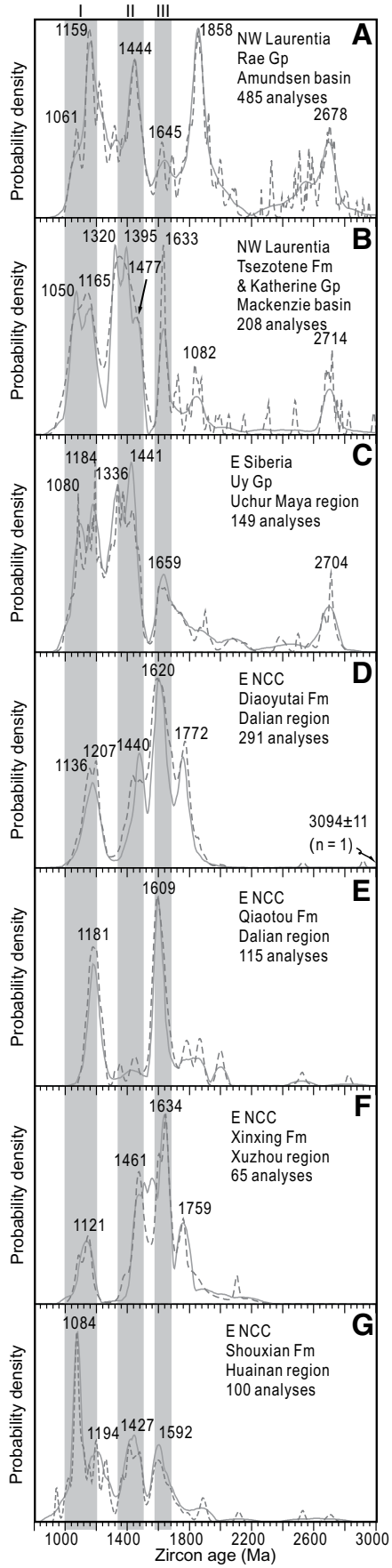


Figure 16. Probability density plots (dashed lines) and kernel density estimates (solid lines) of detrital zircon age distributions for different late Mesoproterozoic to early Neoproterozoic strata from the Laurentia, Siberia, and North China Craton (NCC). The age peaks (Ma) are marked. Data source: Rae Group (Gp)—Turner et al. (2016); Rainbird et al. (2017); Tsezotene Formation (Fm) and Katherine Gp—Rainbird et al. (2017); Uy Gp—Rainbird et al. (1998); Khudoley et al. (2015); Diaoyutai Fm—Yang et al. (2012); this study; Qiaotou Fm—this study; Xinxing Fm—Yang et al. (2012); Shouxian Fm—this study. For ages younger than 1500 Ma, the $^{206}Pb/^{238}U$ age is used, whereas the $^{207}Pb/^{206}Pb$ age is used for zircon with ages older than 1500 Ma. All data based on analyses with ≤5% discordance. The patterned polygons depict three major age ranges: I—1.00–1.20 Ga; II—1.34–1.50 Ga; III—1.58–1.68 Ga. Probability density plots—Ludwig (2003); kernel density estimates—Vermeesch (2012).

model between Laurentia and Siberia, in which the NCC–NW Laurentia–Baltica landmass did not connect with Siberia at that time.

A Neoproterozoic proximity between the São Francisco (SF)/Congo Craton and the NCC was speculated based on the correlation of igneous activities (Peng et al., 2011a; Cederberg et al., 2016; De Oliveira Chaves et al., 2019). However, recently published U-Pb baddeleyite dating and paleomagnetic data from the ca. 1100 Ma Huila-Epembe swarms in the Congo Craton support a relatively low latitude position of the SF/Congo Craton at ca. 1100 Ma (Salminen et al., 2018), while our ca. 1100 Ma paleomagnetic results place the NCC in a polar region. The post–1100 Ma motion of SF/Congo Craton and its ensuing location relative to Rodinia are less clear (e.g., Li et al., 2008; Merdith et al., 2017, and references therein), but its moderate-high latitude location at ca. 920 Ma (Evans et al., 2016b) is also incompatible with the low-latitude position of the NCC within Rodinia at that time. Collectively, available paleomagnetic evidence indicates that SF/Congo Craton might not directly connect with the NCC in Rodinia.

The NCC's connections in Rodinia suggested in this study are quite different from its paleomagnetically constrained location in the Nuna supercontinent (e.g., Zhang et al., 2012; Xu et al., 2014; Pisarevsky et al., 2014). If the NCC's connections are correct, the NCC should move an appreciable distance during the transition between Nuna and Rodinia.

CONCLUSIONS

In the eastern NCC, three sills were newly dated at ca. 945 Ma and one at ca. 920 Ma by using zircon U-Pb SIMS method. Geochronologic analyses of the detrital zircons and the mafic sill intrusions constrain the Nanfen Fm and its equivalents between ca. 1120 and ca. 945 Ma. A high-quality pole at (28.2 °S, 141.9 °E, A_{95} = 10.4°) was obtained from the sills. Primary origin interpretation of the pole was supported by the presence of reversals, a positive baked-contact test, and a secular variation test. Three other high-quality poles were also obtained from the lower part of the Nanfen Fm (and its equivalents) and the upper part of the Lower Member and Middle Member of the Nanfen Fm, respectively. The new results, together with the existing data, support a NCC–NW Laurentia connection between ca. 1120 and 890 Ma, aided by the "right-way-up" connection between Laurentia and Baltica in Rodinia.

ACKNOWLEDGMENTS

This study was supported by the National Science Foundation of China, (NSFC 41830215). The authors thank Wangqi Ren for assistance in the field and lab. We are grateful to Prof. Ian Dalziel for helpful comments and suggestions on earlier versions of the manuscript. We appreciate the discussions with Prof. Ganqing Jiang and Dr. Fred Bowyer. We gratefully acknowledge helpful reviews from Dr. Nick Swanson-Hysell and an anonymous reviewer, and editorial comments from Profs. Wenjiao Xiao and Zheng-Xiang Li. This paper is a contribution to IGCP 648.

REFERENCES CITED

ABGMR (Anhui Bureau of Geology and Mineral Resources), 1979, Geological Map of Shouxian I-50-XXXIII, scale 1:200,000, 1 sheet [in Chinese].

ABGMR (Anhui Bureau of Geology and Mineral Resources), 1987, Regional Geology of Anhui Province: Beijing, Geological Publishing House, 721 p. [in Chinese with English abstract].

Biggin, A.J., Strik, G.H.M.A., and Langereis, C.G., 2008, Evidence for a very-long-term trend in geomagnetic secular variation: Nature Geoscience, v. 1, no. 6, p. 395–398, https://doi.org/10.1038/ngeo181.

Boyden, J., Müller, R.D., Gurnis, M.S., Torsvik, T., Clark, J.A., Turner, M., Ivey-Law, H., Watson, R.J., and Cannon, J.S., 2011, Next-generation plate-tectonic reconstructions using GPlates, *in* Keller, G.R., and Baru, C., eds., Geoinformatics: Cyberinfrastructure for the Solid Earth Sciences: Cambridge, UK, Cambridge University Press, p. 95–113, https://doi.org/10.1017/CBO9780511976308.008.

Brown, L.L., and McEnroe, S.A., 2004, Palaeomagnetism of the Egersund–Ogna anorthosite, Rogaland, Norway, and the position of Fennoscandia in the Late Proterozoic: Geophysical Journal International, v. 158, no. 2, p. 479–488, https://doi.org/10.1111/j.1365-246X.2004.02349.x.

Brown, L.L., and McEnroe, S.A., 2012, Paleomagnetism and magnetic mineralogy of Grenville metamorphic and igneous rocks, Adirondack Highlands, USA: Precambrian Research, v. 212–213, p. 57–74, https://doi.org/10.1016/j.precamres.2012.04.012.

Brown, L.L., and McEnroe, S.A., 2015, 916 Ma pole for southwestern Baltica: palaeomagnetism of the Bjerkreim-Sokndal layered intrusion, Rogaland Igneous

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

Complex, southern Norway: Geophysical Journal International, v. 203, no. 1, p. 567–587, https://doi.org/10.1093/gji/ggv299.

Cao, R., Zhao, W., and Xiao, G., 1985, Late Precambrian stromatolites from north Anhui Province: Memoir of the Nanjing Institute of Geology and Palaeontology, Academia Sinica, v. 21, p. 1–54 [in Chinese with English abstract].

Cawood, P.A., and Pisarevsky, S.A., 2006, Was Baltica right-way-up or upside-down in the Neoproterozoic?: Journal of the Geological Society, v. 163, p. 753–759, https://doi.org/10.1144/0016-76492005-126.

Cawood, P.A., and Pisarevsky, S.A., 2017, Laurentia-Baltica-Amazonia relations during Rodinia assembly: Precambrian Research, v. 292, p. 386–397, https://doi.org/10.1016/j.precamres.2017.01.031.

Cawood, P.A., Strachan, R., Cutts, K., Kinny, P.D., Hand, M., and Pisarevsky, S., 2010, Neoproterozoic orogeny along the margin of Rodinia: Valhalla orogen, North Atlantic: Geology, v. 38, no. 2, p. 99–102, https://doi.org/10.1130/G30450.1.

Cederberg, J., Söderlund, U., Oiliviera, E.P., Ernst, R.E., and Pisarevsky, S.A., 2016, U-Pb baddeleyite dating of the Proterozoic Para de Minas dyke swarm in the Sao Francisco craton (Brazil)-implications for tectonic correlations with Siberia, Congo and North China cratons: GFF (Journal of the Geological Society of Sweden), v. 138, no. 1, p. 219–240, https://doi.org/10.1080/11035897.2015.1093543.

Chough, S.K., 2013, Geology and Sedimentology of the Korean Peninsula: Elsevier, Amsterdam, 363 p.

Cogné, J.P., 2003, PaleoMac: a Macintosh™ application for treating paleomagnetic data and making plate reconstructions: Geochemistry Geophysics Geosystems, v. 4, p. 8, https://doi.org/10.1029/2001GC000227.

Corfu, F., Hanchar, J.M., Hoskin, P.W.O., and Kinny, P., 2003, Atlas of zircon textures: Reviews in Mineralogy and Geochemistry, v. 53, p. 469–500, https://doi.org/10.2113/0530469.

Cumming, V.M., Poulton, S.W., Rooney, A.D., and Selby, D., 2013, Anoxia in the terrestrial environment during the late Mesoproterozoic: Geology, v. 41, p. 583–586, https://doi.org/10.1130/G34299.1.

Davis, D., and Green, J., 1997, Geochronology of the North American Midcontinent Rift in western Lake Superior and implications for its geodynamic evolution: Canadian Journal of Earth Sciences, v. 34, p. 476–488, https://doi.org/10.1139/e17-039.

Davis, D., and Paces, J., 1990, Time resolution of geologic events on the Keweenaw Peninsula and applications for development of the Midcontinent Rift system: Earth and Planetary Science Letters, v. 97, p. 54–64, https://doi.org/10.1016/0012-821X(90)90098-I.

De Oliveira Chaves, A., Ernst, R.E., Söderlund, U., Wang, X., and Naeraa, T., 2019, The 920–900 Ma Bahia-Gangila LIP of the São Francisco and Congo cratons and link with Dashigou-Chulan LIP of North China craton: New insights from U-Pb geochronology and geochemistry: Precambrian Research, v. 329, p. 124–137, https://doi.org/10.1016/j.precamres.2018.08.023.

Didenko, A.N., Vodovozov, V.Y., Pisarevsky, S.A., Gladkochub, D.P., Donskaya, T.V., Mazukabzov, A.M., Stanevich, A.M., Bibikova, E.V., and Kirnozova, T.I., 2009, Palaeomagnetism and U-Pb dates of the Palaeoproterozoic Akitkan Group (South Siberia) and implications for pre-Neoproterozoic tectonics, *in* Reddy, S.M., et al., eds., Palaeoproterozoic Supercontinents and Global Evolution: Geological Society of London Special Publication 323, p. 145–163, https://doi.org/10.1144/SP323.7.

Dong, L., Xiao, S., Shen, B., Yuan, X., Yan, X., and Peng, Y., 2008, Restudy of the worm-like carbonaceous compression fossils Protoarenicola, Pararenicola, and Sinosabellidites from early Neoproterozoic successions in North China: Palaeogeography, Palaeoclimatology, Palaeoecology, v. 258, p. 138–161, https://doi.org/10.1016/j.palaeo.2007.05.019.

Donskaya, T., Gladkochub, D., Ernst, R., Pisarevsky, S., Mazukabzov, A., and Demonterova, E., 2018, Geochemistry and petrogenesis of Mesoproterozoic dykes of the Irkutsk Promontory, southern part of the Siberian Craton: Minerals (Basel), v. 8, no. 12, p. 545, https://doi.org/10.3390/min8120545.

Duan, C., 1982, Late Precambrian algal megafossils *Chuaria* and *Tawuia* in some areas of eastern China: Alcheringa, v. 6, no. 1, p. 57–68, https://doi.org/10.1080/03115518208565420.

Dunlop, D.J., 2002, Theory and application of the Day plot (Mrs/Ms versus Hcr/Hc): 1. Theoretical curves and tests using titanomagnetite data: Journal of Geophysical Research, v. 107, B3, https://doi.org/10.1029/2001JB000486.

Dunlop, D.J., and Özdemir, Ö., 1997, Rock magnetism: Fundamentals and frontiers: Cambridge, UK, Cambridge University Press, 573 p., https://doi.org/10.1017/CBO9780511612794.

Efron, B., 1982, The jackknife, the bootstrap and other resampling plans. Regional Conference Series in Applied Mathematics, volume 38: SIAM, Philadelphia, 93 p, https://doi.org/10.1137/1.9781611970319.

Enkin, R.J., 1990, Formation et deformation de l'Asie depuis la fin de l'ere primaire: les apports de l'etude paleomagnetique des formations secondaires de Chine du Sud [Ph.D. thesis]: Université de Paris, 333 p.

Ernst, R.E., Hamilton, M.A., Söderlund, U., Hanes, J.A., Gladkochub, D.P., Okrugin, A.V., Kolotilina, T., Mekhonoshin, A.S., Bleeker, W., and Lecheminant, A.N., 2016, Long-lived connection between southern Siberia and northern Laurentia in the Proterozoic: Nature Geoscience, v. 9, p. 464–469, https://doi.org/10.1038/ngeo2700.

Evans, D.A.D., 2009, The palaeomagnetically viable, long-lived and all-inclusive Rodinia supercontinent reconstruction, *in* Murphy, J.B., Keppie, J.D., and Hynes, A., eds., Ancient Orogens and Modern Analogues: Geological Society [London] Special Publication 327, p. 371–404, https://doi.org/10.1144/SP327.16.

Evans, D.A.D., and Mitchell, R.N., 2011, Assembly and breakup of the core of Paleoproterozoic–Mesoproterozoic supercontinent Nuna: Geology, v. 39, p. 443–446, https://doi.org/10.1130/G31654.1.

Evans, D.A.D., Veselovsky, R.V., Petrov, P.Y., Shatsillo, A.V., and Pavlov, V.E., 2016a, Paleomagnetism of Mesoproterozoic margins of the Anabar Shield: A hypothesized billion-year partnership of Siberia and northern Laurentia: Precambrian Research, v. 281, p. 639–655, https://doi.org/10.1016/j.precamres.2016.06.017.

Evans, D.A.D., Trindade, R.I.F., Catelani, E.L., D'Agrella-Filho, M.S., Heaman, L.M., Oliveira, E.P., Söderlund, S., Ernst, R.E., Smirnov, A.V., and Salminen, J.M., 2016b, Return to Rodinia? Moderate to high palaeolatitude of the São Francisco/Congo craton at 920 Ma, *in* Li, Z.X, Evans, D.A.D. and Murphy, J.B., eds., Supercontinent Cycles through Earth History: London, Geological Society [London] Special Publication 424, p. 167–190, https://doi.org/10.1144/SP424.1.

Everitt, C.W.F., and Clegg, J.A., 1962, A field test for palaeomagnetic stability: Geophysical Journal of the Royal Astronomical Society, v. 6, p. 312–319, https://doi.org/10.1111/j.1365-246X.1962.tb00354.x.

Fairchild, L.M., Swanson-Hysell, N.L., Ramezani, J., Sprain, C.J., and Bowring, S.A., 2017, The end of Midcontinent Rift magmatism and the paleogeography of Laurentia: Lithosphere, v. 9, no. 1, p. 117–133, https://doi.org/10.1130/L580.1.

Fang, D., Guo, Y., and Zhu, Z., 1983, Study on the paleomagnetism of Upper Precambrian in Northern Jiangsu and correlation between the Upper Precambrian strata in South and North China: Scientia Geologica Sinica, v. 4, p. 324–337.

Fisher, R., 1953, Dispersion on a sphere: Proceedings of the Royal Society of London, v. A217, p. 295–305, https://doi.org/10.1098/rspa.1953.0064.

Frost, B.R., Avechenko, O.V., Chamberlain, K.R., and Frost, C.D., 1998, Evidence for extensive Proterozoic remobilization of the Aldan shield and implications for Proterozoic plate tectonic reconstructions of Siberia and Laurentia: Precambrian Research, v. 89, no. 1-2, p. 1–23, https://doi.org/10.1016/S0301-9268(97)00074-0.

Fu, X., 2016, New Paleomagnetic Results from the Huaibei Group and Neoproterozoic Mafic Sills, North China Craton, and Their Paleogeographic Implications [Ph.D. thesis]: China University of Geosciences, Beijing, 107 p. [in Chinese with English abstract].

Fu, X., Zhang, S., Li, H., Ding, J., Li, H., Yang, T., Wu, H., Yuan, H., and Lv, J., 2015, New paleomagnetic results from the Huaibei Group and Neoproterozoic mafic sills in the North China Craton and their paleogeographic implications: Precambrian Research, v. 269, p. 90–106, https://doi.org/10.1016/j.precamres.2015.08.013.

Gallet, Y., Pavlov, V., Semikhatov, M., and Petrov, P., 2000, Late Mesoproterozoic magnetostratigraphic results from Siberia: Paleogeographic implications and magnetic field behavior: Journal of Geophysical Research, v. 105, p. 16,481–16,499, https://doi.org/10.1029/1999JB900354.

Gallet, Y., Pavlov, V., Halverson, G., and Hulot, G., 2012, Toward constraining the long-term reversing behavior of the geodynamo: A new "Maya" superchron ~1 billion years ago from the magnetostratigraphy of the Kartochka Formation (southwestern Siberia): Earth and Planetary Science Letters, v. 339–340, p. 117–126, https://doi.org/10.1016/j.epsl.2012.04.049.

Gao, L., Zhang, C., Liu, P., Tang, F., Song, B., and Ding, X., 2009, Reclassification of the Meso-and Neoproterozoic chronostratigraphy of North China by SHRIMP zircon ages: Acta Geologica Sinica–English edition, v. 83, no. 6, p. 1074–1084.

Gao, Y.Y., Li, X.H., Griffin, W.L., O'Reilly, S.Y., and Wang, Y.F., 2014, Screening criteria for reliable U-Pb geochronology and oxygen isotope analysis in uranium-rich zircons: a case study from the Suzhou A-type granites, SE China: Lithos, v. 192–195, p. 180–191, https://doi.org/10.1016/j.lithos.2014.02.002.

Gladkochub, D.P., Stanevich, A.M., Mazukabzov, A.M., Donskaya, T.V., Pisarevsky, S.A., Nicoll, G., Motova, Z.L., and Kornilova, T.A., 2013, Early evolution of the Paleoasian ocean: LA-ICP-MS dating of detrital zircon from Late Precambrian sequences of the southern margin of the Siberian craton: Russian Geology and Geophysics, v. 54, p. 1150–1163, https://doi.org/10.1016/j.rgg.2013.09.002.

Gladkochub, D.P., Donskaya, T.V., Stanevich, A.M., Pisarevsky, S.A., Zhang, S., Motova, Z.L., Mazukabzov, A.M., and Li, H., 2019, U-Pb detrital zircon geochronology and provenance of Neoproterozoic sedimentary rocks in southern Siberia: New insights into breakup of Rodinia and opening of Paleo-Asian Ocean: Gondwana Research, v. 65, p. 1–16, https://doi.org/10.1016/j.gr.2018.07.007.

Gong, Z., Evans, D.A.D., Elming, S.-Å., Söderlund, U., and Salminen, J.M., 2018, Paleomagnetism, magnetic anisotropy and U-Pb baddeleyite geochronology of the early Neoproterozoic Blekinge-Dalarna dolerite dykes, Sweden: Precambrian Research, v. 317, p. 14–32, https://doi.org/10.1016/j.precamres.2018.08.019 (corrigendum: http://dx.doi.org/10.1016/j.precamres.2018.10.013).

Halls, H., and Pesonen, L., 1982, Paleomagnetism of Keweenawan rocks, *in* Wold, R.J., and Hinze, W.J., eds., Geology and Tectonics of the Lake Superior Basin: Geological Society of America Memoir 156, p. 173–202, https://doi.org/10.1130/MEM156-p173.

He, T., Zhou, Y., Vermeesch, P., Rittner, M., Miao, L., Zhu, M., Carter, A., von Strandmann, P.A.P., and Shields, G.A., 2017, Measuring the 'Great Unconformity' on the North China Craton using new detrital zircon age data, *in* Turner, J.P., et al., eds., Geomechanics and Geology: Geological Society of London Special Publication 448, p. 145–159, https://doi.org/10.1144/SP448.14, https://doi.org/10.1144/SP448.14.

Heaman, L.M., Bowins, R., and Crocket, J., 1990, The chemical composition of igneous zircon suites: Implications for geochemical tracer studies: Geochimica et Cosmochimica Acta, v. 54, p. 1597–1607, https://doi.org/10.1016/0016-7037(90)90394-Z.

Henry, S., Mauk, F., and der Voo, R.V., 1977, Paleomagnetism of the upper Keweenawan sediments: Nonesuch Shale and Freda Sandstone: Canadian Journal of Earth Science, v. 14, p. 1128–1138.

Hnat, J.S., van der Pluijm, B.A., and Van der Voo, R., 2006, Primary curvature in the Mid-Continent Rift: Paleomagnetism of the Portage Lake Volcanics (northern Michigan, USA): Tectonophysics, v. 425, p. 71–82, https://doi.org/10.1016/j.tecto.2006.07.006.

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

Hoffman, P.F., 1991, Did the breakout of Laurentia turn Gondwana inside out?: Science, v. 252, p. 1409–1412, https://doi.org/10.1126/science.252.5011.1409.

Hong, Z., Huang, Z., and Liu, X., 1991, Upper Precambrian Geology in Southern Liaodong Peninsula: Beijing, Geological Publishing House, 189 p. [in Chinese].

Hoskin, P.W.O., and Schaltegger, U., 2003, The composition of zircon and igneous and metamorphic petrogenesis, *in* Hanchar, J.M., and Hoskin, P.W.O., eds., Zircon: Washington, D.C., Mineralogical Society of America, p. 27–62, https://doi.org/10.1515/9781501509322-005.

Hou, K.J., Li, Y.H., and Tian, Y.R., 2009, In situ U-Pb zircon dating using laser ablation–multi ion counting–ICP-MS: Mineralium Deposita, v. 28, p. 481–492 [in Chinese with English abstract].

Hu, B., Zhai, M., Li, T., Li, Z., Peng, P., Guo, J., and Kusky, T.M., 2012, Mesoproterozoic magmatic events in the eastern North China Craton and their tectonic implications: Geochronological evidence from detrital zircons in the Shandong Peninsula and North Korea: Gondwana Research, v. 22, p. 828–842, https://doi.org/10.1016/j.gr.2012.03.005.

Huang, B.C., Zhou, Y.X., and Zhu, R.X., 2008, Discussion on Phanerozoic evolution and formation of continental China, based on paleomagnetic studies: Earth Science Frontiers, v. 15, no. 3, p. 348–359 [in Chinese with English abstract].

IGCAGS, (Institute of Geology, Chinese Academy of Geological Sciences), 2002, China Geological Map (Second Edition), scale 1:4,000,000, 1 sheet [in Chinese].

JBGMR (Jiangsu Bureau of Geology and Mineral Resources), 1978, Geological Map of Xuzhou I-50-XVI, scale 1:200,000, 1 sheet, 256 p. text [in Chinese].

JBGMR (Jiangsu Bureau of Geology and Mineral Resources), 1984, Regional Geology of Jiangsu Province and Shanghai Municipality: Beijing, Geological Publishing House, 857 p. [in Chinese with English abstract].

Kean, W., Williams, I., and Feeney, J., 1997, Magnetism of the Keweenawan age Chengwatana lava flows, northwest Wisconsin: Geophysical Research Letters, v. 24, p. 1523–1526, https://doi.org/10.1029/97GL00993.

Khudoley, A.K., Rainbird, R.H., Stern, R.A., Kropachev, A.P., Heaman, L.M., Zanin, A.M., Podkovyrov, V.N., Belova, V.N., and Sukhorukov, V.I., 2001, Sedimentary evolution of the Riphean–Vendian basin of southeastern Siberia: Precambrian Research, v. 111, p. 129–163, https://doi.org/10.1016/S0301-9268(01)00159-0.

Khudoley, A., Chamberlain, K., Ershova, V., Sears, J., Prokopiev, A., MacLean, J., Kazakova, G., Malyshev, S., Molchanov, A., Kullerud, K., Toro, J., Miller, E., Veselovskiy, R., Li, A., and Chipley, D., 2015, Proterozoic supercontinental restorations: constraints from provenance studies of Mesoproterozoic to Cambrian clastic rocks, eastern Siberian Craton: Precambrian Research, v. 259, p. 78–94, https://doi.org/10.1016/j.precamres.2014.10.003.

Kirschvink, J., 1980, The least-squares line and plane and the analysis of palaeomagnetic data: Geophysical Journal International, v. 62, p. 699–718, https://doi.org/10.1111/j.1365-246X.1980.tb02601.x.

Kulakov, E.V., Smirnov, A.V., and Diehl, J.F., 2013, Paleomagnetism of 1.09 Ga Lake Shore Traps (Keweenaw Peninsula, Michigan): New results and implications: Canadian Journal of Earth Sciences, v. 50, p. 1085–1096, https://doi.org/10.1139/cjes-2013-0003.

LBGMR (Liaoning Bureau of Geology and Mineral Resources), 1972a, Geological Map of Fuzhou 10–51–2, scale 1:200,000, 1 sheet, 56 p. text [in Chinese].

LBGMR (Liaoning Bureau of Geology and Mineral Resources), 1972b, Geological Map of Luda 10-51-8, scale 1:200,000, 1 sheet, 56 p. text [in Chinese].

LBGMR (Liaoning Bureau of Geology and Mineral Resources), 1975, Geological Map of Liaoyang K-51-28, scale 1:200,000, 1 sheet [in Chinese].

LBGMR (Liaoning Bureau of Geology and Mineral Resources), 1989, Regional Geology of Liaoning Province: Beijing, Geological Publishing House, 856 p. [in Chinese with English abstract].

Li, X., Chen, F., Guo, J., Li, Q., Xie, L., and Siebel, W., 2007, South China provenance of the lower-grade Penglai Group north of the Sulu UHP orogenic belt, eastern China: Evidence from detrital zircon ages and Nd-Hf isotopic composition: Geochemical Journal, v. 41, p. 29–45, https://doi.org/10.2343/geochemj.41.29.

Li, X., Liu, Y., Li, Q., Guo, C., and Chamberlain, K.R., 2009, Precise determination of Phanerozoic zircon Pb/Pb age by multi-collector SIMS without external standardization: Geochemistry Geophysics Geosystems, v. 10, no. 4, https://doi.org/10.1029/2009GC002400.

Li, Z.X., Zhang, L., and Powell, C.M., 1996, Positions of the East Asian cratons in the Neoproterozoic supercontinent Rodinia: Australian Journal of Earth Sciences, v. 43, no. 6, p. 593–604, https://doi.org/10.1080/08120099608728281.

Li, Z.X., Bogdanova, S.V., Collins, A.S., Davidson, A., De Waele, B., Ernst, R.E., Fitzsimons, I.C.W., Fuck, R.A., Gladkochub, D.P., Jacobs, J., Karlstrom, K.E., Lu, S., Natapov, L.M., Pease, V., Pisarevsky, S.A., Thrane, K., and Vernikovsky, V., 2008, Assembly, configuration, and break-up history of Rodinia: a synthesis: Precambrian Research, v. 160, p. 179–210, https://doi.org/10.1016/j.precamres.2007.04.021.

Likhanov, I., Nozhkin, A., Reverdatto, V., and Kozlov, P., 2014, Grenville tectonic events and evolution of the Yenisei Ridge at the western margin of the Siberian Craton: Geotectonics, v. 48, p. 371–389, https://doi.org/10.1134/S0016852114050045.

Likhanov, I., Reverdatto, V., Kozlov, P., Khiller, V., and Sukhorukov, V., 2015, P–T–t constraints on polymetamorphic complexes of the Yenisey Ridge, East Siberia: implications for Neoproterozoic paleocontinental reconstructions: Journal of Asian Earth Sciences, v. 113, p. 391–410, https://doi.org/10.1016/j.jseaes.2014.10.026.

Lin, J., 1984, The Apparent Polar Wander Paths for the North and South China Blocks [Ph.D. thesis]: University of California, Santa Barbara, 264 p.

Lin, W., and Xing, Y., 1984, Study on the Sinian Metazoan Fossils in Liaonan Group: Bulletin, Shenyang Institute of Geology and Mineral Resources: Chinese Academy of Geological Science v. 10, p. 51–67 [in Chinese].

Liu, A., Feng, Q., and Tian, L., 2018, The Qingbaikou Macroalgae biota and their implications for Rodinia reconstruction: Earth Science, v. 43, no. 2, p. 475–490 [in Chinese with English abstract].

Liu, C., Zhao, G., Liu, F., and Shi, J., 2017, Detrital zircon U-Pb and Hf isotopic and whole-rock geochemical study of the Bayan Obo Group, northern margin of the North China Craton: implications for Rodinia reconstruction: Precambrian Research, v. 303, p. 372–391, https://doi.org/10.1016/j.precamres.2017.04.033.

Liu, Y., Gao, L., Liu, Y., Song, B., and Wang, Z., 2006, Zircon U–Pb dating for the earliest Neoproterozoic mafic magmatism in the southern margin of the North China Block: Chinese Science Bulletin, v. 51, p. 2375–2382, https://doi.org/10.1007/s11434-006-2114-0.

Liu, Y., Gao, S., Hu, Z., Gao, C., Zong, K., and Wang, D., 2010, Continental and oceanic crust recycling-induced melt-peridotite interactions in the Trans-North China Orogen: U-Pb dating, Hf isotopes and trace elements in zircons from mantle xenoliths: Journal of Petrology, v. 51, p. 537–571, https://doi.org/10.1093/petrology/egp082.

Lowrie, W., 1990, Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties: Geophysical Research Letters, v. 17, p. 159–162, https://doi.org/10.1029/GL017i002p00159.

Lu, S., Zhao, G., Wang, H., and Hao, G., 2008, Precambrian metamorphic basement and sedimentary cover of the North China Craton: a review: Precambrian Research, v. 160, no. 1-2, p. 77–93, https://doi.org/10.1016/j.precamres.2007.04.017.

Ludwig, K.R., 2003, User's Manual for Isoplot 3.00: A Geochronologic Toolkit for Microsoft Excel: Berkeley Geochronology Center Special Publication 4, 70 p.

Luo, Y., Sun, M., Zhao, G., Li, S., and Xia, X., 2006, LA-ICP-MS U-Pb zircon geochronology of the Yushulazi Group in the eastern block, North China craton: International Geology Review, v. 48, p. 828–840, https://doi.org/10.2747/0020-6814.48.9.828.

McCabe, C., and Van der Voo, R., 1983, Paleomagnetic results from the upper Keweenawan Chequamegon Sandstone: Implications for red bed diagenesis and Late Precambrian apparent polar wander of North America: Canadian Journal of Earth Sciences, v. 20, p. 105–112, https://doi.org/10.1139/e83-010.

McFadden, P.L., and McElhinny, M.W., 1990, Classification of the reversal test in palaeomagnetism: Geophysical Journal International, v. 103, no. 3, p. 725–729, https://doi.org/10.1111/j.1365-246X.1990.tb05683.x.

Merdith, A.S., Collins, A.S., Williams, S.E., Pisarevsky, S., Foden, J.F., Archibald, D., Blades, M.L., Alessio, B.L., Armistead, S., and Plavsa, D., 2017, A full-plate global reconstruction of the Neoproterozoic: Gondwana Research, v. 50, p. 84–134, https://doi.org/10.1016/j.gr.2017.04.001.

Mertanen, S., Pesonen, L., and Huhma, H., 1996, Paleomagnetism and Sm-Nd ages of the Neoproterozoic diabase dykes in Laanila and Kautokeino, northern Fennoscandia, *in* Brewer, T., ed., Precambrian crustal evolution in the North Atlantic region: Geological Society [London] Special Publication 112, p. 331–358, https://doi.org/10.1144/GSL.SP.1996.112.01.18.

Metelkin, D.V., Vernikovsky, V.A., and Kazansky, A.Yu., 2007, Neoproterozoic evolution of Rodinia: constraints from new paleomagnetic data on the western margin of the Siberian craton: Russian Geology and Geophysics, v. 48, p. 32–45, https://doi.org/10.1016/j.rgg.2006.12.004.

Nozhkin, A.D., Popov, N.V., Dmitrieva, N.V., Storozhenko, A.A., and Vasil'ev, N.F., 2015, Neoproterozoic collisional S-type granitoids of the Yenisei Ridge: petrogeochemical composition and U–Pb, Ar–Ar, and Sm–Nd isotope data: Russian Geology and Geophysics, v. 56, p. 689–695, https://doi.org/10.1016/j.rgg.2015.04.001.

Palmer, H., and Davis, D., 1987, Paleomagnetism and U-Pb geochronology of volcanic rocks from Michipicoten Island, Lake Superior, Canada: precise calibration of the Keweenawan polar wander track: Precambrian Research, v. 37, p. 157–171, https://doi.org/10.1016/0301-9268(87)90077-5.

Pavlov, V.E., and Gallet, Y., 2010, Variations in geomagnetic reversal frequency during the Earth's middle age: Geochemistry Geophysics Geosystems, v. 11, Q01Z10, https://doi.org/10.1029/2009GC002583.

Pavlov, V.E., Gallet, Y., Petrov, P.Yu., Zhuravlev, D.Z., and Shatsillo, A.V., 2002, Uy series and late Riphean sills of the Uchur-Maya area: isotopic and palaeomagnetic data and the problem of the Rodinia supercontinent: Geotectonics, v. 36, p. 278–292.

Pavlov, V.E., Gallet, Y., and Petrov, P.Yu., 2019, A new Siberian record of the ~1.0 Gyr-old Maya superchron: Precambrian Research, v. 320, p. 350–370, https://doi.org/10.1016/j.precamres.2018.11.005.

Peng, P., Bleeker, W., Ernst, R.E., Söderlund, U., and McNicoll, V., 2011a, U–Pb baddeleyite ages, distribution and geochemistry of 925 Ma mafic dykes and 900 Ma sills in the North China craton: Evidence for a Neoproterozoic mantle plume: Lithos, v. 127, p. 210–221, https://doi.org/10.1016/j.lithos.2011.08.018.

Peng, P., Zhai, M., Li, Q., Wu, F., Hou, Q., Li, Z., Li, T., and Zhang, Y., 2011b, Neoproterozoic (900 Ma) Sariwon sills in North Korea: geochronology, geochemistry and implications for the evolution of the south-eastern margin of the North China craton: Gondwana Research, v. 20, p. 243–254, https://doi.org/10.1016/j.gr.2010.12.011.

Piper, J.D.A., and Zhang, Q.R., 1997, Palaeomagnetism of Neoproterozoic glacial rocks of the Huabei shield: the North China block in Gondwana: Tectonophysics, v. 283, no. 1-4, p. 145–171, https://doi.org/10.1016/S0040-1951(97)00152-2.

Pisarevsky, S.A., and Natapov, L.M., 2003, Siberia and Rodinia: Tectonophysics, v. 375, p. 221–245, https://doi.org/10.1016/j.tecto.2003.06.001.

Pisarevsky, S.A., Wingate, M.T.D., Powell, C.M., Johnson, S., and Evans, D.A.D., 2003, Models of Rodinia assembly and fragmentation, *in* Yoshida, M., et al., eds., Proterozoic East Gondwana: Supercontinent assembly and breakup: Geological Society of London Special Publication 206, p. 35–55, https://doi.org/10.1144/GSL.SP.2003.206.01.04.

Pisarevsky, S.A., Natapov, L.M., Donskaya, T.V., Gladkochub, D.P., and Vernikovsky, V.A., 2008, Proterozoic Siberia: A promontory of Rodinia: Precambrian Research, v. 160, p. 66–76, https://doi.org/10.1016/j.precamres.2007.04.016.

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

Pisarevsky, S.A., Elming, S.-Å., Pesonen, L.J., and Li, Z.-X., 2014, Mesoproterozoic paleo-geography: supercontinent and beyond: Precambrian Research, v. 244, p. 207–225, https://doi.org/10.1016/j.precamres.2013.05.014.

Pivarunas, A.F., Meert, J.G., and Miller, S.R., 2018, Assessing the intersection/remagnetization puzzle with synthetic apparent polar wander paths: Geophysical Journal International, v. 214, no. 2, p. 1164–1172, https://doi.org/10.1093/gji/ggy216.

Priyatkina, N., Collins, W.J., Khudoley, A.K., Zastrozhnov, D., Ershova, V., Chamberlain, K., Shatsillo, A., and Proskurnin, V., 2017, The Proterozoic evolution of northern Siberian Craton margin: a comparison of U-Pb–Hf signatures from sedimentary units of the Taimyr orogenic belt and the Siberian platform: International Geology Review, v. 59, no. 13, p. 1632–1656, https://doi.org/10.1080/00206814.2017.1289341.

Rainbird, R.H., Stern, R.A., Khudoley, A.K., Kropachev, A.P., Heaman, L.M., and Sukhorukov, V.I., 1998, U–Pb geochronology of Riphean sandstone and gabbro from southeast Siberia and its bearing on the Laurentia–Siberia connection: Earth and Planetary Science Letters, v. 164, p. 409–420, https://doi.org/10.1016/S0012-821X(98)00222-2.

Rainbird, R.H., Cawood, P.A., and Gehrels, G.E., 2012, The Great Grenvillian sedimentation episode: Record of supercontinent Rodinia's assembly, *in* Busby, C., and Azor, A., eds., Recent Advances in Tectonics of Sedimentary Basins: Chichester, UK, Wiley-Blackwell Publishing Ltd., p. 583–601, https://doi.org/10.1002/9781444347166.ch29.

Rainbird, R.H., Rayner, N.M., Hadlari, T., Heaman, L.M., Ielpi, A., Tuner, E.C., and MacNaughton, R.B., 2017, Zircon provenance data record the lateral extent of pan-continental, early Neoproterozoic rivers and erosional unroofing history of the Grenville orogeny: Geological Society of America Bulletin, v. 129, no. 11-12, p. 1408–1423, 10.1130/B31695.1.

Ren, Q., Zhang, S., Li, H., Wu, H., Liang, Z., Miao, X., Zhao, H., Li, H., Yang, T., Pei, J., and Davis, G.A., 2016, Further paleomagnetic results from the ~155 Ma Tiaojishan formation, North China, and their tectonic implications: Gondwana Research, v. 35, p. 180–191, https://doi.org/10.1016/j.gr.2015.05.002.

Ren, Q., Zhang, S., Wu, Y., Yang, T., Gao, Y., Turbold, S., Zhao, H., Wu, H., Li, H., Fu, H., Xu, B., Zhang, J., and Tomurtogoo, O., 2018, New Late Jurassic to Early Cretaceous paleomagnetic results from North China and southern Mongolia and their implications for the evolution of the Mongol-Okhotsk suture: Journal of Geophysical Research. Solid Earth, v. 123, p. 10,370–10,398, https://doi.org/10.1029/2018JB016703.

Roy, J., and Robertson, W., 1978, Paleomagnetism of the Jacobsville Formation and the apparent polar wander path for the interval ~1100 to ~670 m.y. for North America: Journal of Geophysical Research, v. 83, p. 1289–1304, https://doi.org/10.1029/JB083iB03p01289.

Salminen, J., Hanson, R., Evans, D.A.D., Gong, Z., Larson, T., Walker, O., Gumsley, A., Söderlund, U., and Ernst, R., 2018, Direct Mesoproterozoic connection of the Congo and Kalahari cratons in proto-Africa: Strange attractors across supercontinental cycles: Geology, v. 46, no. 11, p. 1011–1014, https://doi.org/10.1130/G45294.1.

SBGMR (Shandong Bureau of Geology and Mineral Resources), 1961, Geological Map of Zaozhuang I-50-X, scale 1:200,000, 1 sheet [in Chinese].

SBGMR (Shandong Bureau of Geology and Mineral Resources), 1991, Regional Geology of Shandong Province: Beijing, Geological Publishing House, 533 p. [in Chinese with English abstract].

Sláma, J., Košler, J., Condon, D.J., Crowley, J.L., Gerdes, A., Hanchar, J.M., Horstwood, M.S.A., Morris, G.A., Nasdala, L., Norberg, N., Schaltegger, U., Schoene, B., Tubrett, M.N., and Whitehouse, M.J., 2008, Plešovice zircon—A new natural reference material for U-Pb and Hf isotopic microanalysis: Chemical Geology, v. 249, p. 1–35, https://doi.org/10.1016/j.chemgeo.2007.11.005.

Spencer, C.J., Kirkland, C.L., and Taylor, R.J.M., 2016, Strategies towards statistically robust interpretations of *in situ* U-Pb zircon geochronology: Geoscience Frontiers, v. 7, no. 4, p. 581–589, https://doi.org/10.1016/j.gsf.2015.11.006.

Stacey, J.S., and Kramers, J.D., 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221, https://doi.org/10.1016/0012-821X(75)90088-6.

Sun, W., Wang, G., and Zhou, B., 1986, Macroscopic worm-like body fossils from the Upper Precambrian (900–700 Ma), Huainan district, Anhui, China and their stratigraphic and evolutionary significance: Precambrian Research, v. 31, p. 377–403, https://doi.org/10.1016/0301-9268(86)90041-0.

Swanson-Hysell, N.L., Burgess, S.D., Maloof, A.C., and Bowring, S.A., 2014a, Magmatic activity and plate motion during the latent stage of Midcontinent Rift development: Geology, v. 42, p. 475–478, https://doi.org/10.1130/G35271.1.

Swanson-Hysell, N.L., Vaughan, A.A., Mustain, M.R., and Asp, K.E., 2014b, Confirmation of progressive plate motion during the Midcontinent Rift's early magmatic stage from the Osler Volcanic Group, Ontario, Canada: Geochemistry Geophysics Geosystems, v. 15, p. 2039–2047, https://doi.org/10.1002/2013GC005180.

Swanson-Hysell, N.L., Ramezani, J., Fairchild, L.M., and Rose, I.R., 2019, Failed rifting and fast drifting: Midcontinent Rift development, Laurentia's rapid motion and the driver of Grenvillian orogenesis: Geological Society of America Bulletin, v. 131, p. 913–940, https://doi.org/10.1130/B31944.1.

Symons, D.T.A., Kawasaki, K., and Diehl, J.F., 2013, Age and genesis of the White Pine stratiform copper mineralization, northern Michigan, USA, from paleomagnetism: Geofluids, v. 13, p. 112–126, https://doi.org/10.1111/gfl.12024.

Tang, Q., Pang, K., Xiao, S., Yuan, X., Ou, Z., and Wan, B., 2013, Organic-walled microfossils from the early Neoproterozoic Liulaobei Formation in the Huainan region of North China and their biostratigraphic significance: Precambrian Research, v. 236, p. 157–181, https://doi.org/10.1016/j.precamres.2013.07.019.

Tang, Q., Pang, K., Yuan, X., and Xiao, S., 2017, Electron microscopy reveals evidence for simple multicellularity in the Proterozoic fossil *Chuaria*: Geology, v. 45, p. 75–78, https://doi.org/10.1130/G38680.1.

Tarling, D.H., and Hrouda, F., 1993, The Magnetic Anisotropy of Rock: London, Chapman and Hall, 217 p.

Tauxe, L., 1993, Sedimentary records of relative paleointensity of the geomagnetic field: theory and practice: Reviews of Geophysics, v. 31, p. 319–354, https://doi.org/10.1029/93RG01771.

Tauxe, L., 2010, Essentials of Paleomagnetism: Berkeley, California, University of California Press, 512 p.

Tauxe, L., and Kodama, K., 2009, Paleosecular variation models for ancient times: Clues from Keweenawan lava flows: Physics of the Earth and Planetary Interiors, v. 177, p. 31–45, https://doi.org/10.1016/j.pepi.2009.07.006.

Torsvik, T., 2003, The Rodinia jigsaw puzzle: Science, v. 300, no. 5624, p. 1379–1381, https://doi.org/10.1126/science.1083469.

Turner, E.C., Long, D.G.F., Rainbird, R.H., Petrus, J.A., and Rayner, N.M., 2016, Late Mesoproterozoic rifting in Arctic Canada during Rodinia assembly: Impactogens, trans-continental far-field stress and zinc mineralization: Terra Nova, v. 28, p. 188–194, https://doi.org/10.1111/ter.12207.

Van der Voo, R., 1990, The reliability of paleomagnetic data: Tectonophysics, v. 184, p. 1–9, https://doi.org/10.1016/0040-1951(90)90116-P.

Veikkolainen, T., and Pesonen, L.J., 2014, Palaeosecular variation, field reversals and the stability of the geodynamo in the Precambrian: Geophysical Journal International, v. 199, p. 1515–1526, https://doi.org/10.1093/gji/ggu348.

Vermeesch, P., 2012, On the visualisation of detrital age distributions: Chemical Geology, v. 312, p. 190–194, https://doi.org/10.1016/j.chemgeo.2012.04.021.

Walderhaug, H.J., Torsvik, T.H., Eide, E.A., Sundvoll, B., and Bingen, B., 1999, Geochronology and palaeomagnetism of the Hunnedalen dykes, SW Norway: implications for the Sveconorwegian apparent polar wander loop: Earth and Planetary Science Letters, v. 169, p. 71–83, https://doi.org/10.1016/S0012-821X(99)00066-7.

Walderhaug, H.J., Torsvik, T., and Halvorsen, E., 2007, The Egersund dykes (SW Norway): a robust Early Ediacaran (Vendian) palaeomagnetic pole from Baltica: Geophysical Journal International, v. 168, p. 935–948, https://doi.org/10.1111/j.1365-246X.2006.03265.x.

Wang, G., Zhang, S., Li, S., Yan, Y., Dou, S., and Fang, D., 1984, Research on the Upper Precambrian of Northern Jiangsu and Anhui Provinces, Hefei: Anhui Press of Science and Technology, 209 p. [in Chinese].

Wang, Q., Yang, D., and Xu, W., 2012, Neoproterozoic basic magmatism in the southeast margin of North China Craton: evidence from whole-rock geochemistry, U–Pb and Hf isotopic study of zircons from diabase swarms in the Xuzhou-Huaibei area of China: Science China. Earth Sciences, v. 55, p. 1461–1479, https://doi.org/10.1007/s11430-011-4237-7.

Warnock, A.C., Kodama, K.P., and Zeitler, P.K., 2000, Using thermochronometry and low-temperature demagnetization to accurately date Precambrian paleomagnetic poles: Journal of Geophysical Research, v. 105, p. 19,435–19,453, https://doi.org/10.1029/2000JB900114.

Weil, A., Geissman, J., Heizler, M., and Van der Voo, R., 2003, Paleomagnetism of Middle Proterozoic mafic intrusions and Upper Proterozoic (Nankoweap) red beds from the Lower Grand Canyon Supergroup, Arizona: Tectonophysics, v. 375, p. 199–220, https://doi.org/10.1016/S0040-1951(03)00339-1.

White, L.T., and Ireland, T.R., 2012, High-uranium matrix effect in zircon and its implications for SHRIMP U-Pb age determinations: Chemical Geology, v. 306–307, p. 78–91, https://doi.org/10.1016/j.chemgeo.2012.02.025.

Wiedenbeck, M., Alle, P., Corfu, F., Griffin, W.L., Meier, M., Oberli, F., Vonquadt, A., Roddick, J.C., and Speigel, W., 1995, Three natural zircon standards for U–Th–Pb, Lu–Hf, trace-element and REE analyses: Geostandards Newsletter, v. 19, p. 1–23, https://doi.org/10.1111/j.1751-908X.1995.tb00147.x.

Wingate, M.T.D., Pisarevsky, S.A., Gladkochub, D.P., Donskaya, T.V., Konstantinov, K.M., Mazukabzov, A.M., and Stanevich, A.M., 2009, Geochronology and paleomagnetism of mafic igneous rocks in the Olenek Uplift, northern Siberia: Implications for Mesoproterozoic supercontinents and paleogeography: Precambrian Research, v. 170, p. 256–266, https://doi.org/10.1016/j.precamres.2009.01.004.

Xiao, S., Shen, B., Tang, Q., Kaufman, A.J., Yuan, X., Li, J., and Qian, M., 2014, Biostratigraphic and chemostratigraphic constraints on the age of early Neoproterozoic carbonate successions in North China: Precambrian Research, v. 246, p. 208–225, https://doi.org/10.1016/j.precamres.2014.03.004.

Xiao, S., and Tang, Q., 2018, After the boring billion and before the freezing millions: evolutionary patterns and innovations in the Tonian Period: Emerging Topics in Life Sciences, v. 2, no. 2, p. 161–171, https://doi.org/10.1042/ETLS20170165.

Xing, Y., 1989, The Upper Precambrian of China, Stratigraphy of China, No. 3: Beijing, Geological Publishing House, 314 p. [in Chinese].

Xu, H., Yang, Z., Peng, P., Meert, J.G., and Zhu, R., 2014, Paleo-position of the North China Craton within the supercontinent Columbia: constraints from new paleomagnetic results: Precambrian Research, v. 255, p. 276–293, https://doi.org/10.1016/j.precamres.2014.10.004.

Yang, D.B., Xu, W.L., Xu, Y.G., Wang, Q.H., Pei, F.P., and Wang, F., 2012, U–Pb ages and Hf isotope data from detrital zircons in the Neoproterozoic sandstones of northern Jiangsu and southern Liaoning Provinces, China: implications for the-late Precambrian evolution of the southeastern North China Craton: Precambrian Research, v. 216, p. 162–176, https://doi.org/10.1016/j.precamres.2012.07.002.

Yang, T.N., Peng, Y., Leech, M.L., and Lin, H.Y., 2011, Fold patterns indicating Triassic constrictional deformation on the Liaodong peninsula, eastern China, and tectonic implications: Journal of Asian Earth Sciences, v. 40, p. 72–83, https://doi.org/10.1016/j.jseaes.2010.08.017.

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user

Zang, W., and Walter, M., 1992, Late Proterozoic and Early Cambrian microfossils and biostratigraphy, northern Anhui and Jiangsu, central-eastern China: Precambrian Research, v. 57, p. 243–323, https://doi.org/10.1016/0301-9268(92)90004-8.

Zartman, R., Nicholson, S., Cannon, W., and Morey, G., 1997, U-Th-Pb zircon ages of some Keweenawan Supergroup rocks from the south shore of Lake Superior: Canadian Journal of Earth Sciences, v. 34, p. 549–561, https://doi.org/10.1139/e17-044.

Zhai, M., Shao, J., Hao, J., and Peng, P., 2003, Geological signature and possible position of the North China Block in the supercontinent Rodinia: Gondwana Research, v. 6, no. 2, p. 171–183, https://doi.org/10.1016/S1342-937X(05)70968-0.

Zhang, S., Li, Z., Wu, H., and Wang, H., 2000, New paleomagnetic results from the Neoproterozoic successions in southern North Block and paleogeographic implications: Science China Series D, v. 43, no. 1, p. 233–244, https://doi.org/10.1007/BF02911948.

Zhang, S., Li, Z.X., and Wu, H., 2006, New Precambrian palaeomagnetic constraints on the position of the North China Block in Rodinia: Precambrian Research, v. 144, p. 213–238, https://doi.org/10.1016/j.precamres.2005.11.007.

Zhang, S., Li, Z., Evans, D.A.D., Wu, H., Li, H., and Dong, J., 2012, Pre-Rodinia supercontinent Nuna shaping up: a global synthesis with new paleomagnetic results from North China: Earth and Planetary Science Letters, v. 353, p. 145–155, https://doi.org/10.1016/j.epsl.2012.07.034.

Zhang, S.H., Zhao, Y., Ye, H., and Hu, G.H., 2016, Early Neoproterozoic emplacement of the diabase sill swarms in the Liaodong Peninsula and pre-magmatic uplift of the southeastern North China Craton: Precambrian Research, v. 272, p. 203–225, https://doi.org/10.1016/j.precamres.2015.11.005.

Zhang, Y., Wu, F., Wilde, S.A., Zhai, M., Lu, X., and Zhang, H., 2008, Geochronology and tectonic implications of the "Proterozoic" Seluohe Group at the northern margin of the North China Craton: International Geology Review, v. 50, no. 2, p. 135–153, https://doi.org/10.2747/0020-6814.50.2.135.

Zhao, G., Cao, L., Wilde, S.A., Sun, M., Choe, W., and Li, S., 2006, Implications based on the first SHRIMP U-Pb zircon dating on Precambrian granitoid rocks in North Korea: Earth and Planetary Science Letters, v. 251, p. 365–379, https://doi.org/10.1016/j.epsl.2006.09.021.

Zhao, X.X., Coe, R.S., Gilder, S., and Frost, G., 1996, Palaeomagnetic constraints on the palaeogeography of China: implications for Gondwanaland: Australian Journal of Earth Sciences, v. 43, p. 643–672, https://doi.org/10.1080/08120099608728285.

Zhu, M., Yang, A., Yuan, J., Li, G., Zhang, J., Zhao, F., Ahn, S.Y., and Miao, L., 2019, Cambrian integrative stratigraphy and timescale of China: Science China. Earth Sciences, v. 62, p. 25–60, https://doi.org/10.1007/s11430-017-9291-0.

Zijderveld, J.D.A., 1967, A.C. demagnetization of rocks: Analysis of results, *in* Collinson, D.W., Creer, K.M., and Runcorn, S.K., eds., Methods in Paleomagnetism: Elsevier, Amsterdam, p. 254–286.

Science Editor: Wenjiao Xiao
Associate Editor: Zheng-Xiang Li

Manuscript Received 3 December 2018
Revised Manuscript Received 28 April 2019
Manuscript Accepted 10 June 2019

Printed in the USA

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/132/3-4/739/4946380/739.pdf
by Curtin University Library user